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(54) **METHODS AND DEVICES FOR UTILIZING BONDABLE MATERIALS**

(56) **References Cited**

U.S. PATENT DOCUMENTS

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319,296 A 6/1885 Molesworth

668,878 A 2/1901 Jensen

(Continued)

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FOREIGN PATENT DOCUMENTS

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CA 2641580 8/2007

CA 2680827 9/2008

(Continued)

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OTHER PUBLICATIONS

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Wong et al, Case Report: Proper Insertion Angle Is Essential to Prevent Intra-Articular Protrusion of a Knotless Suture Anchor in Shoulder Rotator Cuff Repair, Arthroscopy: The Journal of Arthroscopic and Related Surgery, vol. 26, No. 2 Feb. 2010: pp. 286-290.

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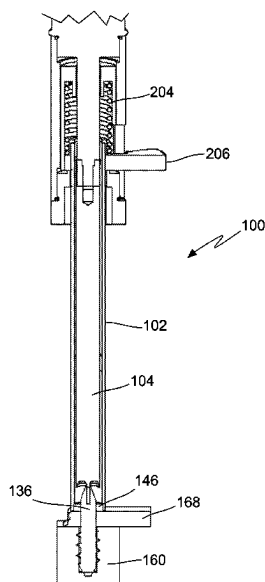
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(57) **ABSTRACT**

The invention primarily relates to fastening and stabilizing tissues, implants, and/or bondable materials, such as the fastening of a tissue and/or implant to a bondable material, the fastening of an implant to tissue, and/or the fastening of an implant to another implant. This may involve using an energy source to bond and/or mechanically to stabilize a tissue, an implant, a bondable material, and/or other bio-compatible material. The invention may also relate to the use of an energy source to remove and/or install an implant and/or bondable material or to facilitate solidification and/or polymerization of bondable material.

20 Claims, 34 Drawing Sheets



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2002/30971 (2013.01); *A61F 2002/30973* (2013.01); *A61F 2/32* (2013.01); *A61F 2/34* (2013.01); *A61F 2002/3401* (2013.01); *A61F 2002/3403* (2013.01); *A61F 2/36* (2013.01); *A61F 2/3603* (2013.01); *A61F 2/38* (2013.01); *A61F 2/3804* (2013.01); *A61F 2/3859* (2013.01); *A61F 2/389* (2013.01); *A61F 2/40* (2013.01); *A61F 2/4081* (2013.01); *A61F 2/4202* (2013.01); *A61F 2/44* (2013.01); *A61F 2002/444* (2013.01); *A61F 2/4455* (2013.01); *A61F 2002/4619* (2013.01); *A61F 2/468* (2013.01); *A61F 2002/4683* (2013.01); *A61F 2210/0004* (2013.01); *A61F 2210/0071* (2013.01); *A61F 2220/0058* (2013.01); *A61F 2230/0015* (2013.01); *A61F 2250/0068* (2013.01); *A61F 2310/00011* (2013.01); *A61F 2310/00365* (2013.01); *A61F 2310/00383* (2013.01); *A61F 2310/00952* (2013.01)

(56)

References Cited**U.S. PATENT DOCUMENTS**

668,879	A	2/1901	Miller
702,789	A	6/1902	Gibson
862,712	A	8/1907	Collins
2,121,193	A	6/1938	Hanicke
2,178,840	A	11/1939	Lorenian
2,187,852	A	1/1940	Friddle
2,199,025	A	4/1940	Conn
2,235,419	A	3/1941	Callahan
2,248,054	A	7/1941	Becker
2,270,188	A	1/1942	Longfellow
2,518,276	A	8/1950	Braward
2,557,669	A	6/1951	Lloyd
2,566,499	A	9/1951	Richter
2,621,653	A	12/1952	Briggs
2,725,053	A	11/1955	Bambara
2,830,587	A	4/1958	Everett
3,204,635	A	9/1965	Voss
3,347,234	A	10/1967	Voss
3,367,809	A	2/1968	Soloff
3,391,690	A	7/1968	Armao
3,400,805	A	9/1968	Thompson
3,477,429	A	11/1969	Sampson
3,513,848	A	5/1970	Winston
3,518,993	A	7/1970	Blake
3,577,991	A	5/1971	Wilkinson
3,596,292	A	8/1971	Erb
3,608,539	A	9/1971	Miller
3,625,220	A	12/1971	Engelsher
3,648,705	A	3/1972	Lary
3,653,388	A	4/1972	Tenckhoff
3,656,476	A	4/1972	Swinney
3,657,056	A	4/1972	Winston
3,678,980	A	7/1972	Gutshall
3,709,218	A	1/1973	Halloran
3,711,347	A	1/1973	Wagner
3,760,808	A	9/1973	Bleuer
3,788,318	A	1/1974	Kim
3,789,852	A	2/1974	Kim
3,802,438	A	4/1974	Wolvek
3,807,394	A	4/1974	Attenborough
3,809,075	A	5/1974	Matles
3,811,449	A	5/1974	Gravlee
3,825,010	A	7/1974	McDonald
3,830,524	A	8/1974	Abildgaard
3,833,003	A	9/1974	Taricco
3,835,849	A	9/1974	McGuire
3,842,824	A	10/1974	Neufeld
3,857,396	A	12/1974	Hardwick
3,867,932	A	2/1975	Huene
3,875,652	A	4/1975	Arnold
3,888,405	A	6/1975	Jones
3,898,992	A	8/1975	Balamuth

(56)

References Cited

U.S. PATENT DOCUMENTS

3,918,442	A	11/1975	Nikolaev	4,722,948	A	2/1988	Sanderson
3,968,800	A	7/1976	Vilasi	4,724,584	A	2/1988	Kasai
4,023,559	A	5/1977	Gaskell	4,738,255	A	4/1988	Goble
4,064,566	A	12/1977	Fletcher	4,739,751	A	4/1988	Sapega
4,089,071	A	5/1978	Kainberz	4,741,330	A	5/1988	Hayhurst
4,102,421	A	7/1978	Ozaki et al.	4,749,585	A	6/1988	Greco
4,156,574	A	5/1979	Boben	4,750,492	A	6/1988	Jacobs
4,164,794	A	8/1979	Spector	4,768,507	A	9/1988	Fischell
4,171,544	A	10/1979	Hench	4,772,286	A	9/1988	Goble
4,183,102	A	1/1980	Guiset	4,773,406	A	9/1988	Spector et al.
4,199,864	A	4/1980	Ashman	4,776,328	A	10/1988	Frey
4,200,939	A	5/1980	Oser	4,776,738	A	10/1988	Winston
4,210,148	A	7/1980	Stivala	4,776,851	A	10/1988	Bruchman
4,213,816	A	7/1980	Morris	4,781,182	A	11/1988	Purnell
4,235,233	A	11/1980	Mouwen	4,790,303	A	12/1988	Steffee
4,235,238	A	11/1980	Ogiu	4,792,336	A	12/1988	Hiavacek
4,257,411	A	3/1981	Cho	4,817,591	A	4/1989	Klalue
4,265,231	A	5/1981	Scheller	4,822,224	A	4/1989	Carl
4,281,649	A	8/1981	Derweduwen	4,823,794	A	4/1989	Pierce
4,291,698	A	9/1981	Fuchs	4,832,025	A	5/1989	Coates
4,309,488	A	1/1982	Heide	4,832,026	A	5/1989	Jones
4,320,762	A	3/1982	Bentov	4,834,752	A	5/1989	Van Kampen
4,341,574	A	7/1982	Landes	4,841,960	A	6/1989	Garner
4,351,069	A	9/1982	Ballintyn	4,843,112	A	6/1989	Gerhart
4,364,381	A	12/1982	Sher	4,846,812	A	7/1989	Walker
4,365,356	A	12/1982	Broemer	4,869,242	A	9/1989	Galluzzo
4,388,921	A	6/1983	Sutter	4,870,957	A	10/1989	Goble
4,395,798	A	8/1983	McVey	4,883,048	A	11/1989	Purnell
4,409,974	A	10/1983	Freedland	4,890,612	A	1/1990	Kensey
4,414,166	A	11/1983	CHarlson	4,895,148	A	1/1990	Bays
4,437,191	A	3/1984	Van der Zat	4,898,156	A	2/1990	Gattuma
4,437,362	A	3/1984	Hurst	4,899,729	A	2/1990	Giii
4,444,180	A	4/1984	Schneider	4,899,744	A	2/1990	Fujitsuka
4,448,194	A	5/1984	DiGiovanni	4,901,721	A	2/1990	Hakki
4,456,005	A	6/1984	Lichty	4,911,718	A	3/1990	Lee et al.
4,461,281	A	7/1984	Carson	4,921,479	A	5/1990	Grayzel
4,493,317	A	1/1985	Klaue	4,922,897	A	5/1990	Sapega
4,495,664	A	1/1985	Bianquaert	4,924,866	A	5/1990	Yoon
4,501,031	A	2/1985	McDaniel	4,932,960	A	6/1990	Green
4,504,268	A	3/1985	Herlitze	4,935,026	A	6/1990	Drews
4,506,681	A	3/1985	Mundell	4,935,028	A	6/1990	Drews
4,514,125	A	4/1985	Stol	4,945,625	A	8/1990	Winston
4,526,173	A	7/1985	Sheehan	4,946,468	A	8/1990	Libarid
4,532,926	A	8/1985	O'Holla	4,954,126	A	9/1990	Wallsten
4,535,772	A	8/1985	Sheehan	4,955,910	A	9/1990	Bolesky
4,547,327	A	10/1985	Bruins	4,957,498	A	9/1990	Caspari
4,549,684	A	10/1985	Telly et al.	4,961,741	A	10/1990	Hayhurst
4,556,350	A	12/1985	Bernhardt	4,962,882	A	10/1990	Sarazen, Jr. et al.
4,566,138	A	1/1986	Lewis	4,963,151	A	10/1990	Ducheyne
4,589,868	A	5/1986	Dretler	4,966,583	A	10/1990	Debbas
4,590,928	A	5/1986	Hunt	4,968,315	A	11/1990	Gattuma
4,597,379	A	7/1986	Kihn	4,969,888	A	11/1990	Scholten
4,599,085	A	7/1986	Riess	4,969,892	A	11/1990	Burton
4,601,893	A	7/1986	Cardinal	4,990,161	A	2/1991	Kampner
4,606,335	A	8/1986	Wedeen	4,994,071	A	2/1991	MacGregor
4,621,640	A	11/1986	Mulhollan	4,997,445	A	3/1991	Hodorek
4,630,609	A	12/1986	Chin	4,998,539	A	3/1991	Delsanti
4,632,101	A	12/1986	Freedland	5,002,550	A	3/1991	Li
4,645,503	A	2/1987	Lin	5,002,563	A	3/1991	Pyka
4,653,487	A	3/1987	Maale	5,009,652	A	4/1991	Morgan
4,657,460	A	4/1987	Bien	5,009,663	A	4/1991	Broome
4,659,268	A	4/1987	Del Mundo	5,009,664	A	4/1991	Sievers
4,662,063	A	5/1987	Collins	5,013,316	A	5/1991	Goble
4,662,068	A	5/1987	Polonsky	5,019,090	A	5/1991	Pinchuk
4,662,887	A	5/1987	Turner	5,021,059	A	6/1991	Kensey
4,669,473	A	6/1987	Richards	5,035,713	A	7/1991	Friis
4,685,458	A	8/1987	Leckrone	5,037,404	A	8/1991	Gold
4,691,741	A	9/1987	Affa et al.	5,037,422	A	8/1991	Hayhurst
4,705,040	A	11/1987	Mueller	5,041,093	A	8/1991	Chu
4,706,670	A	11/1987	Andersen	5,041,114	A	8/1991	Chapman
4,708,139	A	11/1987	Dunbar	5,041,129	A	8/1991	Hayhurst
4,713,077	A	12/1987	Small	5,046,513	A	9/1991	Gattuma
4,716,901	A	1/1988	Jackson	5,047,055	A	9/1991	Bao
4,718,909	A	1/1988	Brown	5,051,049	A	9/1991	Wills
4,722,331	A	2/1988	Fox	5,053,046	A	10/1991	Janese
				5,053,047	A	10/1991	Kuslich
				5,059,193	A	10/1991	Kuslich
				5,059,206	A	10/1991	Winters
				5,061,274	A	10/1991	Kensey

(56)

References Cited

U.S. PATENT DOCUMENTS

5,061,286	A	10/1991	Lyle	5,330,486	A	7/1994	Wilkinson
5,069,674	A	12/1991	Fearnot	5,336,231	A	8/1994	Adair
5,078,731	A	1/1992	Hayhurst	5,336,240	A	8/1994	Metzler
5,078,744	A	1/1992	Chvapil	5,339,799	A	8/1994	Kami
5,078,745	A	1/1992	Rhenter	5,349,956	A	9/1994	Bonutti
5,084,050	A	1/1992	Draenert	5,352,229	A	10/1994	Goble
5,084,051	A	1/1992	Tormala	5,354,298	A	10/1994	Lee
5,085,660	A	2/1992	Lin	5,354,302	A	10/1994	Ko
5,085,661	A	2/1992	Moss	5,365,784	A	11/1994	Morrissey
5,098,433	A	3/1992	Freedland	5,366,480	A	11/1994	Corriveau
5,098,434	A	3/1992	Serbousek	5,370,646	A	12/1994	Reese
5,098,436	A	3/1992	Ferrante	5,370,660	A	12/1994	Weinstein
5,100,405	A	3/1992	McLaren	5,372,146	A	12/1994	Branch
5,100,417	A	3/1992	Cerier	5,374,235	A	12/1994	Ahrens
5,102,417	A	4/1992	Palmaz	5,376,126	A	12/1994	Lin
5,108,399	A	4/1992	Eitenmuller	5,382,254	A	1/1995	McGarry
5,120,175	A	6/1992	Arbegast	5,383,883	A	1/1995	Wilk
5,123,520	A	6/1992	Schmid	5,383,905	A	1/1995	Golds
5,123,914	A	6/1992	Cope	5,391,173	A	2/1995	Wilk
5,123,941	A	6/1992	Lauren	5,395,308	A	3/1995	Fox
5,133,732	A	7/1992	Wiktor	5,397,311	A	3/1995	Walker
RE34,021	E	8/1992	Mueller	5,403,312	A	4/1995	Yates
5,141,520	A	8/1992	Goble	5,403,348	A	4/1995	Bonutti
5,147,362	A	9/1992	Goble	5,405,359	A	4/1995	Pierce
5,154,720	A	10/1992	Trott	5,411,523	A	5/1995	Goble
5,156,613	A	10/1992	Sawyer	5,413,585	A	5/1995	Pagedas
5,156,616	A	10/1992	Meadows	5,417,691	A	5/1995	Hayhurst
5,158,566	A	10/1992	Pianetti	5,417,701	A	5/1995	Holmes
5,158,934	A	10/1992	Ammann	5,417,712	A	5/1995	Whittaker
5,163,960	A	11/1992	Bonutti	5,423,796	A	6/1995	Skikhman
5,171,251	A	12/1992	Bregen	5,431,670	A	7/1995	Holmes
5,176,682	A	1/1993	Chow	5,439,470	A	8/1995	Li
5,179,964	A	1/1993	Cook	5,441,538	A	8/1995	Bonutti
5,180,388	A	1/1993	DiCarlo	5,443,512	A	8/1995	Parr
5,183,464	A	2/1993	Dubrul	5,447,503	A	9/1995	Miller
5,192,287	A	3/1993	Fournier	5,449,372	A	9/1995	Schmaltz
5,192,326	A	3/1993	Bao	5,449,382	A	9/1995	Dayton
5,197,166	A	3/1993	Meier	5,451,235	A	9/1995	Lock
5,197,971	A	3/1993	Bonutti	5,453,090	A	9/1995	Martinez
5,203,784	A	4/1993	Ross	5,456,722	A	10/1995	McLeod
5,203,787	A	4/1993	Noblitt	5,458,653	A	10/1995	Davison
5,208,950	A	5/1993	Merritt	5,462,561	A	10/1995	Voda
5,209,776	A	5/1993	Bass	5,464,424	A	11/1995	O'Donnell
5,217,493	A	6/1993	Raad	5,464,426	A	11/1995	Bonutti
5,219,359	A	6/1993	McQuilkin	5,464,427	A	11/1995	Curtis
5,226,899	A	7/1993	Lee	5,470,337	A	11/1995	Moss
5,234,006	A	8/1993	Eaton	5,472,444	A	12/1995	Huebner
5,234,425	A	8/1993	Fogarty	5,474,554	A	12/1995	Ku
5,236,438	A	8/1993	Wilk	5,478,351	A	12/1995	Meade
5,236,445	A	8/1993	Hayhurst	5,478,353	A	12/1995	Yoon
5,242,902	A	9/1993	Murphy	5,480,403	A	1/1996	Lee
5,254,113	A	10/1993	Wilk	5,486,197	A	1/1996	Le
5,258,007	A	11/1993	Spetzler	5,487,844	A	1/1996	Fujita
5,258,015	A	11/1993	Li	5,488,958	A	2/1996	Topel
5,258,016	A	11/1993	DiPoto	5,496,292	A	3/1996	Burnham
5,266,325	A	11/1993	Kuzma	5,496,335	A	3/1996	Thomason
5,269,783	A	12/1993	Sander	5,496,348	A	3/1996	Bonutti
5,269,785	A	12/1993	Bonutti	5,499,382	A	3/1996	Nusinov et al.
5,269,809	A	12/1993	Hayhurst	5,500,000	A	3/1996	Feagin
5,281,235	A	1/1994	Habermeyer	5,501,700	A	3/1996	Hirata
5,282,832	A	2/1994	Toso	5,504,977	A	4/1996	Weppner
5,290,281	A	3/1994	Tschakaloff	5,505,735	A	4/1996	Li
5,304,119	A	4/1994	Balaban	5,507,754	A	4/1996	Green
5,306,280	A	4/1994	Bregen	5,522,844	A	6/1996	Johnson
5,306,301	A	4/1994	Graf	5,522,845	A	6/1996	Wenstrom
5,306,309	A	4/1994	Wagner et al.	5,522,846	A	6/1996	Bonutti
5,315,741	A	5/1994	Dubberke	5,527,341	A	6/1996	Gogolewski
5,318,570	A	6/1994	Hood	5,527,342	A	6/1996	Pietrzak
5,318,588	A	6/1994	Horzewski	5,527,343	A	6/1996	Bonutti
5,320,611	A	6/1994	Bonutti	5,529,075	A	6/1996	Clark
5,329,846	A	7/1994	Bonutti	5,531,759	A	7/1996	Kensey
5,329,924	A	7/1994	Bonutti	5,534,012	A	7/1996	Bonutti
5,330,446	A	7/1994	Weldon et al.	5,534,028	A	7/1996	Bao
5,330,468	A	7/1994	Burkhart	5,540,718	A	7/1996	Bartlett
5,330,476	A	7/1994	Hiot	5,542,423	A	8/1996	Bonutti
				5,545,178	A	8/1996	Kensey
				5,545,180	A	8/1996	Le
				5,545,206	A	8/1996	Carson
				5,549,630	A	8/1996	Bonutti

(56)

References Cited

U.S. PATENT DOCUMENTS

5,549,631	A	8/1996	Bonutti	5,735,875	A	4/1998	Bonutti
5,556,402	A	9/1996	Xu	5,735,877	A	4/1998	Pagedas
5,569,252	A	10/1996	Justin	5,735,899	A	4/1998	Schwartz
5,569,305	A	10/1996	Bonutti	5,741,282	A	4/1998	Anspach
5,569,306	A	10/1996	Thal	5,752,952	A	5/1998	Adamson
5,571,190	A	11/1996	Ulrich et al.	5,752,974	A	5/1998	Rhee
5,573,517	A	11/1996	Bonutti	5,755,809	A	5/1998	Cohen
5,573,538	A	11/1996	Laboureau	5,762,458	A	6/1998	Wang
5,573,542	A	11/1996	Stevens	5,766,221	A	6/1998	Benderev
5,575,801	A	11/1996	Habermeyer	5,769,894	A	6/1998	Ferragamo
5,580,344	A	12/1996	Hasson	5,772,672	A	6/1998	Toy
5,584,835	A	12/1996	Greenfield	5,776,151	A	7/1998	Chan
5,584,860	A	12/1996	Goble	5,779,706	A	7/1998	Tschakaloff
5,584,862	A	12/1996	Bonutti	5,782,862	A	7/1998	Bonutti
5,591,206	A	1/1997	Moufarrege	5,785,713	A	7/1998	Jobe
5,593,422	A	1/1997	Muijs Van De Moer	5,792,096	A	8/1998	Rentmeester
5,593,425	A	* 1/1997	Bonutti B29C 66/8322 606/228	5,797,931	A	8/1998	Bito
5,593,625	A	1/1997	Rieble	5,800,537	A	9/1998	Bell
5,601,557	A	2/1997	Hayhurst	5,807,403	A	9/1998	Beyar
5,601,558	A	2/1997	Torrie	5,810,849	A	9/1998	Slavros
5,601,595	A	2/1997	Schwartz	5,810,853	A	9/1998	Yoon
5,607,427	A	3/1997	Tschakaloff	5,810,884	A	9/1998	Kim
5,609,595	A	3/1997	Pennig	5,814,072	A	9/1998	Bonutti
5,618,314	A	4/1997	Harwin	5,814,073	A	9/1998	Bonutti
5,620,461	A	4/1997	Muijs Van De Moer	5,817,107	A	10/1998	Schaller
5,626,612	A	5/1997	Bartlett	5,823,994	A	10/1998	Sharkey
5,626,614	A	5/1997	Hart	5,824,009	A	10/1998	Fukuda
5,626,718	A	5/1997	Philippe	5,830,125	A	11/1998	Scribner
5,628,751	A	5/1997	Sander	5,836,897	A	11/1998	Sakural
5,630,824	A	5/1997	Hart	5,839,899	A	11/1998	Robinson
5,634,926	A	6/1997	Jobe	5,843,178	A	12/1998	Vanney
5,643,274	A	7/1997	Sander	5,845,645	A	12/1998	Bonutti
5,643,293	A	7/1997	Kogasaka	5,851,185	A	12/1998	Berns
5,643,295	A	7/1997	Yoon	5,865,834	A	2/1999	McGuire
5,643,321	A	7/1997	McDevitt	5,866,634	A	2/1999	Tokushige
5,645,553	A	7/1997	Kolesa	5,868,749	A	2/1999	Reed
5,645,597	A	7/1997	Krapiva	5,874,235	A	2/1999	Chan
5,645,599	A	7/1997	Samani	5,879,372	A	3/1999	Bartlett
5,649,955	A	7/1997	Hashimoto	5,891,166	A	4/1999	Schervinsky
5,649,963	A	7/1997	McDevitt	5,891,168	A	4/1999	Thal
5,651,377	A	7/1997	O'Donnell	5,893,880	A	4/1999	Egan
5,658,313	A	8/1997	Thal	5,897,574	A	4/1999	Bonutti
5,658,408	A	8/1997	Frantz	5,899,911	A	5/1999	Carter
5,660,225	A	8/1997	Saffran	5,899,921	A	5/1999	Caspari
5,662,658	A	9/1997	Wenstrom	5,906,579	A	5/1999	Vander Salm
5,665,089	A	9/1997	Dall	5,906,625	A	5/1999	Bito
5,665,109	A	9/1997	Yoon	5,908,429	A	6/1999	Yoon
5,667,513	A	9/1997	Torrie	5,911,721	A	6/1999	Nicholson
5,669,917	A	9/1997	Sauer	5,918,604	A	7/1999	Whelan
5,674,240	A	10/1997	Bonutti	5,919,193	A	7/1999	Slavitt
5,681,310	A	10/1997	Yuan	5,919,194	A	7/1999	Anderson
5,681,333	A	10/1997	Burkhart	5,919,208	A	7/1999	Valenti
5,681,351	A	10/1997	Jamiolkowski	5,919,215	A	7/1999	Wiklund
5,681,352	A	10/1997	Clancy	5,921,986	A	7/1999	BOnutti
5,685,820	A	11/1997	Riek	5,925,064	A	7/1999	Meyers
5,688,283	A	11/1997	Knapp	5,928,244	A	7/1999	Tovey
5,690,654	A	11/1997	Ovil	5,928,267	A	7/1999	Bonutti
5,690,655	A	11/1997	Hart	5,931,838	A	8/1999	Vito
5,690,676	A	11/1997	DiPoto	5,931,869	A	8/1999	Boucher
5,693,055	A	12/1997	Zahiri	5,940,942	A	8/1999	Fong
5,697,950	A	12/1997	Fucci	5,941,900	A	8/1999	Bonutti
5,702,397	A	12/1997	Gonie	5,941,901	A	8/1999	Egan
5,702,462	A	12/1997	Oberlander	5,945,002	A	8/1999	Leukes et al.
5,707,395	A	1/1998	Li	5,947,982	A	9/1999	Duran
5,713,903	A	2/1998	Sander	5,948,000	A	9/1999	Larsen
5,713,921	A	2/1998	Bonutti	5,948,001	A	9/1999	Larsen
5,718,717	A	2/1998	Bonutti	5,948,002	A	9/1999	Bonutti
5,720,747	A	2/1998	Burke	5,951,590	A	9/1999	Goldfarb
5,720,753	A	2/1998	Sander	5,957,953	A	9/1999	DiPoto
5,725,541	A	3/1998	Anspach	5,961,499	A	10/1999	Bonutti
5,725,556	A	3/1998	Moser	5,961,521	A	10/1999	Roger
5,725,582	A	3/1998	Bevan	5,961,554	A	10/1999	Janson
5,730,747	A	3/1998	Ek	5,964,765	A	10/1999	Fenton
5,733,306	A	3/1998	Bonutti	5,964,769	A	10/1999	Wagner
				5,968,046	A	10/1999	Castleman
				5,968,047	A	10/1999	Reed
				5,980,520	A	11/1999	Vancaillie
				5,980,559	A	11/1999	Bonutti
				5,984,929	A	11/1999	Bashiri

(56)	References Cited			6,290,644 B1 *	9/2001	Green, II	A61B 17/0218 600/209
	U.S. PATENT DOCUMENTS			6,293,961 B2	9/2001	Schwartz	
				6,306,159 B1	10/2001	Schwartz	
5,989,282 A	11/1999	Bonutti		6,309,405 B1	10/2001	Bonutti	
5,993,458 A	11/1999	Vaitekunas		6,312,448 B1	11/2001	Bonutti	
5,993,477 A	11/1999	Vaitekunas		6,324,308 B1	11/2001	Chen et al.	
6,007,567 A	12/1999	Bonutti		6,328,480 B1	12/2001	Strike	
6,007,580 A	12/1999	Lehto		6,338,730 B1	1/2002	Bonutti	
6,010,525 A	1/2000	Bonutti		6,340,365 B2	1/2002	Dittrich	
6,010,526 A	1/2000	Sandstrom		6,348,056 B1	2/2002	Bates	
6,017,321 A	1/2000	Boone		6,358,271 B1	3/2002	Egan	
6,033,429 A	3/2000	Magovern		6,364,897 B1	4/2002	Bonutti	
6,033,430 A	3/2000	Bonutti		6,364,977 B1	4/2002	Simon	
6,045,551 A	4/2000	Bonutti		6,368,325 B1	4/2002	McKinley	
6,050,998 A	4/2000	Fletcher		6,368,343 B1	4/2002	Bonutti	
6,056,751 A *	5/2000	Fenton, Jr.	A61F 2/0811 606/232	6,371,957 B1	4/2002	Amrein	
				6,409,716 B1	6/2002	Sahatjian et al.	
6,056,772 A	5/2000	Bonutti		6,409,742 B1	6/2002	Fulton	
6,056,773 A	5/2000	Bonutti		6,409,743 B1	6/2002	Fenton	
6,059,797 A	5/2000	Mears		6,409,768 B1	6/2002	Tepic et al.	
6,059,817 A	5/2000	Bonutti		6,419,704 B1	7/2002	Ferree	
6,059,827 A	5/2000	Fenton		6,423,088 B1	7/2002	Fenton	
6,063,095 A	5/2000	Wang		6,425,919 B1	7/2002	Lambrecht	
6,066,151 A	5/2000	Miyawaki		6,428,562 B2	8/2002	Bonutti	
6,066,160 A	5/2000	Colvin		6,432,115 B1	8/2002	Mollenauer	
6,066,166 A	5/2000	Bischoff		6,447,516 B1	9/2002	Bonutti	
6,068,637 A	5/2000	Popov		6,450,985 B1	9/2002	Schoelling	
6,077,277 A	6/2000	Mollenauer		6,458,228 B1	10/2002	Yoshimoto	
6,077,292 A	6/2000	Bonutti		6,461,360 B1	10/2002	Adam	
6,080,161 A *	6/2000	Eaves, III	A61B 17/88 606/76	6,468,293 B2	10/2002	Bonutti	
				6,475,230 B1	11/2002	Bonutti	
6,083,522 A	7/2000	Chu		6,488,196 B1	12/2002	Fenton	
6,086,593 A	7/2000	Bonutti		6,500,195 B2	12/2002	Bonutti	
6,086,608 A	7/2000	Ek		6,503,259 B2	1/2003	Huxel	
6,090,072 A	7/2000	Kratoska		6,527,774 B2	3/2003	Lieberman	
6,099,531 A	8/2000	Bonutti		6,530,933 B1	3/2003	Yeung	
6,099,537 A	8/2000	Sugai		6,535,764 B2	3/2003	Imran	
6,099,550 A	8/2000	Yoon		6,544,267 B1	4/2003	Cole	
6,099,552 A	8/2000	Adams		6,545,390 B1	4/2003	Hahn	
6,102,850 A	8/2000	Wang		6,547,792 B1	4/2003	Tsuji	
6,106,545 A	8/2000	Egan		6,551,304 B1	4/2003	Whalen	
6,117,160 A	9/2000	Bonutti		6,554,852 B1	4/2003	Oberlander	
6,120,536 A	9/2000	Ding		6,557,426 B2	5/2003	Reinemann	
6,123,792 A	9/2000	Samida et al.		6,558,390 B2	5/2003	Cragg	
6,125,574 A	10/2000	Ganaja		6,568,313 B2	5/2003	Fukui	
6,126,677 A	10/2000	Ganaja		6,569,187 B1	5/2003	Bonutti	
6,139,320 A	10/2000	Hahn		6,572,635 B1	6/2003	Bonutti	
RE36,974 E	11/2000	Bonutti		D477,776 S	7/2003	Pontaoe	
6,149,669 A	11/2000	Li		6,585,750 B2	7/2003	Bonutti	
6,152,350 A	11/2000	Hayashi		6,585,764 B2	7/2003	Wright	
6,152,949 A	11/2000	Bonutti		6,592,609 B1	7/2003	Bonutti	
6,155,756 A	12/2000	Mericle		6,594,517 B1	7/2003	Nevo	
6,159,224 A	12/2000	Yoon		6,605,090 B1	8/2003	Trieu	
6,159,234 A	12/2000	Bonutti		6,610,080 B2	8/2003	Morgan	
6,171,307 B1	1/2001	Orlich		6,618,910 B1	9/2003	Pontaoe	
6,174,324 B1	1/2001	Egan		6,623,486 B1	9/2003	Weaver	
6,179,840 B1	1/2001	Bowman		6,623,487 B1 *	9/2003	Goshert	A61B 17/68 606/76
6,179,850 B1	1/2001	Goradia					
6,187,008 B1	2/2001	Hamman		6,626,944 B1	9/2003	Taylor	
6,190,400 B1	2/2001	Van de Moer		6,632,245 B2	10/2003	Kim	
6,190,401 B1	2/2001	Green		6,635,073 B2	10/2003	Bonutti	
6,200,322 B1	3/2001	Branch		6,638,279 B2	10/2003	Bonutti	
6,214,049 B1	4/2001	Gayer et al.		6,641,592 B1	11/2003	Sauer	
6,217,591 B1	4/2001	Egan		6,645,227 B2	11/2003	Fallin	
6,224,593 B1	5/2001	Ryan		6,652,585 B2	11/2003	Lang	
6,224,630 B1	5/2001	Bao		6,666,877 B2	12/2003	Morgan	
6,228,086 B1	5/2001	Wahl		6,669,705 B2	12/2003	Westhaver	
6,231,592 B1	5/2001	Bonutti		6,679,888 B2	1/2004	Green	
6,238,395 B1	5/2001	Bonutti		6,685,750 B1	2/2004	Plos	
6,238,396 B1	5/2001	Bonutti		6,699,240 B2	3/2004	Fracischelli	
6,258,091 B1	7/2001	Sevrain		6,702,821 B2	3/2004	Bonutti	
6,264,675 B1	7/2001	Brotz		6,705,179 B1	3/2004	Mohtasham	
6,267,761 B1	7/2001	Ryan		6,709,436 B1	3/2004	Hover et al.	
6,273,717 B1	8/2001	Hahn		6,709,457 B1	3/2004	Otte	
6,280,474 B1	8/2001	Cassidy		6,719,765 B2	4/2004	Bonutti	
6,286,746 B1	9/2001	Egan		6,719,797 B1	4/2004	Ferree	
6,287,325 B1	9/2001	Bonutti		6,722,552 B2	4/2004	Fenton	
				6,733,531 B1	5/2004	Trieu	

(56)	References Cited				2002/0029083	A1	3/2002	Zucherman	
	U.S. PATENT DOCUMENTS				2002/0029084	A1	3/2002	Paul	
					2002/0045902	A1	4/2002	Bonutti	
					2002/0062153	A1	5/2002	Paul	
6,764,514	B1	7/2004	Li		2002/0082529	A1	6/2002	Suorsa	
6,770,078	B2	8/2004	Bonutti		2002/0099379	A1	7/2002	Adam	
6,780,198	B1	8/2004	Gregoire		2002/0103495	A1	8/2002	Cole	
6,786,989	B2	9/2004	Torriani		2002/0120336	A1	8/2002	Santilli	
6,796,003	B1	9/2004	Marvel		2002/0123750	A1	9/2002	Eisermann	
6,818,010	B2	11/2004	Eichhorn		2002/0165529	A1	11/2002	Danek	
6,823,871	B2	11/2004	Schmieding		2002/0183762	A1	12/2002	Anderson	
6,860,885	B2	3/2005	Bonutti		2002/0188301	A1	12/2002	Dallara	
6,878,167	B2	4/2005	Ferree		2003/0009224	A1	1/2003	Kuras	
6,890,334	B2	5/2005	Brace et al.		2003/0032982	A1 *	2/2003	Bonutti	A61B 17/0487
6,893,434	B2	5/2005	Fenton et al.						606/232
6,899,722	B2	5/2005	Bonutti		2003/0039196	A1	2/2003	Nakamura	
6,913,666	B1 *	7/2005	Aeschlimann	 B29C 66/1142	2003/0040758	A1	2/2003	Wang	
				411/908	2003/0065361	A1	4/2003	Dreyfuss	
6,916,321	B2	7/2005	TenHuisen		2003/0069605	A1 *	4/2003	Bonutti	A61F 2/38
6,921,264	B2	7/2005	Mayer						606/232
6,923,824	B2	8/2005	Morgan		2003/0105474	A1	6/2003	Bonutti	
6,932,835	B2	8/2005	Bonutti		2003/0118518	A1	6/2003	Hahn	
6,942,684	B2	9/2005	Bonutti		2003/0158582	A1	8/2003	Bonutti	
6,944,111	B2	9/2005	Nakamura		2003/0167072	A1	9/2003	Oberlander	
6,955,540	B2	10/2005	Mayer		2003/0181800	A1	9/2003	Bonutti	
6,955,683	B2	10/2005	Bonutti		2003/0187506	A1	10/2003	Ross et al.	
6,958,077	B2	10/2005	Suddaby		2003/0195530	A1	10/2003	Thill	
6,979,376	B2	12/2005	Nuss		2003/0195565	A1	10/2003	Bonutti	
6,981,983	B1	1/2006	Rosenblatt		2003/0204204	A1	10/2003	Bonutti	
6,997,940	B2	2/2006	Bonutti		2003/0216742	A1	11/2003	Wetzler	
7,001,411	B1	2/2006	Dean		2003/0225438	A1	12/2003	Bonutti	
7,004,959	B2	2/2006	Bonutti		2003/0229361	A1	12/2003	Jackson	
7,008,226	B2	3/2006	Mayer		2004/0010287	A1	1/2004	Bonutti	
7,018,380	B2	3/2006	Cole		2004/0030341	A1 *	2/2004	Aeschlimann	 B29C 66/61
7,033,379	B2	4/2006	Peterson						606/76
7,048,755	B2	5/2006	Bonutti		2004/0034357	A1	2/2004	Beane	
7,066,960	B1	6/2006	Dickman		2004/0068320	A1	4/2004	Robie et al.	
7,087,073	B2	8/2006	Bonutti		2004/0087902	A1	5/2004	Richter	
7,090,111	B2	8/2006	Egan		2004/0097939	A1	5/2004	Bonutti	
7,094,251	B2	8/2006	Bonutti		2004/0098050	A1	5/2004	Foerster	
7,104,996	B2	9/2006	Bonutti		2004/0138703	A1	7/2004	Alleyne	
7,128,761	B2	10/2006	Kuras et al.		2004/0143334	A1	7/2004	Ferree	
7,128,763	B1	10/2006	Blatt		2004/0167548	A1	8/2004	Bonutti	
7,147,652	B2	12/2006	Bonutti		2004/0215111	A1	10/2004	Bonutti et al.	
7,160,405	B2 *	1/2007	Aeschlimann	 B29C 66/472	2004/0220616	A1	11/2004	Bonutti	
				156/298	2004/0225325	A1	11/2004	Bonutti	
7,179,259	B1	2/2007	Gibbs		2004/0225371	A1	11/2004	Roger	
7,192,448	B2	3/2007	Ferree		2004/0230223	A1	11/2004	Bonutti	
7,217,279	B2	5/2007	Reese		2004/0230309	A1	11/2004	DiMauro et al.	
7,217,290	B2	5/2007	Bonutti		2004/0236374	A1 *	11/2004	Bonutti	A61B 17/0487
7,241,297	B2	7/2007	Shaolian						606/232
7,250,051	B2	7/2007	Francischelli		2004/0243241	A1	12/2004	Istephanous et al.	
7,252,685	B2	8/2007	Bindseil		2005/0033366	A1	2/2005	Cole	
7,273,497	B2	9/2007	Ferree		2005/0038514	A1	2/2005	Helm	
7,329,263	B2	2/2008	Bonutti		2005/0043796	A1	2/2005	Grant	
7,335,205	B2	2/2008	Aeschlimann		2005/0071012	A1	3/2005	Serhan	
7,429,266	B2	9/2008	Bonutti		2005/0085817	A1	4/2005	Ringeisen	
7,445,634	B2	11/2008	Trieu		2005/0090827	A1	4/2005	Gedebou	
7,481,825	B2	1/2009	Bonutti		2005/0096699	A1	5/2005	Wixey	
7,481,831	B2	1/2009	Bonutti		2005/0113928	A1	5/2005	Cragg	
7,510,895	B2	3/2009	Rateman		2005/0126680	A1 *	6/2005	Aeschlimann	 B29C 66/472
7,632,272	B2	12/2009	Munro et al.						403/270
7,669,749	B2	3/2010	Frank		2005/0143826	A1	6/2005	Zucherman	
7,854,750	B2	12/2010	Bonutti		2005/0149024	A1	7/2005	Ferrante	
7,879,072	B2	2/2011	Bonutti		2005/0149029	A1	7/2005	Bonutti	
7,891,691	B2	2/2011	Bearey		2005/0177162	A1	8/2005	McLeod et al.	
7,967,820	B2	6/2011	Bonutti		2005/0203521	A1	9/2005	Bonutti	
8,128,669	B2	3/2012	Bonutti		2005/0216008	A1	9/2005	Zwirnmann et al.	
8,140,982	B2	3/2012	Hamilton		2005/0216059	A1	9/2005	Bonutti	
8,147,514	B2	4/2012	Bonutti		2005/0216087	A1	9/2005	Zucherman	
8,162,977	B2	4/2012	Bonutti		2005/0222620	A1	10/2005	Bonutti	
11,278,331	B2	3/2022	Bonutti et al.		2005/0240190	A1	10/2005	Gall	
2001/0002440	A1	5/2001	Bonutti		2005/0240227	A1	10/2005	Bonutti	
2001/0009250	A1	7/2001	Herman		2005/0246021	A1	11/2005	Rigeisen	
2001/0041916	A1	11/2001	Bonutti		2005/0261684	A1	11/2005	Shaolian	
2002/0016593	A1	2/2002	Hearn		2005/0267481	A1	12/2005	Carl	
2002/0016633	A1	2/2002	Lin		2005/0267534	A1	12/2005	Bonutti	
2002/0019649	A1	2/2002	Sikora		2006/0009855	A1	1/2006	Goble	
2002/0026244	A1	2/2002	Trieu		2006/0015101	A1	1/2006	Warburton	

(56)

References Cited**U.S. PATENT DOCUMENTS**

2006/0015108	A1	1/2006	Bonutti
2006/0026244	A1	2/2006	Watson
2006/0036203	A1	2/2006	Ouchene et al.
2006/0064095	A1	3/2006	Senn
2006/0089646	A1	4/2006	Bonutti
2006/0122600	A1	6/2006	Cole
2006/0122704	A1	6/2006	Vresilovic
2006/0142799	A1	6/2006	Bonutti
2006/0167495	A1	7/2006	Bonutti
2006/0200199	A1	9/2006	Bonutti
2006/0212073	A1	9/2006	Bonutti
2006/0217765	A1	9/2006	Bonutti
2006/0224357	A1	10/2006	Taware et al.
2006/0229623	A1	10/2006	Bonutti
2006/0229715	A1	10/2006	Istephanous et al.
2006/0235470	A1	10/2006	Bonutti
2006/0241695	A1	10/2006	Bonutti
2006/0264950	A1	11/2006	Nelson
2006/0265009	A1	11/2006	Bonutti
2006/0265011	A1	11/2006	Bonutti
2007/0032825	A1	2/2007	Bonutti
2007/0088362	A1	4/2007	Bonutti
2007/0118129	A1*	5/2007	Fraser A61F 2/0811 606/279
2007/0123878	A1	5/2007	Shaver
2007/0198555	A1	8/2007	Friedman
2007/0265561	A1	11/2007	Yeung
2007/0270833	A1	11/2007	Bonutti
2008/0021474	A1	1/2008	Bonutti
2008/0039845	A1	2/2008	Bonutti
2008/0039873	A1	2/2008	Bonutti
2008/0046090	A1	2/2008	Paul
2008/0097448	A1	4/2008	Binder
2008/0108897	A1	5/2008	Bonutti
2008/0108916	A1	5/2008	Bonutti
2008/0114399	A1	5/2008	Bonutti
2008/0132950	A1	6/2008	Lange
2008/0140116	A1	6/2008	Bonutti
2008/0140117	A1	6/2008	Bonutti
2008/0195145	A1	8/2008	Bonutti
2008/0269753	A1	10/2008	Cannestra
2008/0269808	A1	10/2008	Gall
2009/0024161	A1	1/2009	Bonutti
2009/0093684	A1	4/2009	Schorer
2009/0138014	A1	5/2009	Bonutti
2009/0194969	A1	8/2009	Bearey
2010/0211120	A1	8/2010	Bonutti
2011/0060375	A1	3/2011	Bonutti
2011/0295253	A1	12/2011	Bonutti
2012/0165841	A1	6/2012	Bonutti
2012/0191140	A1	7/2012	Bonutti
2012/0197316	A1	8/2012	Mayer
2012/0215233	A1	8/2012	Bonutti

FOREIGN PATENT DOCUMENTS

CA	2698057	3/2009
DE	1903316	10/1964
DE	3517204	11/1986
DE	3722538	1/1989
DE	9002844	U1 1/1991
EP	784454	5/1996
EP	773004	5/1997
EP	1346752	A1 9/2003
EP	1614525	1/2006
EP	1988837	8/2007
EP	2134294	12/2009
FR	2717368	3/1994
FR	2696338	4/1994
FR	2728779	1/1995
FR	2736257	7/1995
FR	2750031	6/1996
FR	2771621	11/1997
FR	2785171	10/1998

GB	2093701	9/1982
GB	2306110	A 4/1997
JP	H07156907	A 6/1995
JP	8140982	A 6/1996
JP	3020554	B2 3/2000
JP	2004209659	A 7/2004
SU	184396	7/1966
WO	199112779	9/1991
WO	199323094	4/1994
WO	1994008642	4/1994
WO	199516398	6/1995
WO	199531941	11/1995
WO	1996014802	5/1996
WO	1997012779	4/1997
WO	1997049347	12/1997
WO	1998011838	3/1998
WO	1998026720	6/1998
WO	2002053011	7/2002
WO	2007092869	8/2007
WO	2008116203	9/2008
WO	2009029908	3/2009
WO	2010099222	2/2010

OTHER PUBLICATIONS

Cobb et al, Late Correction of Malunited Intercondylar Humeral Fractures Intra-Articular Osteotomy and Tricortical Bone Grafting, J BoneJointSurg [Br] 1994; 76-B:622-6.

Fellinger, et al, Radial avulsion of the triangular fibrocartilage complex in acute wrist trauma: a new technique for arthroscopic repair, Jun. 1997, Arthroscopy vol. 13 No. 3 p. 370-4.

Hecker et al, Pull-out strength of suture anchors for rotator cuff and Bankart lesion repairs, Nov.-Dec. 1993, The American Journal of Sports Medicine, vol. 21 No. 6 p. 874-9.

Hernigou et al, Proximal Tibial Osteotomy for Osteoarthritis with Varus Deformity a Ten to Thirteen-Year Follow-Up Study, J Bone Joint Surg, vol. 69-A, No. 3. Mar. 1987, p. 332-354.

Ibarra et al, Glenoid Replacement in Total Shoulder Arthroplasty, the Orthopedic Clinics of Northamerica: Total Shoulder Arthroplasty, vol. 29 No. 3, Jul. 1998 p. 403-413.

Mosca et al, Calcaneal Lengthening for Valgus Deformity of the Hindfoot: Results in Children who Had Severe, Symptomatic fLATFOOT and Skewfoot, J Bone Joint Surg., 1195—p. 499-512.

Murphy et al, Radial Opening Wedge Osteotomy in Madelung's Deformity, J. Hand Surg, vol. 21 A No. 6 Nov. 1996, p. 1035-44.

Biomet, Stanmore Modular Hip, J. Bone Joint Surg., vol. 76-B : No. Two, Mar. 1994.

Intl Prelim Rep on Patentability and Written Opinion for PCT/US10/25263 dated Aug. 30, 2011.

Petition for Inter Partes Review of U.S. Pat. No. 5,980,559, IPR 2013-00603, Filing Date Sep. 24, 2013.

Declaration of David Kaplan, Ph.D. Regarding U.S. Pat. No. 5,980,559, IPR 2013-00603, Sep. 24, 2013.

Petition for Inter Partes Review of U.S. Pat. No. 7,087,073, IPR 2013-00604, Filing Date Sep. 24, 2013.

Declaration of Wayne J. Sebastianelli, MD Regarding U.S. Pat. No. 7,087,073, Sep. 24, 2013, IPR 2013-00604.

Petition for Inter Partes Review of U.S. Pat. No. 6,500,195, IPR 2013-00624, Filing Date Oct. 2, 2013.

Declaration of Dr. Philip Hardy in Support of Petition for Inter Partes Review of U.S. Pat. No. 6,500,195, IPR 2013-00624, Sep. 25, 2013.

Petition for Inter Partes Review of U.S. Pat. No. 5,527,343, IPR 2013-00628, Filing Date Sep. 26, 2013.

Declaration of Dr. Philip Hardy in Support of Petition for Inter Partes Review of U.S. Pat. No. 5,527,343, IPR 2013-00628, Sep. 25, 2013.

Corrected Petition for Inter Partes Review of U.S. Pat. No. 5,921,986, IPR 2013-00631, Filing Date Sep. 27, 2013.

Expert Declaration of Steve E. Jordan, MD, for Inter Partes Review of U.S. Pat. No. 5,921,986, IPR 2013-00631, Sep. 24, 2013.

Corrected Petition for Inter Partes Review of U.S. Pat. No. 8,147,514, IPR 2013-00632, Filing Date Sep. 27, 2013.

(56)

References Cited**OTHER PUBLICATIONS**

Declaration of Steve Jordan for U.S. Pat. No. 8,147,514, from IPR 2013-00632, dated Sep. 23, 2013 (exhibit 1009).
 Corrected Petition for Inter Partes Review of U.S. Pat. No. 8,147,514, IPR 2013-00633, Filing Date Sep. 27, 2013.
 Declaration of Steve Jordan for U.S. Pat. No. 8,147,514, from IPR 2013-00633, dated Sep. 23, 2013 (exhibit 1006).
 Flory, Principles of Polymer Chemistry, 1953, selected pages (cited in IPR 2013-00603, exhibit 1012).
 Grizzi, Hydrolytic degradation of devices based on poly(DL-lactic acid) size-dependence, Biomaterials, 1995, vol. 16, No. 4, p. 305-11 (cited in IPR 2013-00603, exhibit 1006).
 Gopferich, Mechanisms of polymer degradation and erosion, Biomaterials, 1996, vol. 17, No. 2, p. 103-114 (cited in IPR 2013-00603, exhibit 1013).
 Gao et al, Swelling of Hydroxypropyl Methylcellulose Matrix Tablets . . . , J. of Pharmaceutical Sciences, vol. 85, No. 7, Jul. 1996, p. 732-740 (cited in IPR 2013-00603, exhibit 1014).
 Linvatec, Impact Suture Anchor brochure, 2004 (cited in IPR 2013-00628, exhibit 1010).
 Seitz et al, Repair of the Tibiofibular Syndesmosis with a Flexible Implant, J. of Orthopaedic Trauma, vol. 5, No. 1, p. 78-82, 1991 (cited in IPR 2013-00631, exhibit 1007) (cited in 2013-00632).
 Translation of FR2696338 with translator's certificate dated Sep. 17, 2013 (cited in IPR 2013-00631, 2013-00632).
 Translation of DE9002844.9 with translator's certificate dated Sep. 26, 2013 (cited in IPR 2013-00631, 2013-00632).
 Declaration of Steve Jordan for U.S. Pat. No. 5,921,986, from IPR 2013-00632, dated Sep. 24, 2013 (exhibit 1010).
 Declaration of Steve Jordan for U.S. Pat. No. 5,921,986, from IPR 2013-00633, dated Sep. 24, 2013 (exhibit 1007).
 Declaration of Dr. Steve E. Jordan for U.S. Pat. No. 8,147,514, from IPR 2013-00631, dated Sep. 23, 2013.
 The Search for the Holy Grail: A Century of Anterior Cruciate Ligament Reconstruction, R. John Naranja, American Journal of Orthopedics, Nov. 1997.
 Femoral Bone Plug Recession in Endoscope Anterior Cruciate Ligament Reconstruction, David E. Taylor, Arthroscopy: The Journal of Arthroscopic and Related Surgery, Aug. 1996.
 Meniscus Replacement with Bone Anchors: A Surgical Technique, Arthroscopy: The Journal of Arthroscopic and Related Surgery, 1994.
 Problem Solving Report Question No. 1014984.066, Ultrasonic Welding, (c) 1999.
 Dukane Guide to Ultrasound Plastic Assembly, Ultrasonic Division Publication, (c) 1995.
 Textured Surface Technology, Branson Technology, Branson Ultrasonics Corp., (c) 1992.
 Enabling Local Drug Delivery-Implant Device Combination Therapies, Surmodics, Inc., (c) 2003.
 Stent Based Delivery of Sirolimus Reduces Neointimal Formation in a Porcine Coronary Model, Takeshi Suzuki, American Heart Association, Inc. (c) 2001.

Why Tie a Knot When You Can Use Y-Knot?, Innovasive Devices Inc., (c) 1998.

Ask Oxford projection, compact oxford english dictionary: projection, Mar. 30, 2009.

Ask Oxford projection, compact oxford english dictionary: slit, Mar. 30, 2009.

Packer et al., Repair of Acute Scapho-Lunate Dissociation Facilitated by the "Tag" Suture Anchor, Journal of Hand Surgery (British and European vol. 1994) 19B: 5: 563-564.

Richmond, Modification of the Bankart reconstruction with a suture anchor, Am J Sports Med, vol. 19, No. 4, p. 343-346, 1991.

Shea et al, Technical Note: Arthroscopic Rotator Cuff Repair Using a Transhumeral Approach to Fixation, Arthroscopy: The Journal of Arthroscopic and Related Surgery, vol. 14, No. 1 Jan.-Feb. 1998: pp. 118-122.

Tfix, Acufex just tied the knot . . . , Am. J. Sports Med., vol. 22, No. 3, May-Jun. 1994.

Arthrex, Protect your graft, Am J Sports Med, vol. 22, No. 4, Jul.-Aug. 1994.

Barrett et al, T-Fix endoscopic meniscal repair: technique and approach to different types of tears, Apr. 1995, Arthroscopy 2 vol. 11 No. 2 p. 245-51.

Cope, Suture Anchor for Visceral Drainage, AJR, vol. 148 p. 160-162, Jan. 1986.

Gabriel, Arthroscopic Fixation Devices, Wiley Enc. of Biomed Eng., 2006.

Innovative, We've got you covered, Am J Sports Med, vol. 26, No. 1, Jan.-Feb. 1998.

510k—TranSet Fracture Fixation System, Feb. 24, 2004, k033717.

510k—Linvatec Biomaterials modification of Duet and impact Suture Anchor, Nov. 19, 2004, k042966.

510k, arthrex pushlock, Jun. 29, 2005, K051219.

510k, mitek micro anchor, Nov. 6, 1996, K962511.

510k, Multitak Suture System, Jan. 10, 1997, K964324.

510k, Modified Mitek 3.5mm Absorbable Suture Anchor System, Jun. 9, 1997, K970896.

510K, Summary for Arthrex Inc.'s Bio-Interference Screw, Jul. 9, 1997, K971358.

510k, Surgicraft Bone Tie, Sep. 25, 1998, K982719.

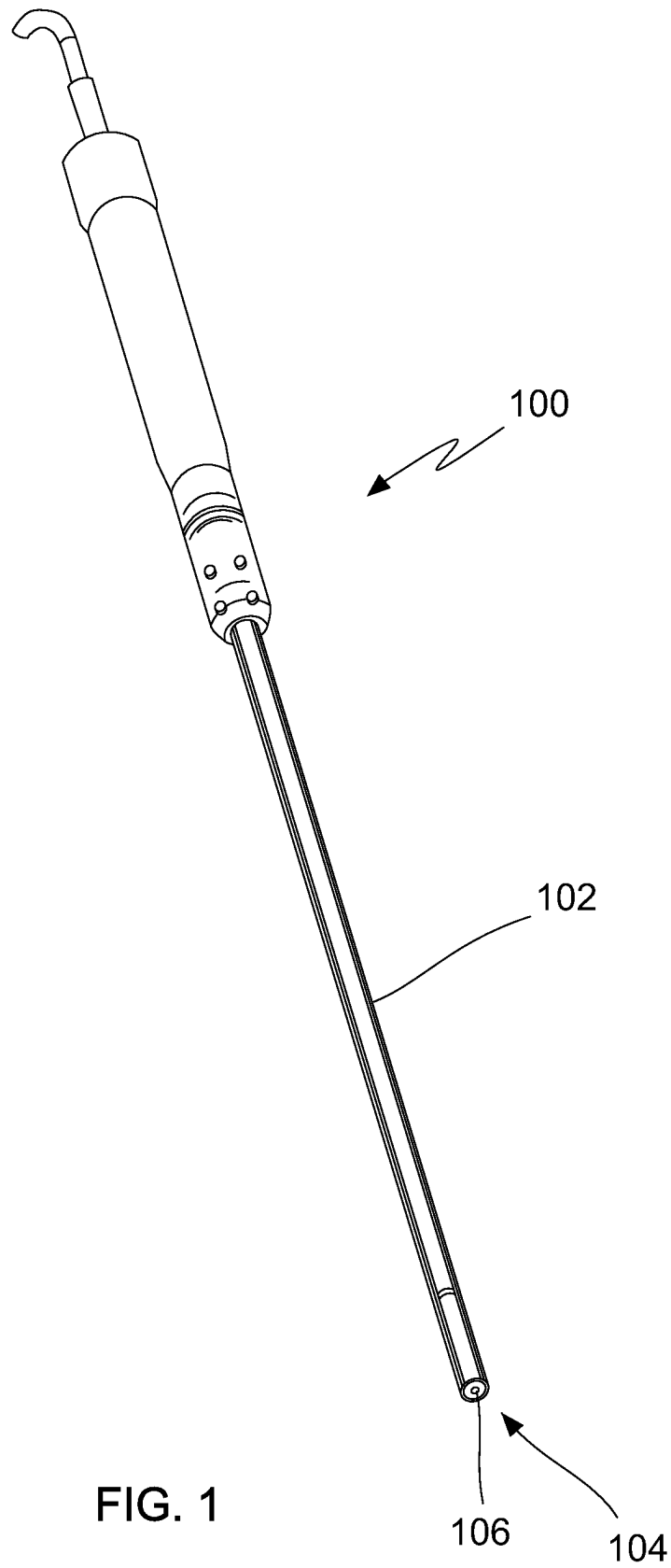
Karlsson et al, Repair of Bankart lesions with a suture anchor in recurrent dislocation of the shoulder, Scand. j. of Med & Science in Sports, 1995, 5:170-174.

Madjar et al, Minimally Invasive Pervaginam Procedures, for the Treatment of Female Stress Incontinence . . . , Artificial Organs, 22 (10) 879-885, 1998.

Nowak et al, Comparative Study of Fixation Techniques in the Open Bankart Operation Using Either a Cannulated Screw or Suture Anchors, Acta Orthopcedica Belgica, vol. 64-2-1998.

Branson, Polymers: Characteristics and Compatibility for Ultrasonic Assembly, Applied Technologies Group, Publication unknown. Decision on Appeal for U.S. Appl. No. 16/930,229 (Affirmed), Appeal 2024-001234, Jul. 18, 2025, 7 pages.

* cited by examiner



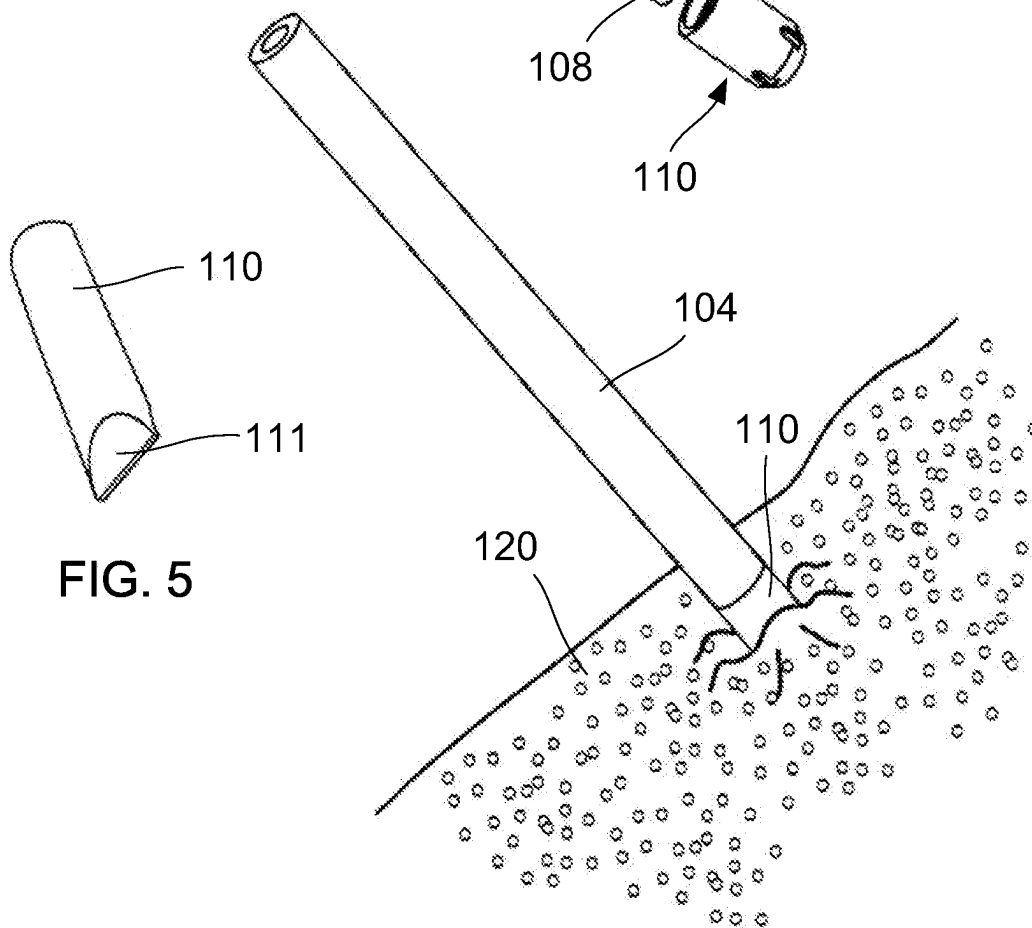
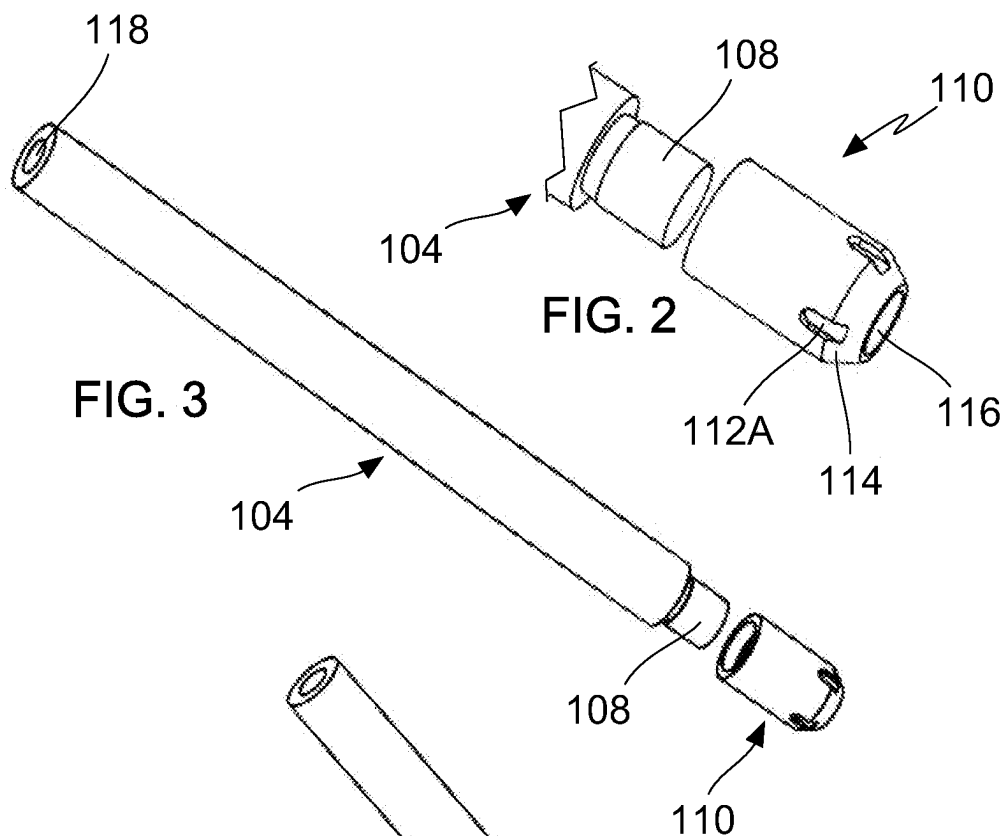


FIG. 4

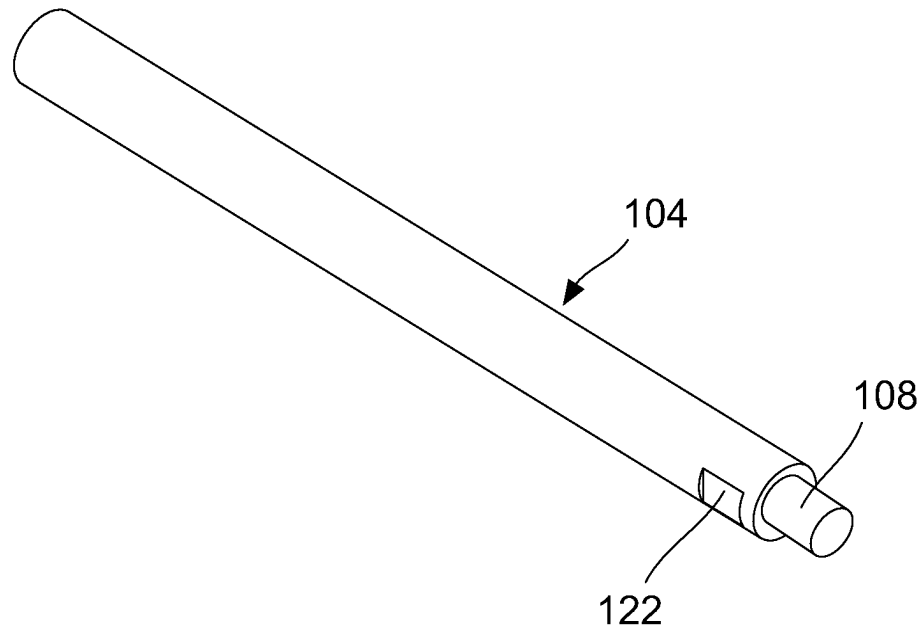


FIG. 6

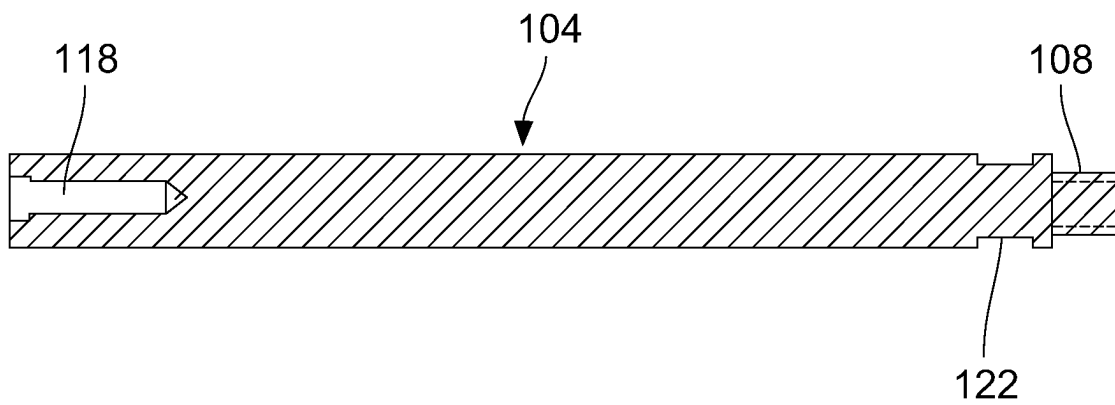


FIG. 7

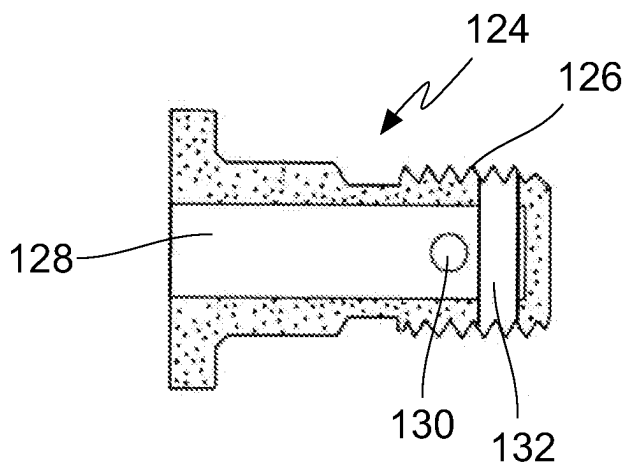


FIG. 8

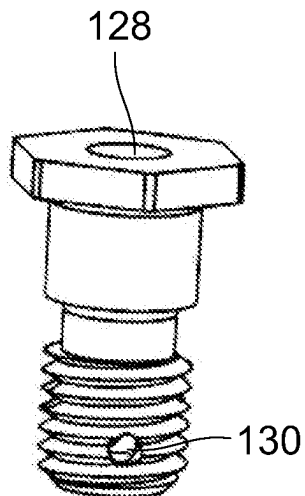


FIG. 9

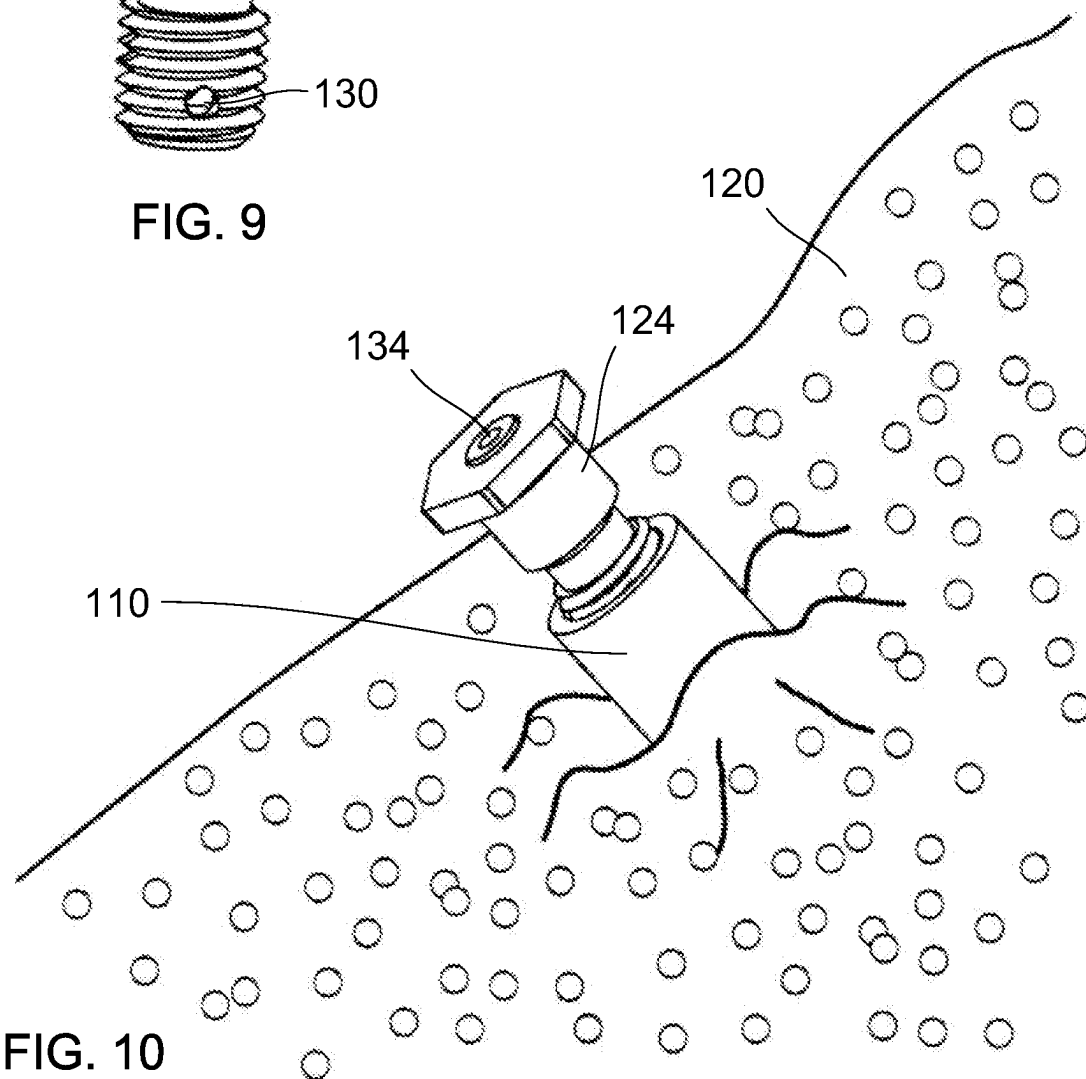


FIG. 10

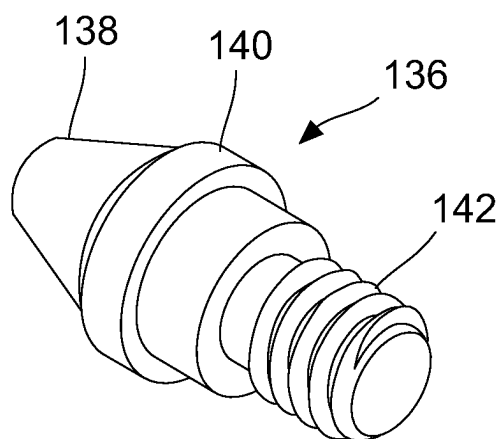


FIG. 11

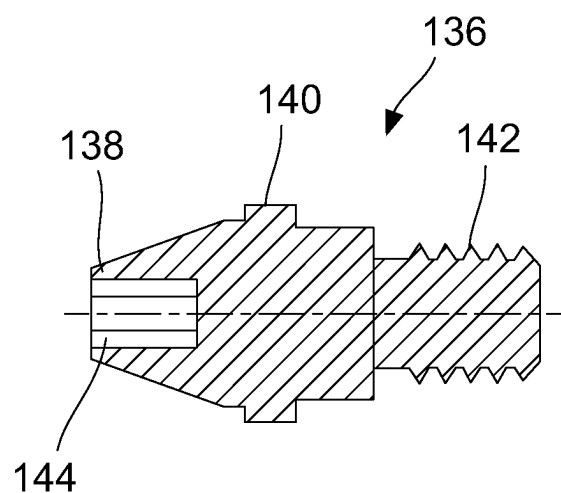


FIG. 12

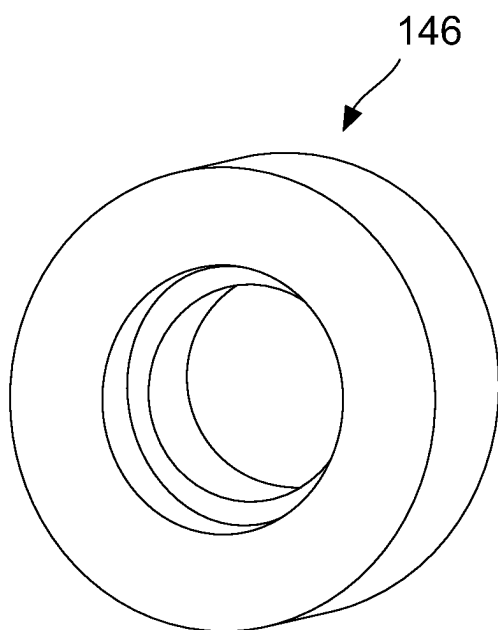


FIG. 13

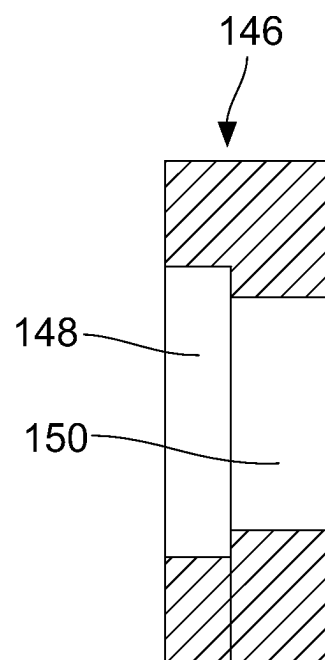


FIG. 14

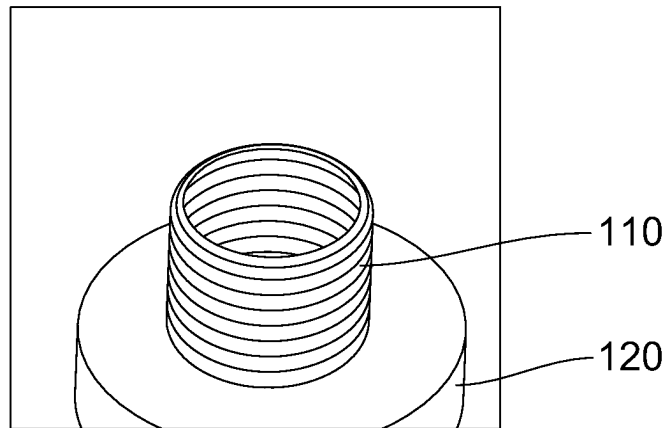


FIG. 15

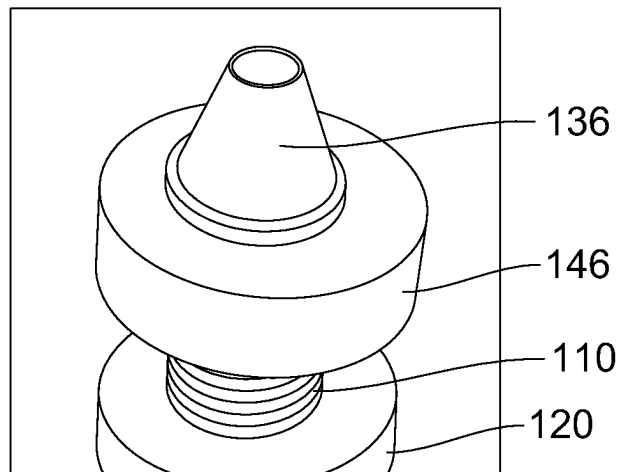


FIG. 16

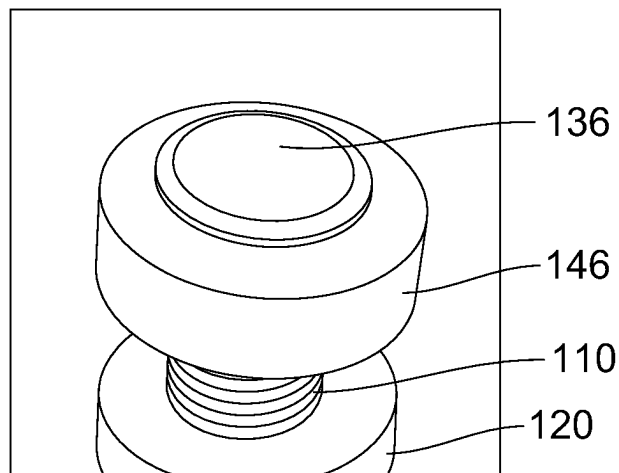


FIG. 17

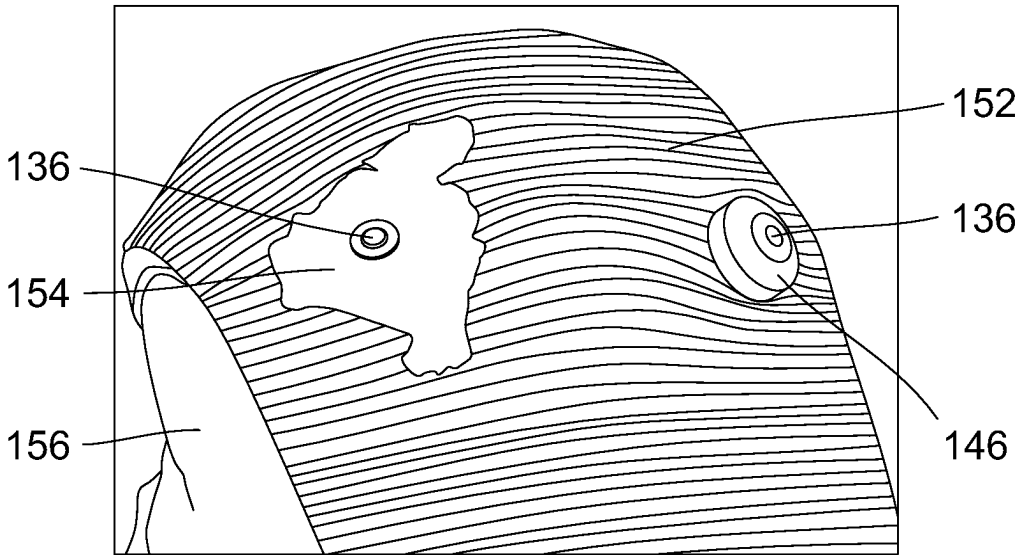


FIG. 18

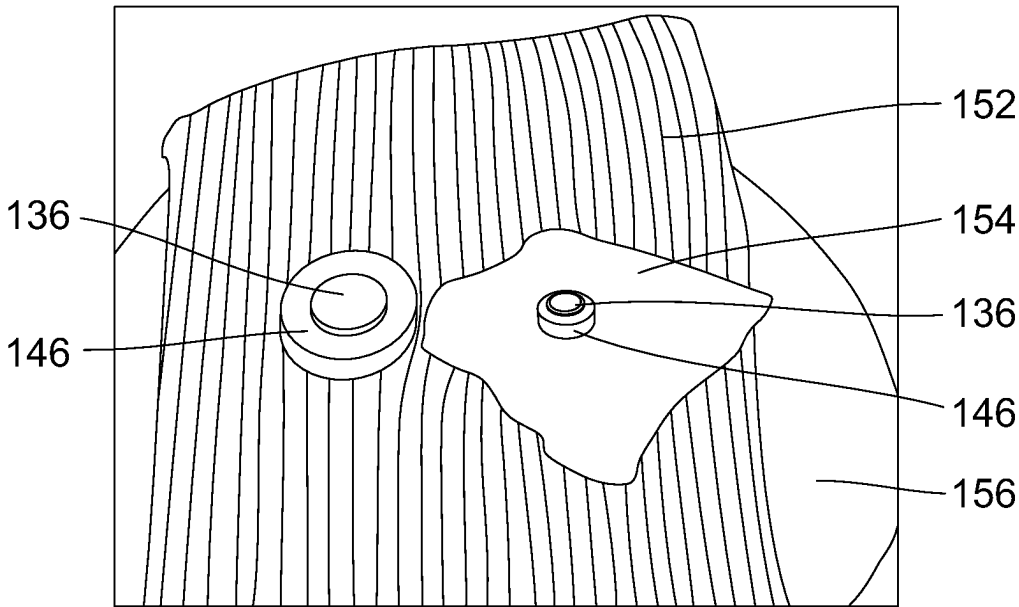


FIG. 19

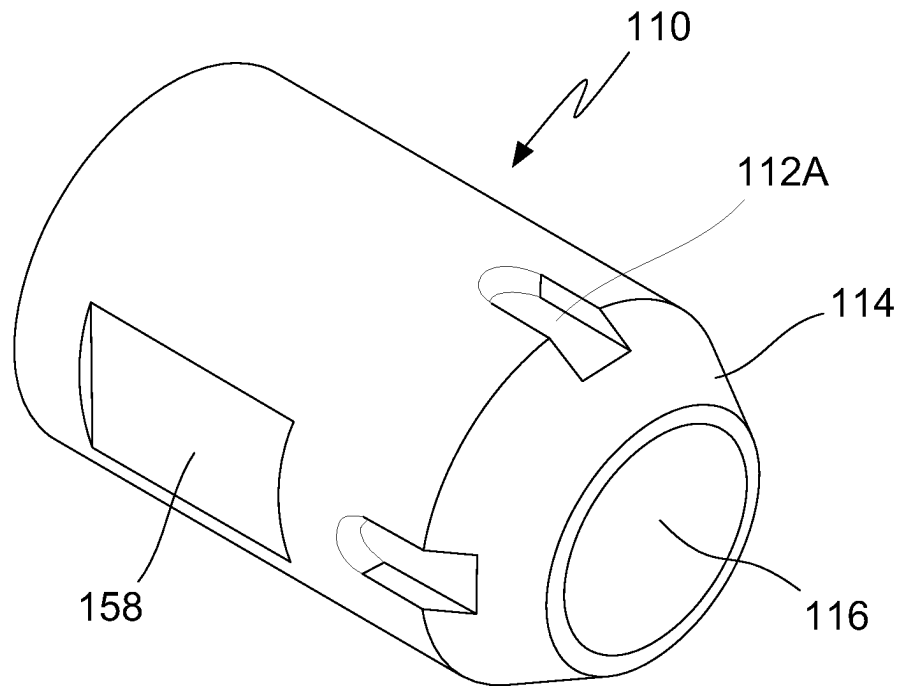


FIG. 20

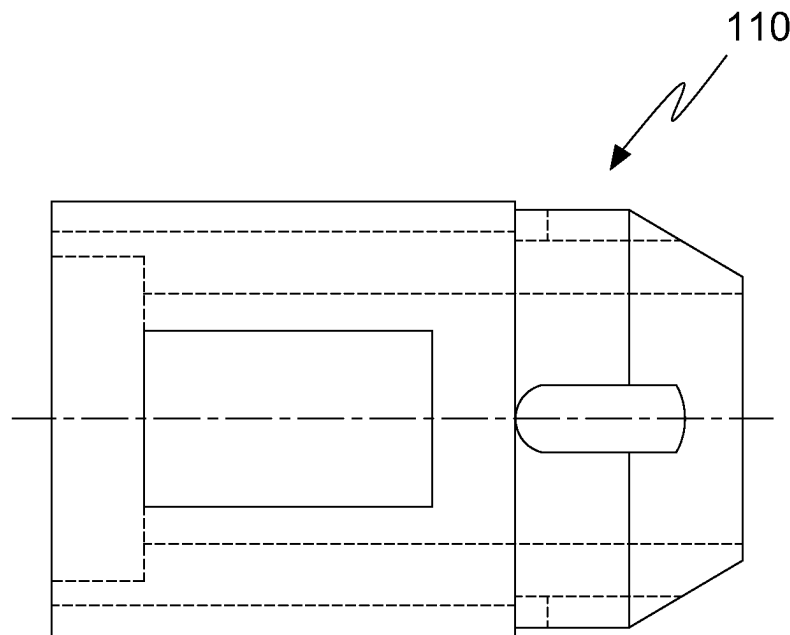


FIG. 21

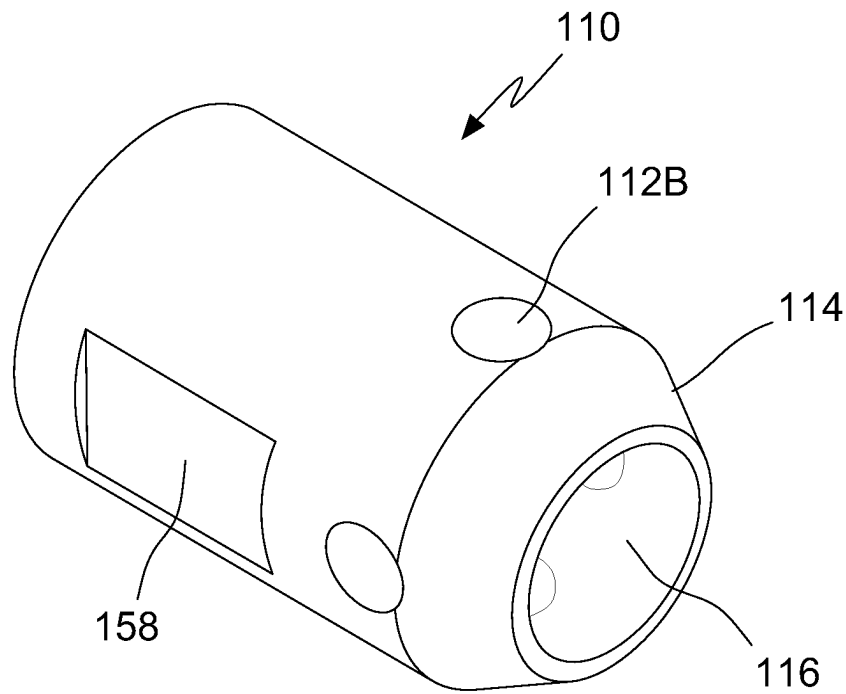


FIG. 22

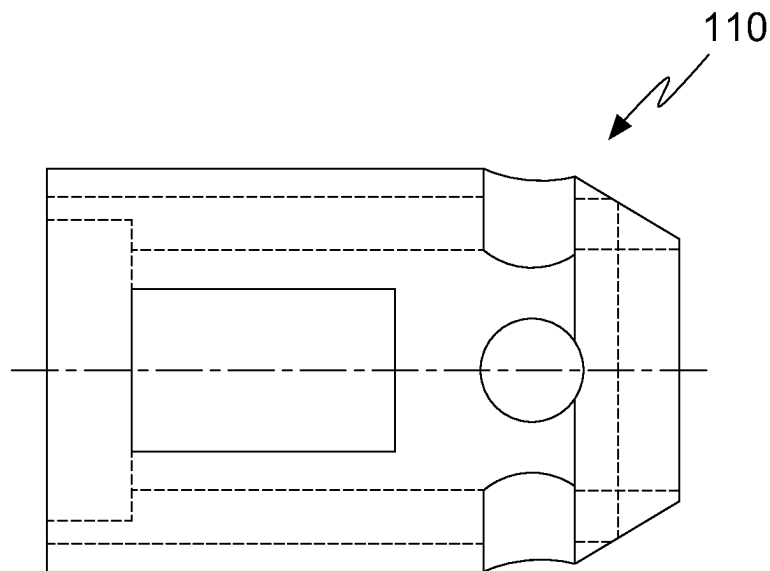


FIG. 23

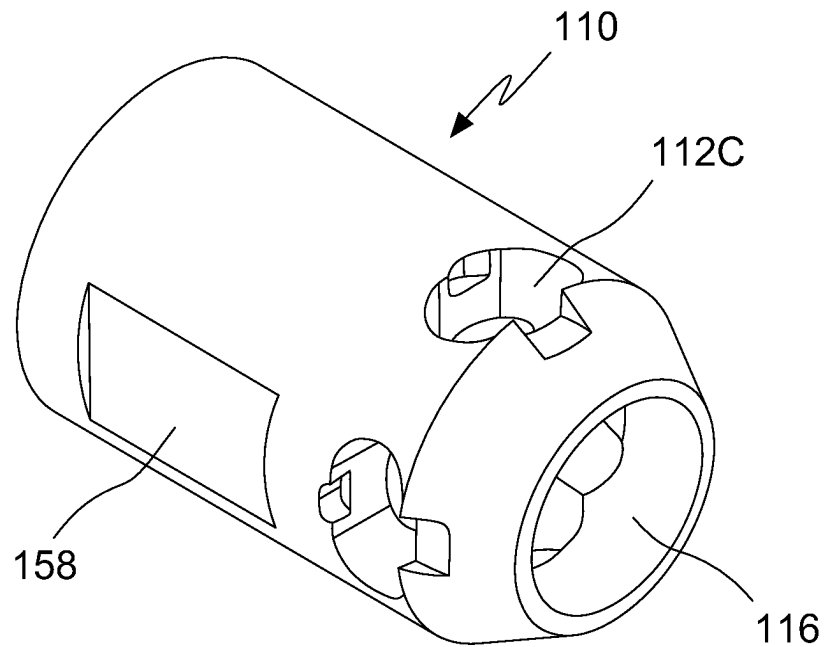


FIG. 24

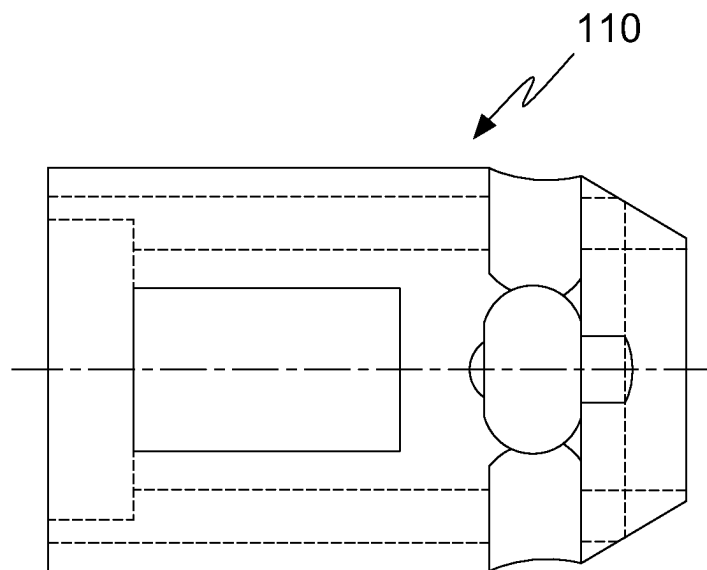


FIG. 25

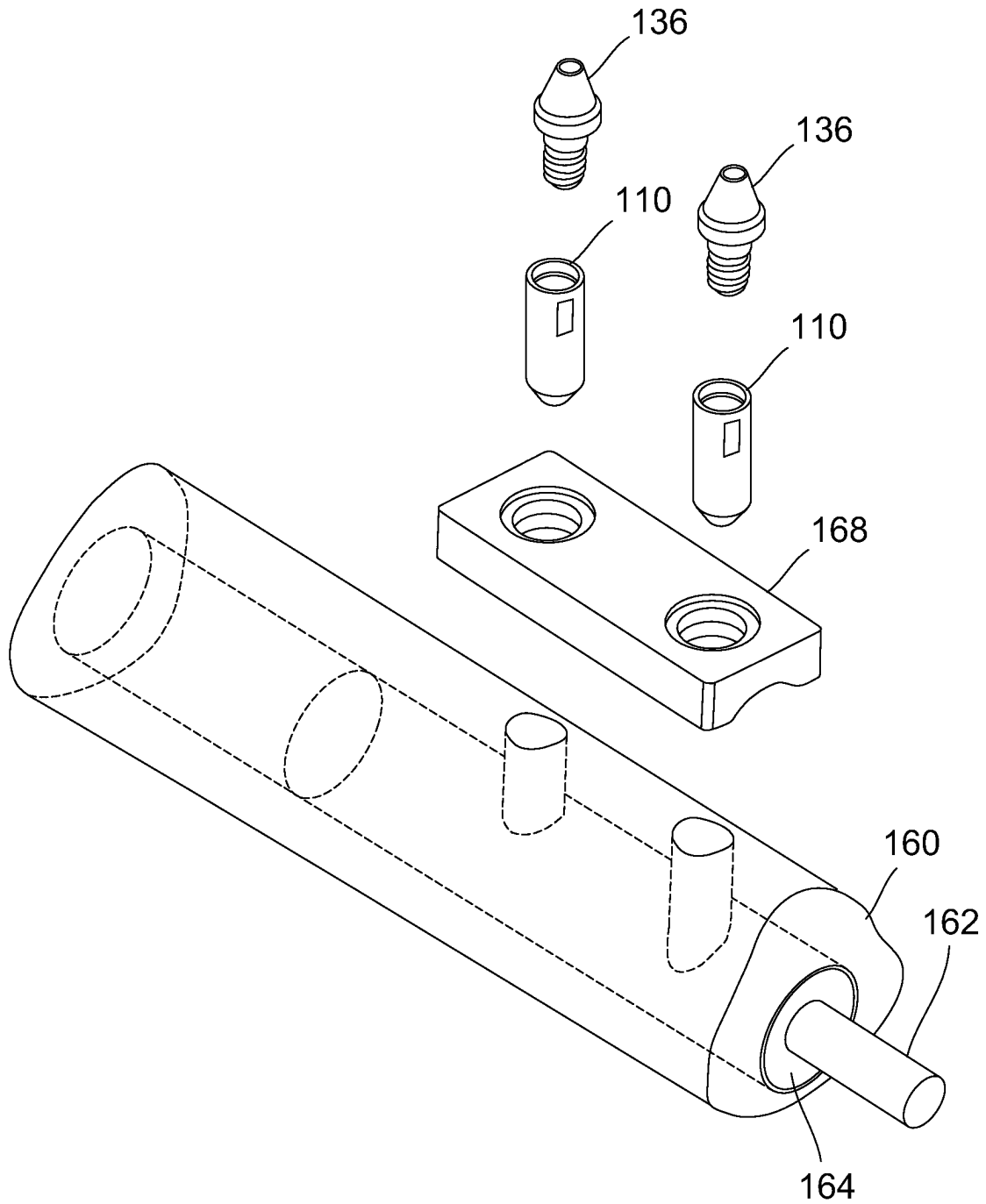


FIG. 26

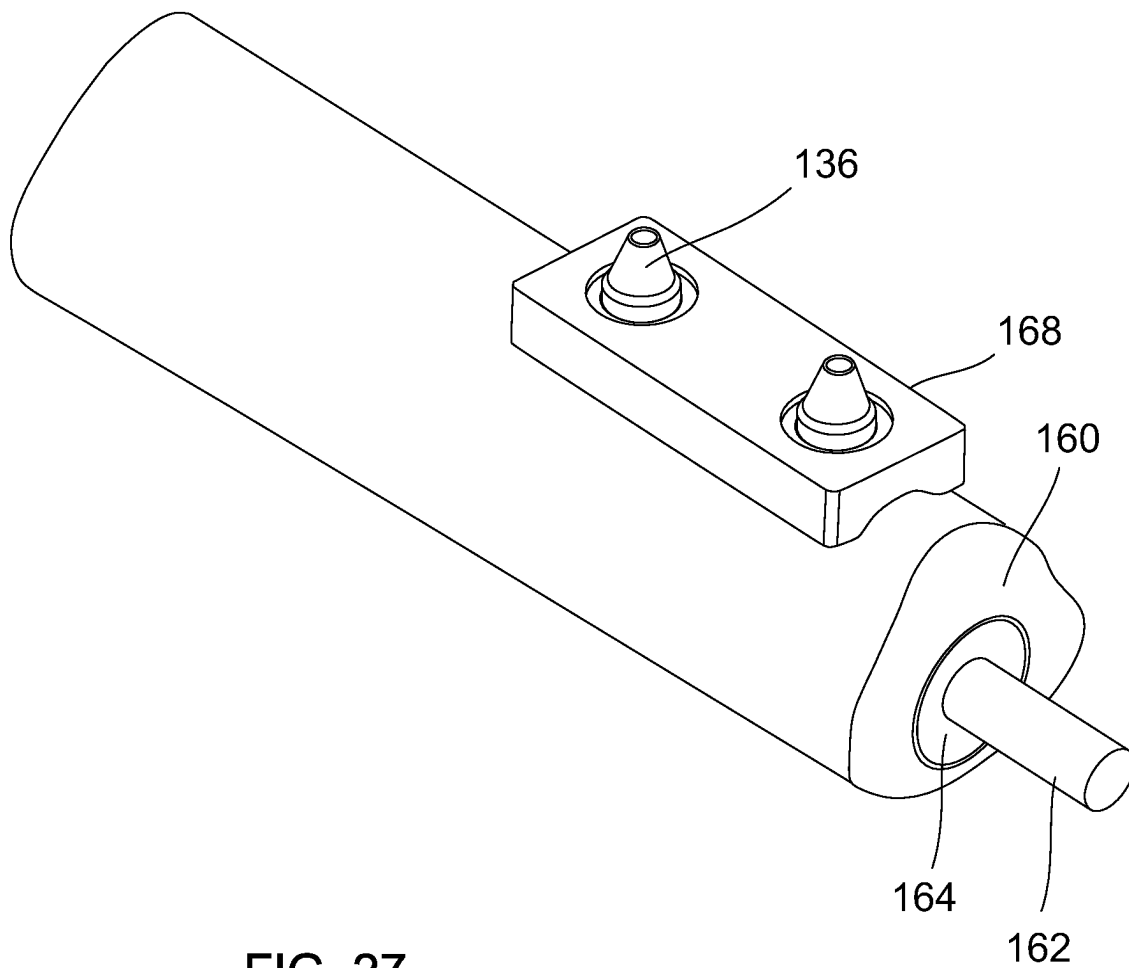


FIG. 27

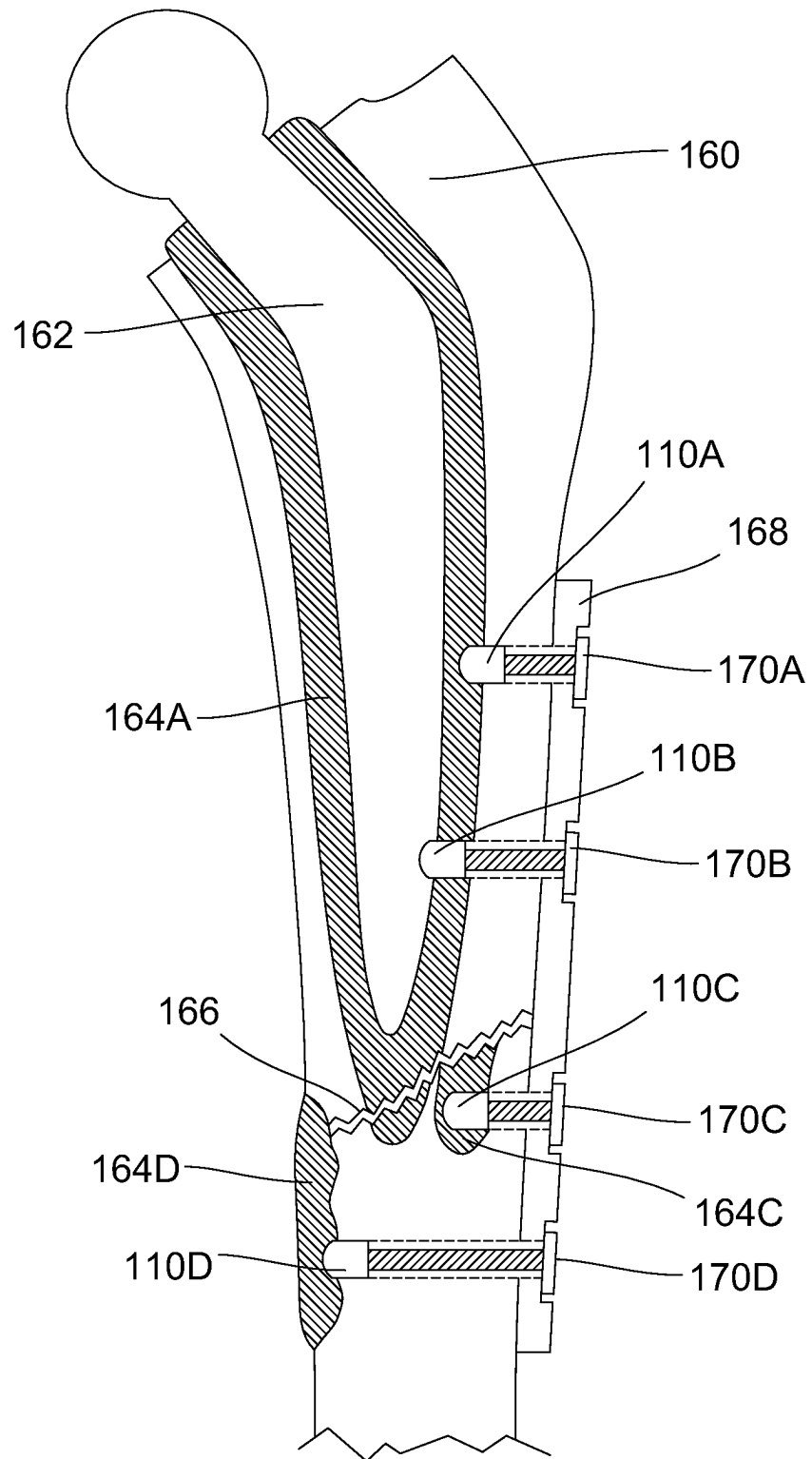


FIG. 28

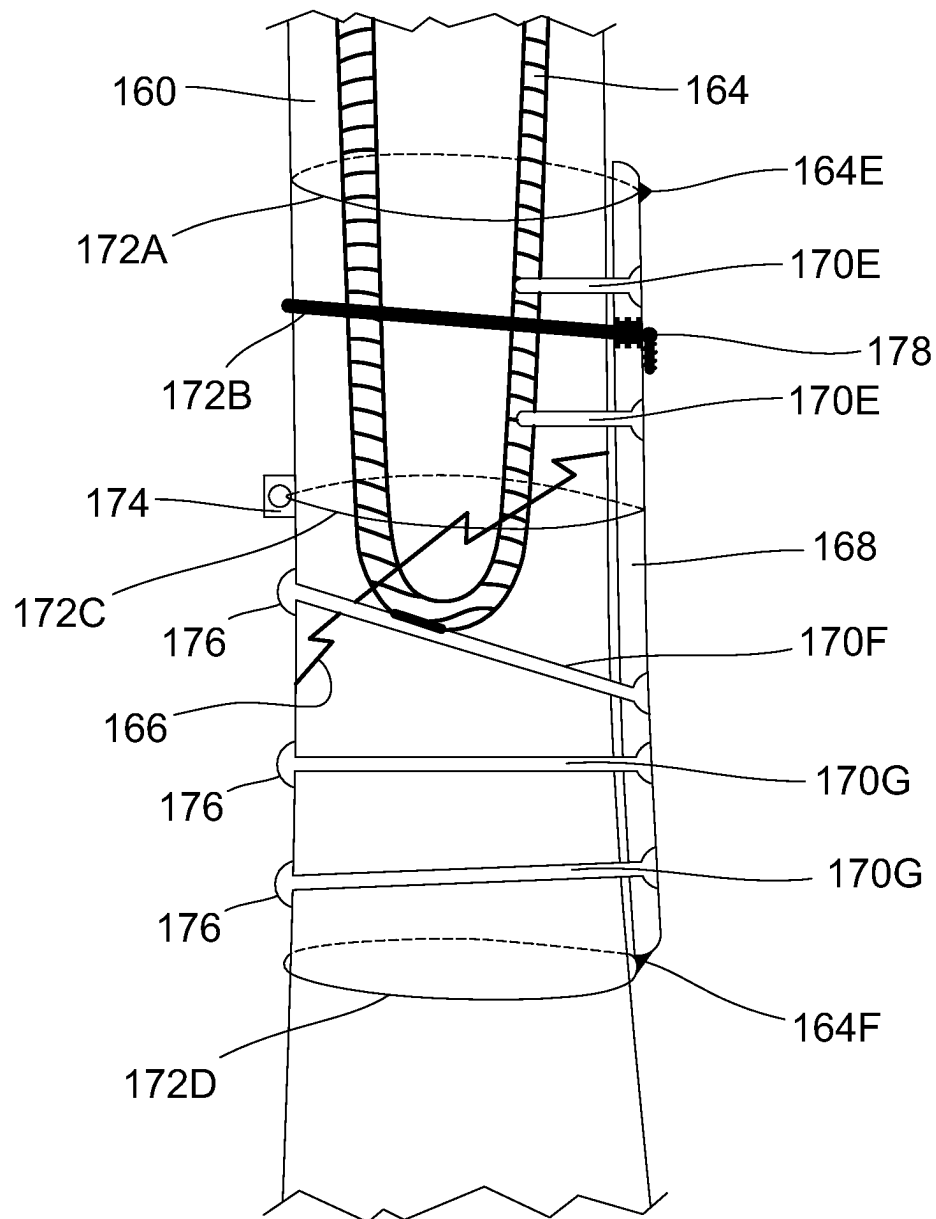


FIG. 29

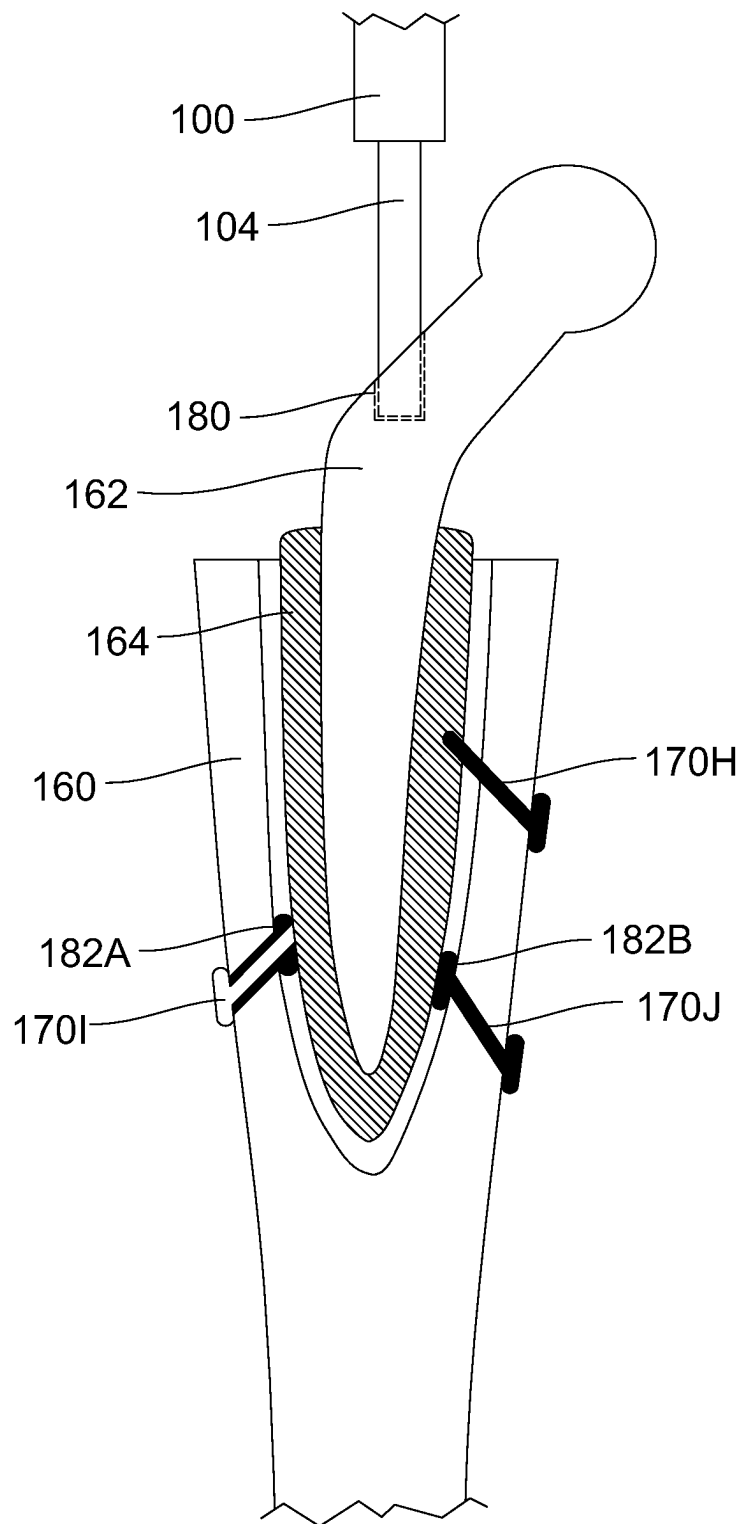


FIG. 30

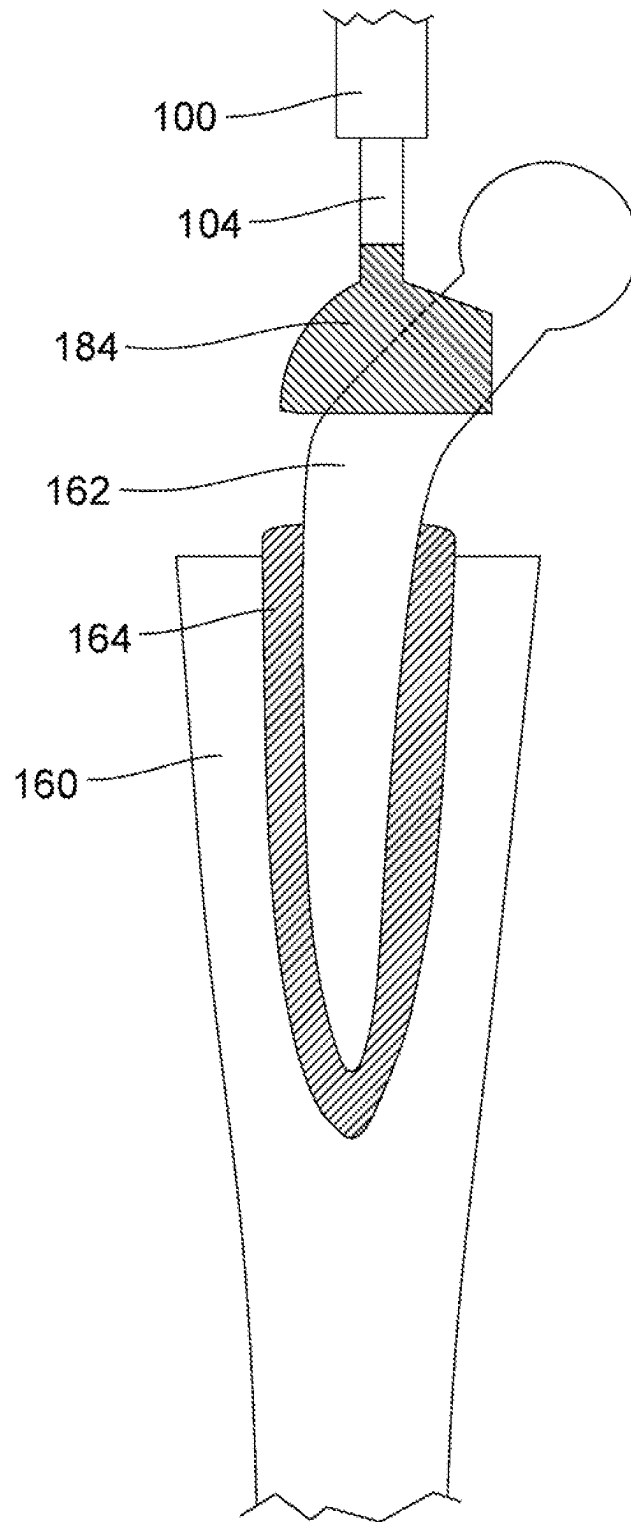


FIG. 31

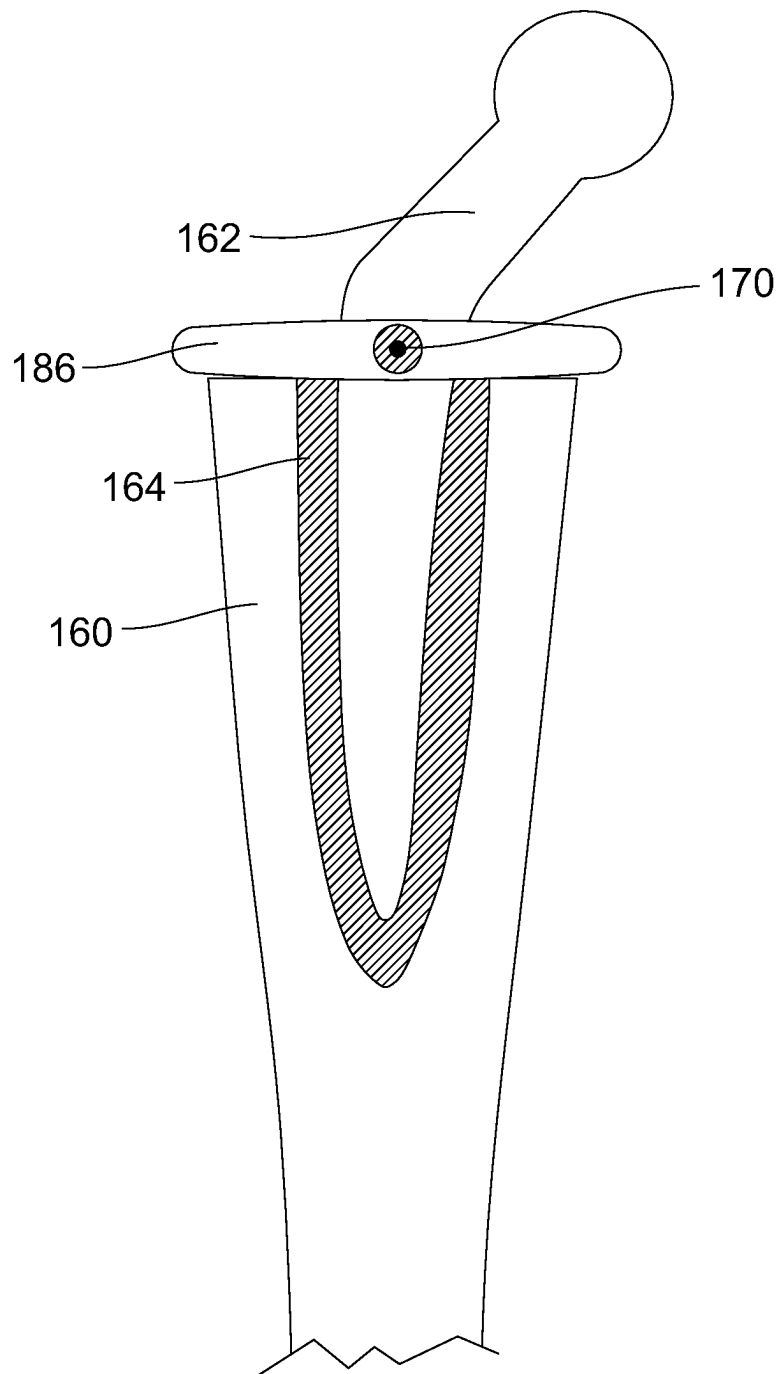


FIG. 32

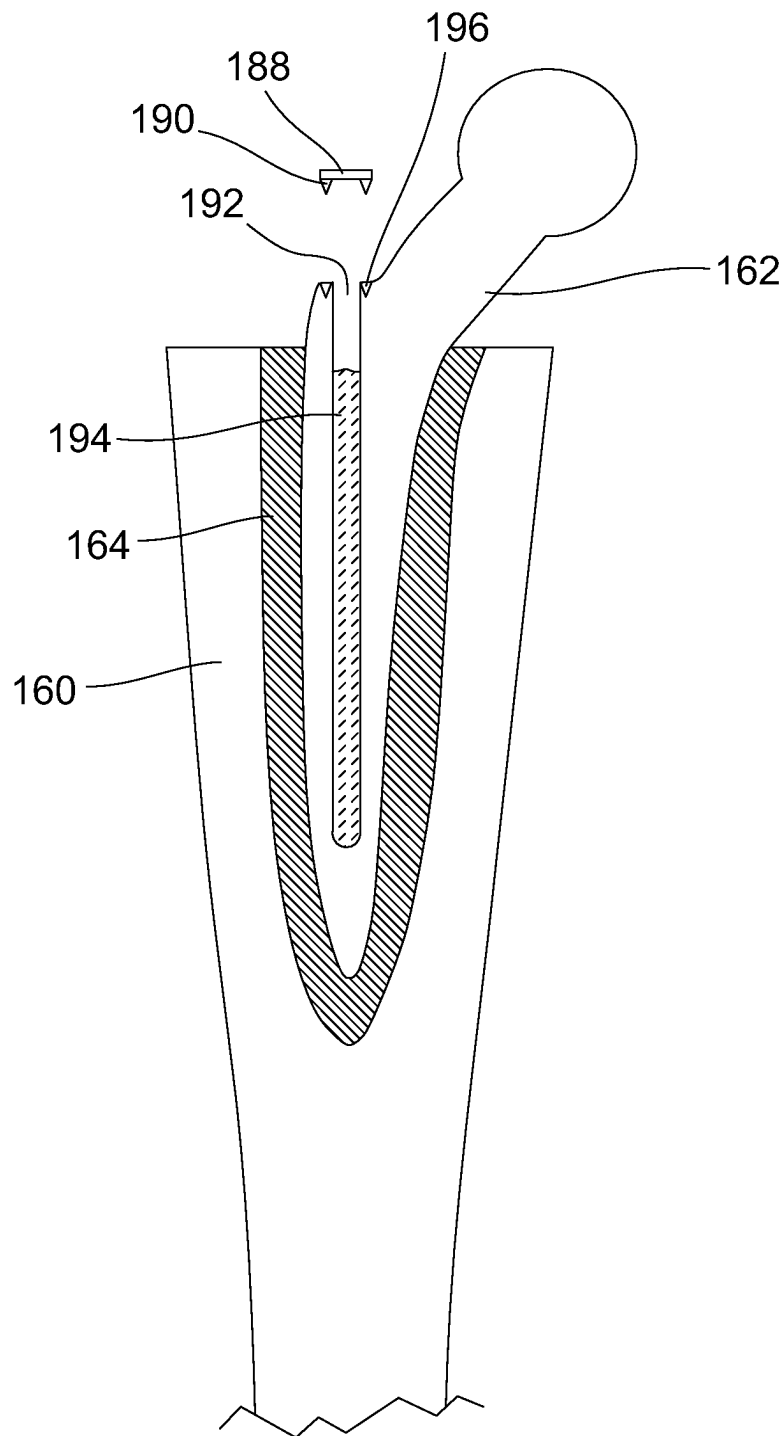


FIG. 33

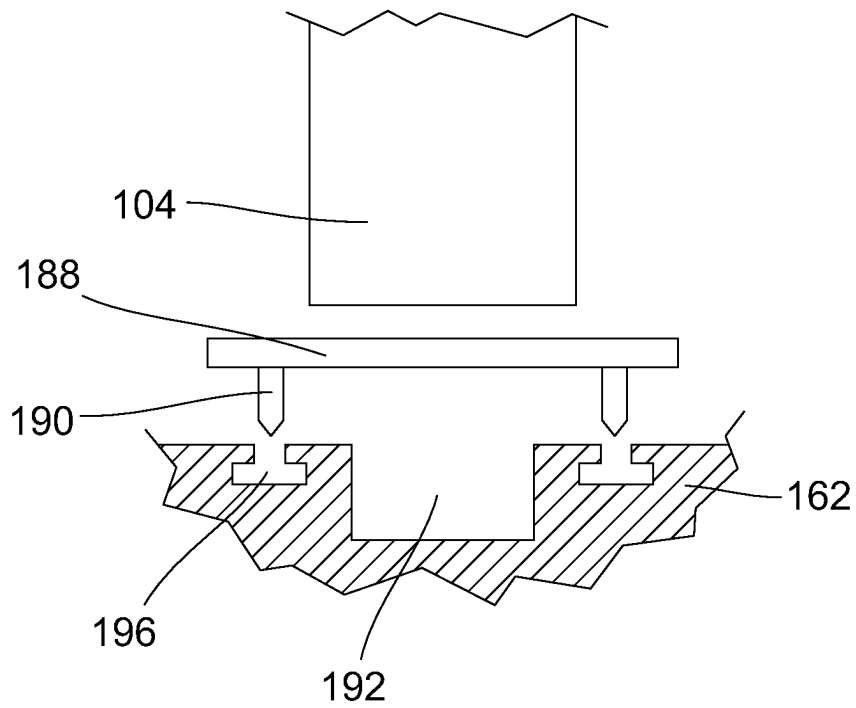


FIG. 34

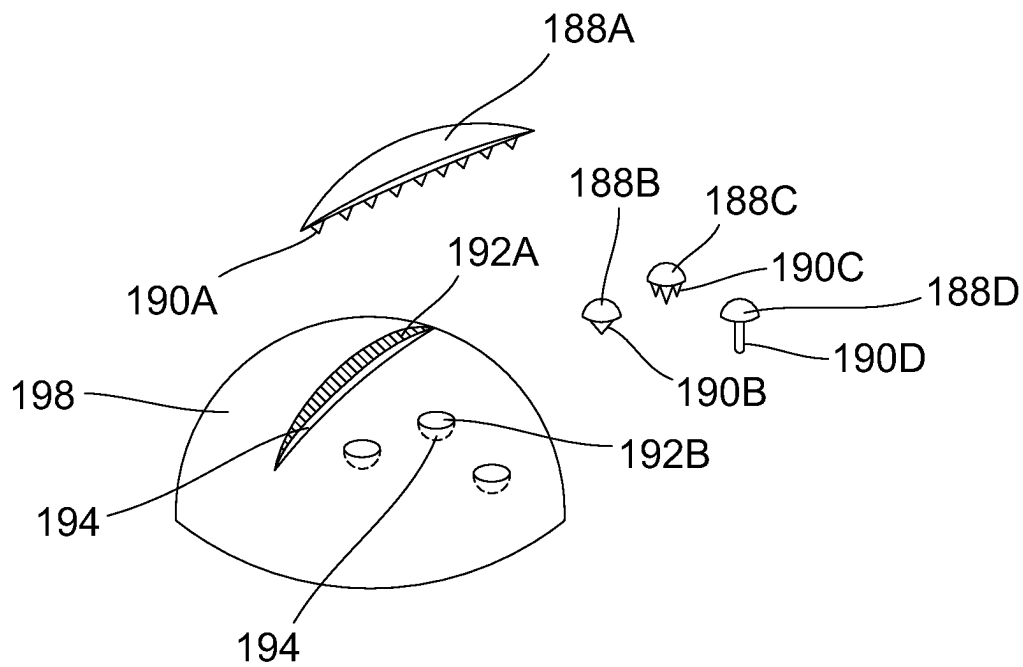


FIG. 35

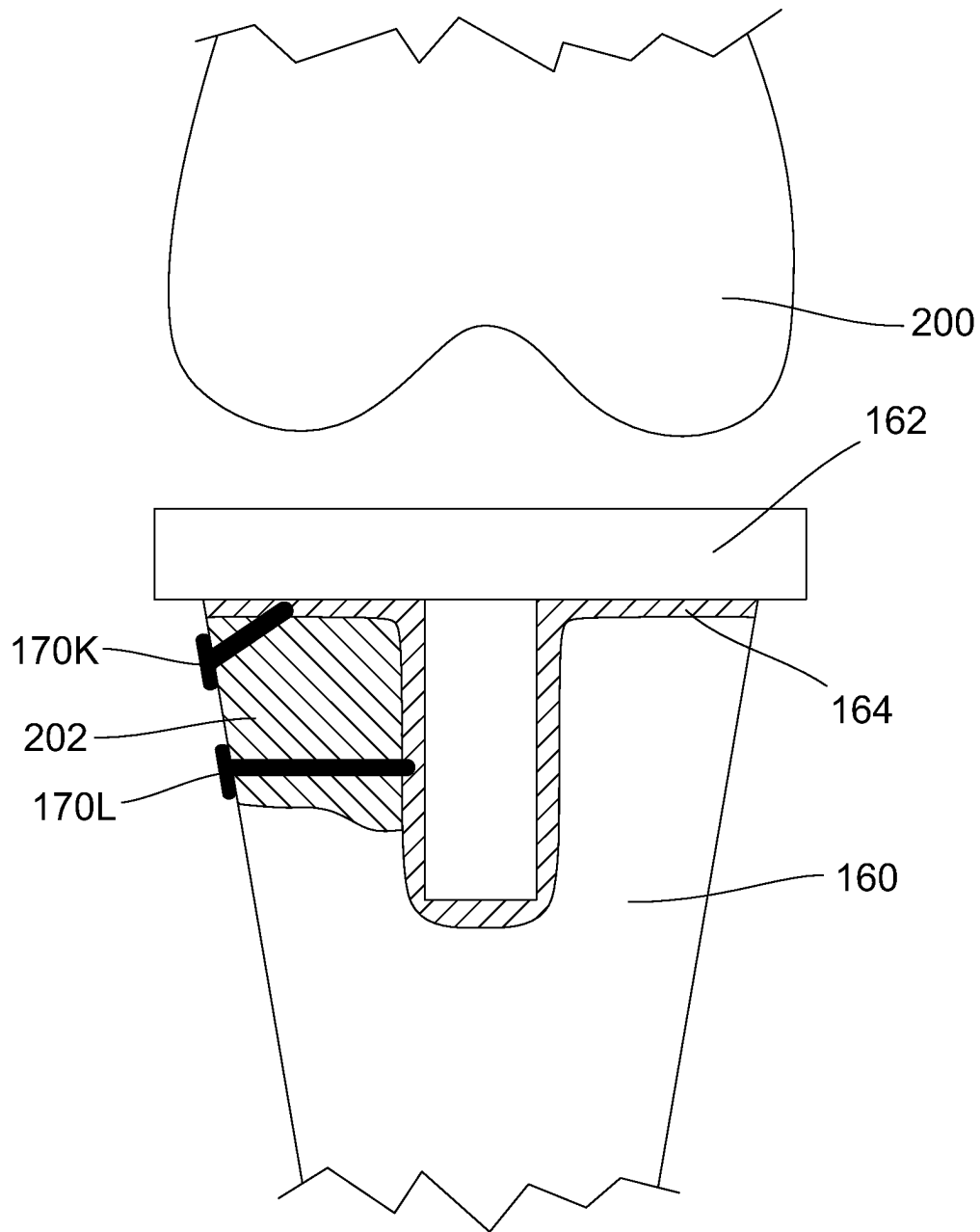


FIG. 36

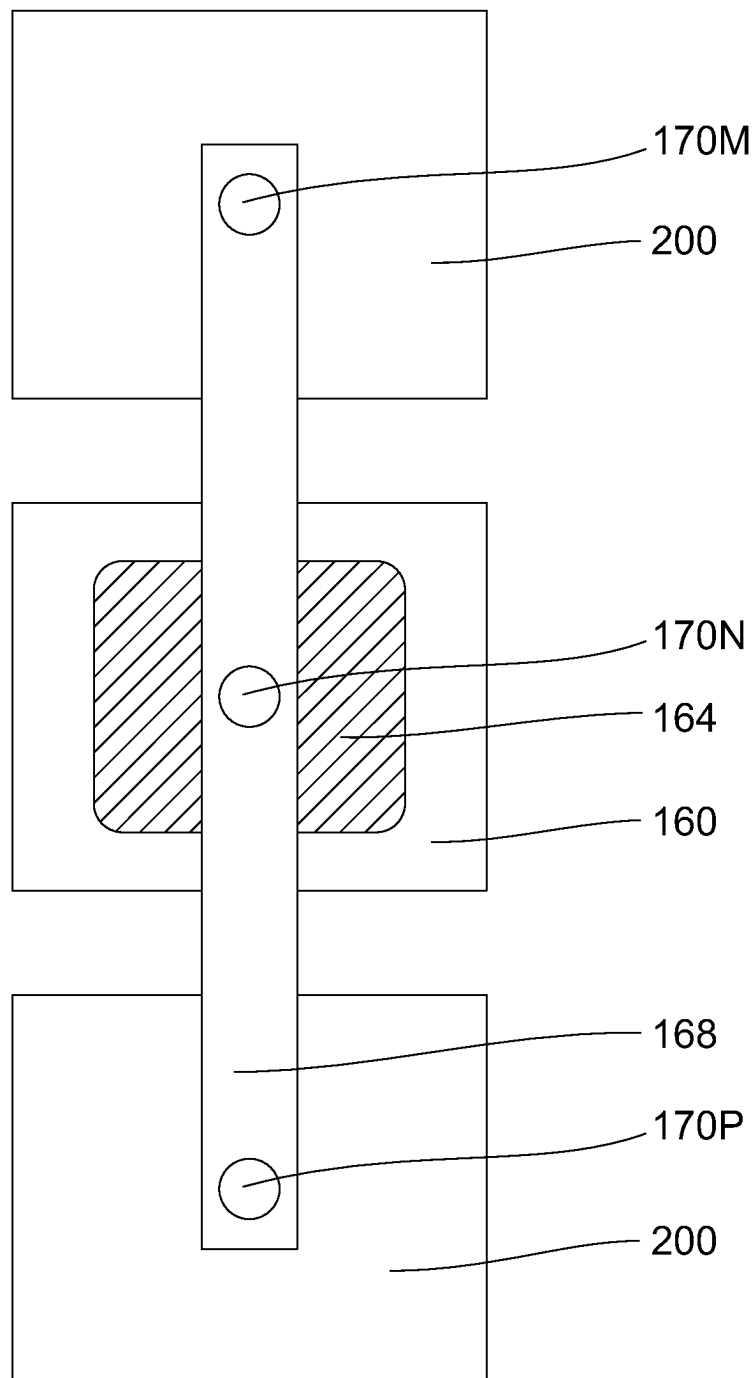


FIG. 37

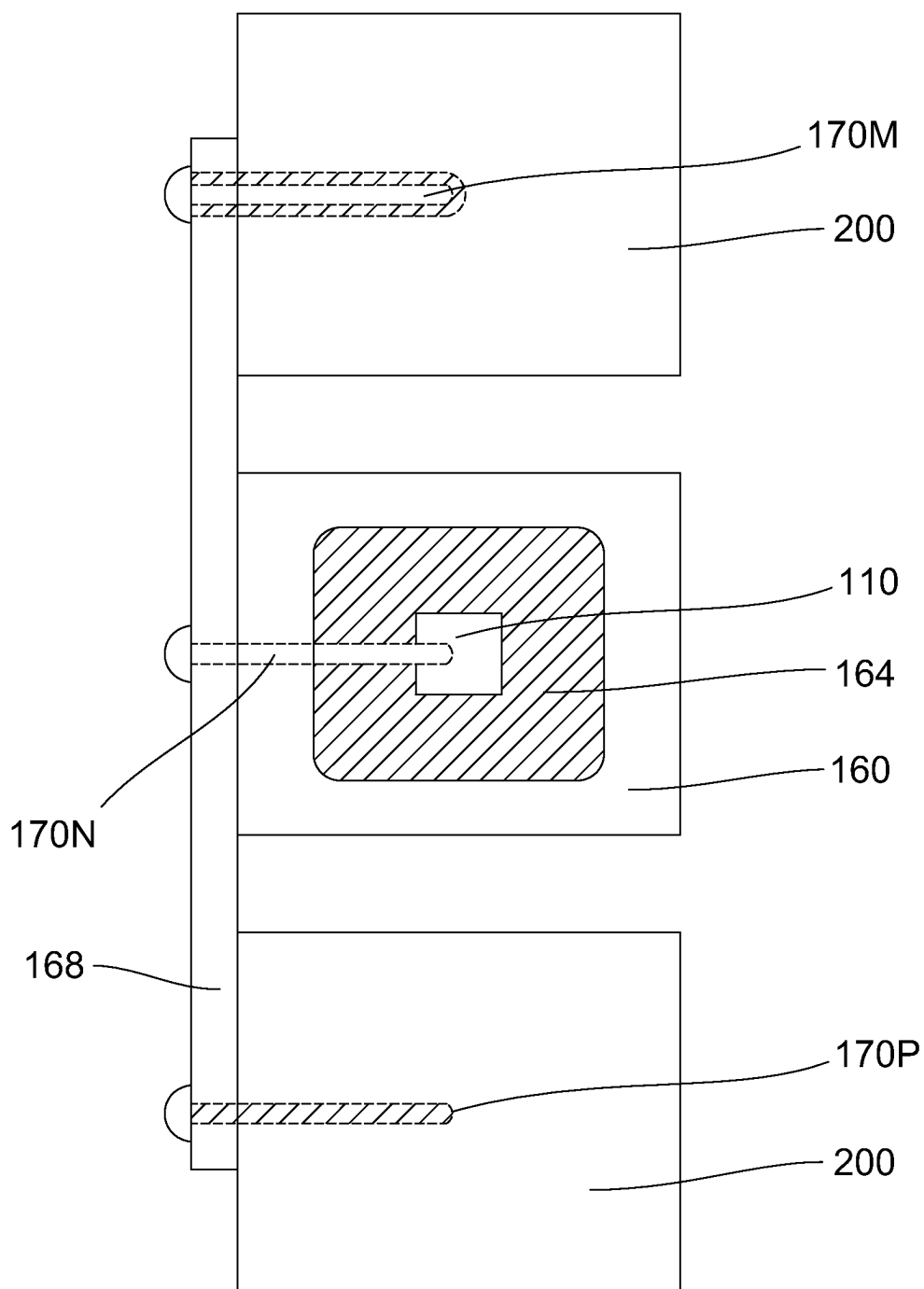


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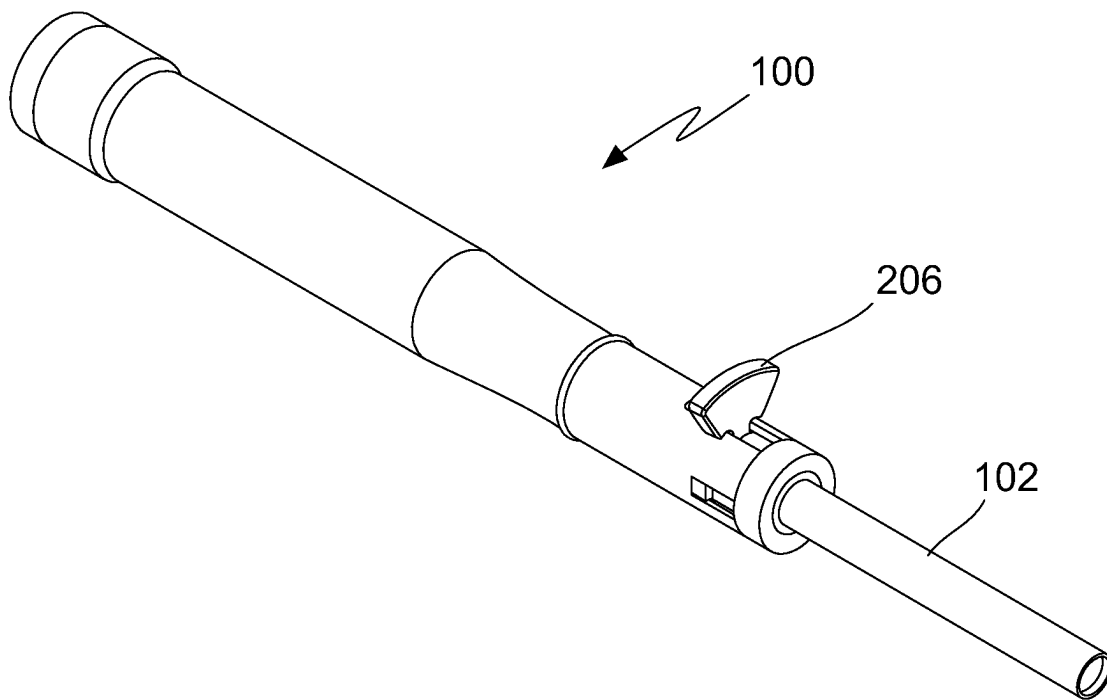


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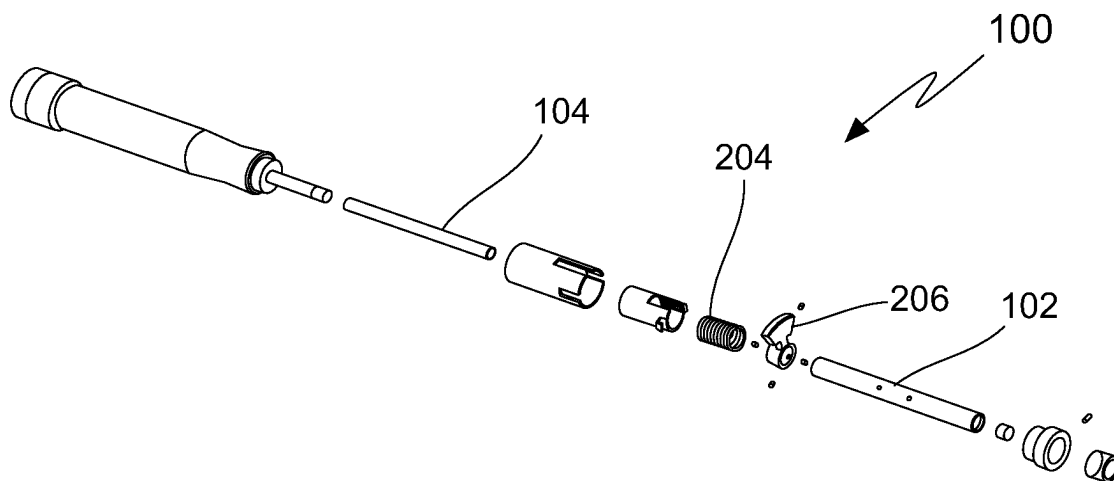


FIG. 40

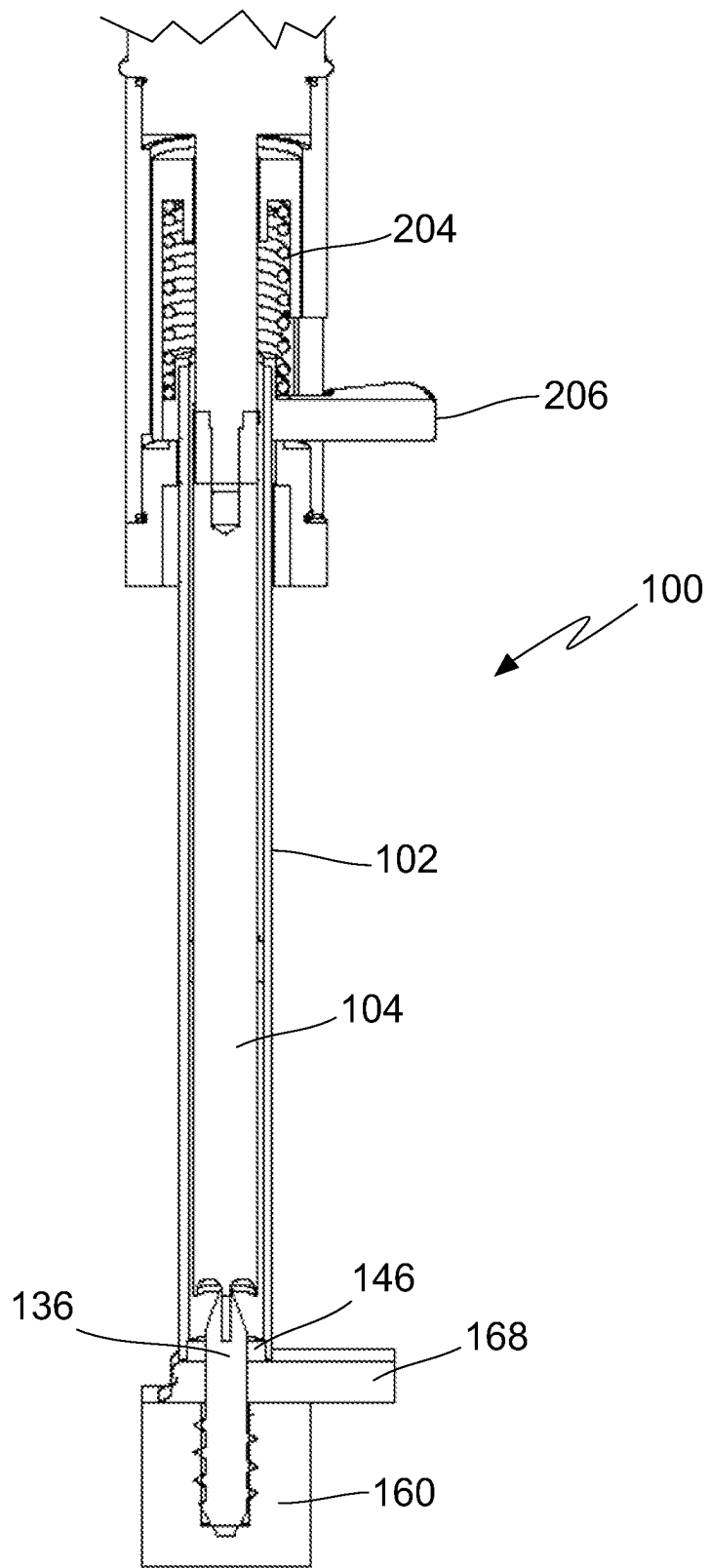


FIG. 41

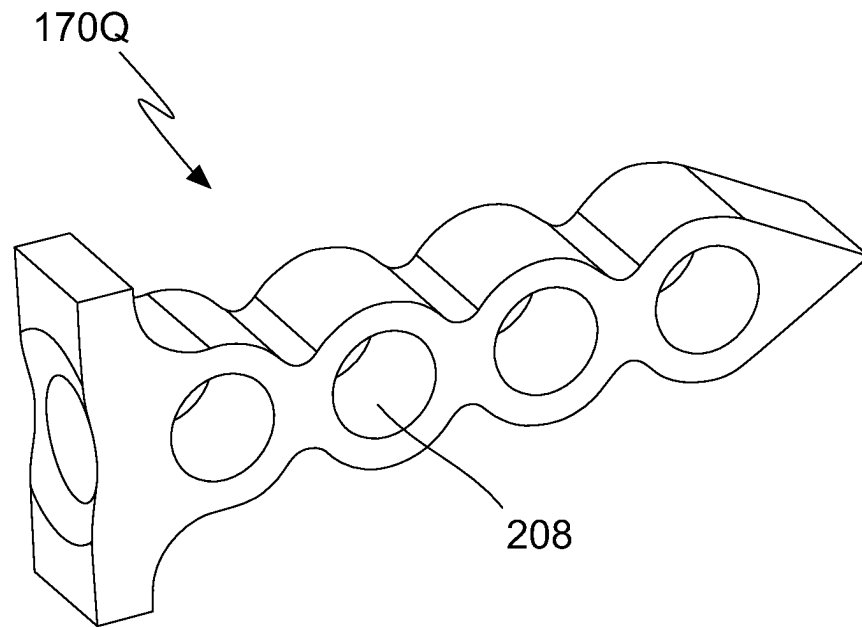


FIG. 42

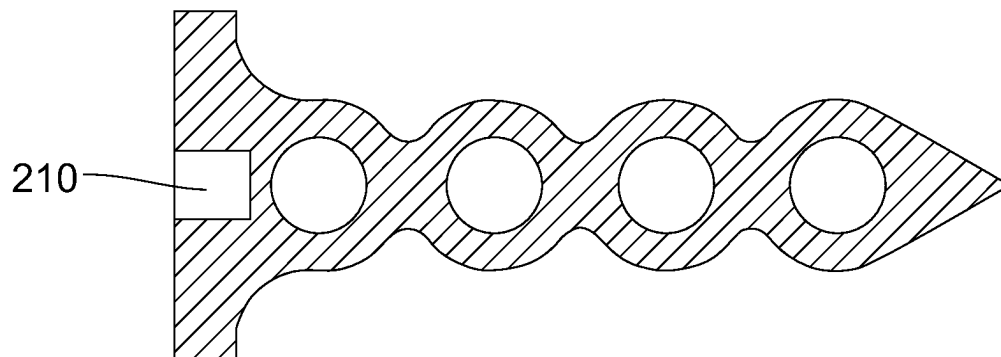


FIG. 43

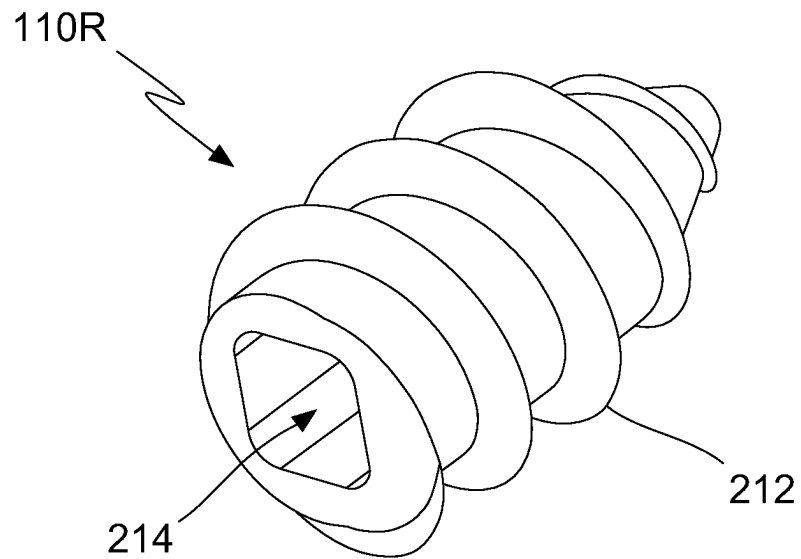


FIG. 44

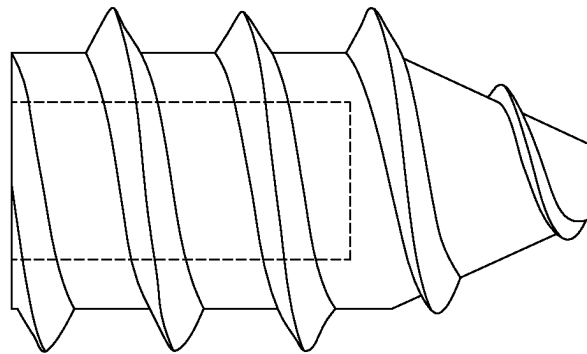


FIG. 45

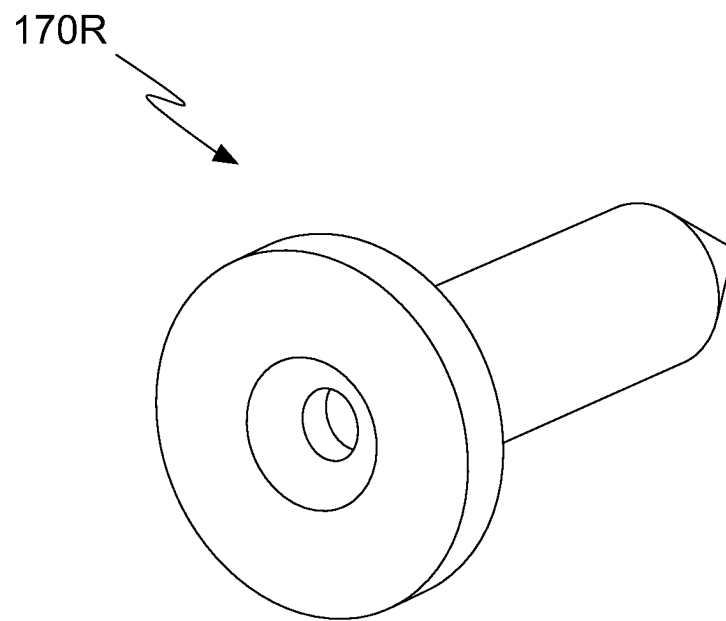


FIG. 46

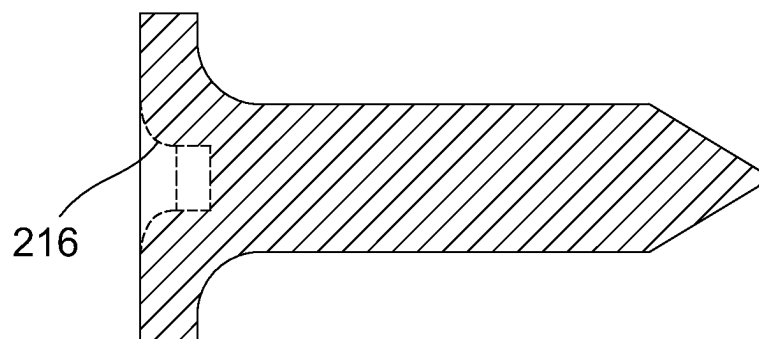


FIG. 47

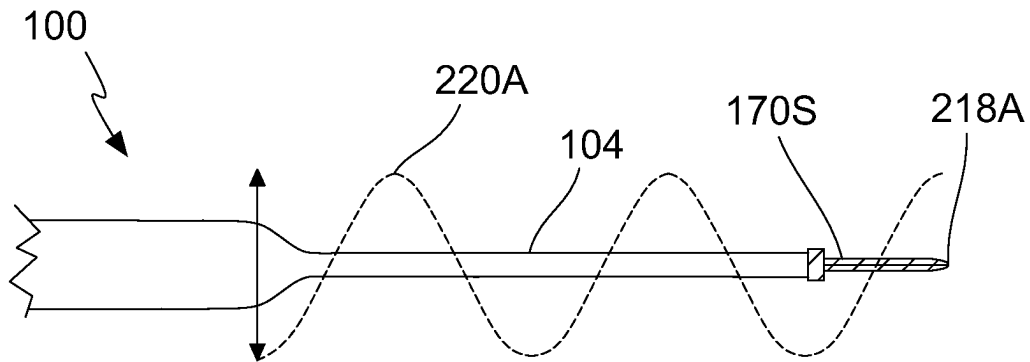


FIG. 48

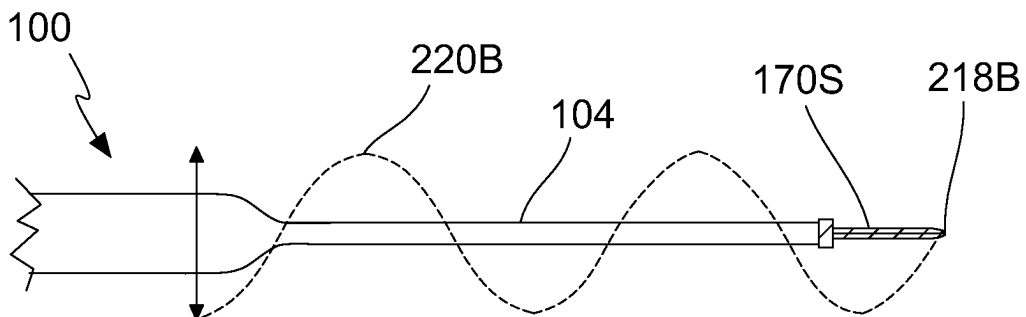


FIG. 49

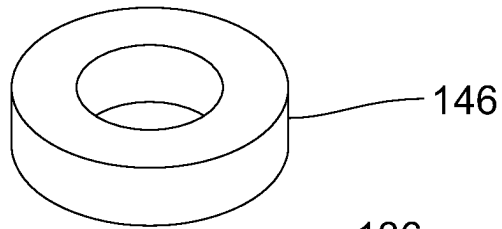


FIG. 50

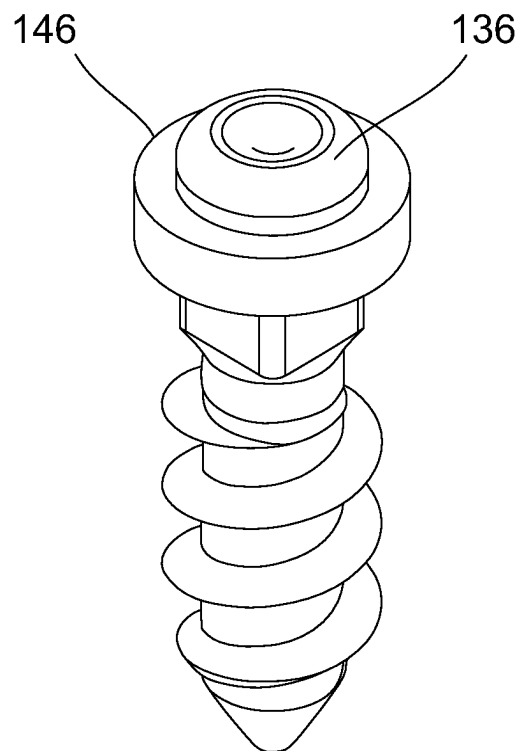


FIG. 51

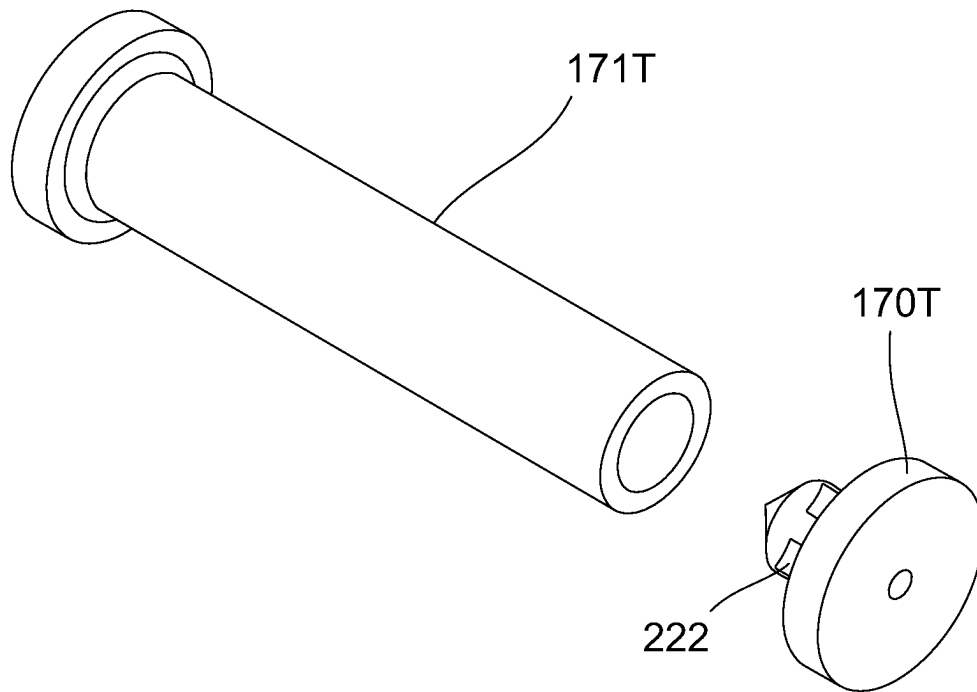


FIG. 52

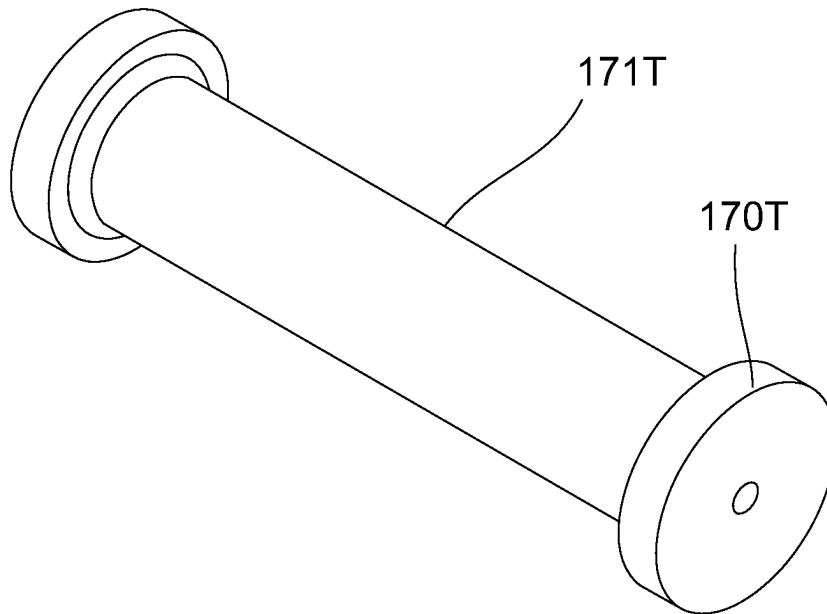


FIG. 53

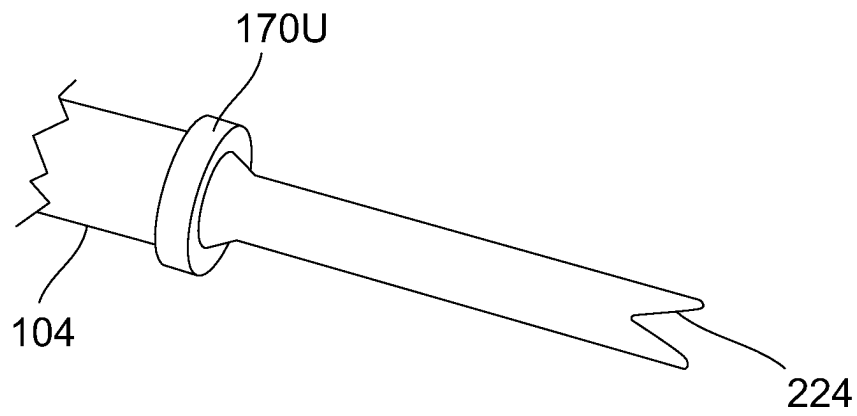


FIG. 54

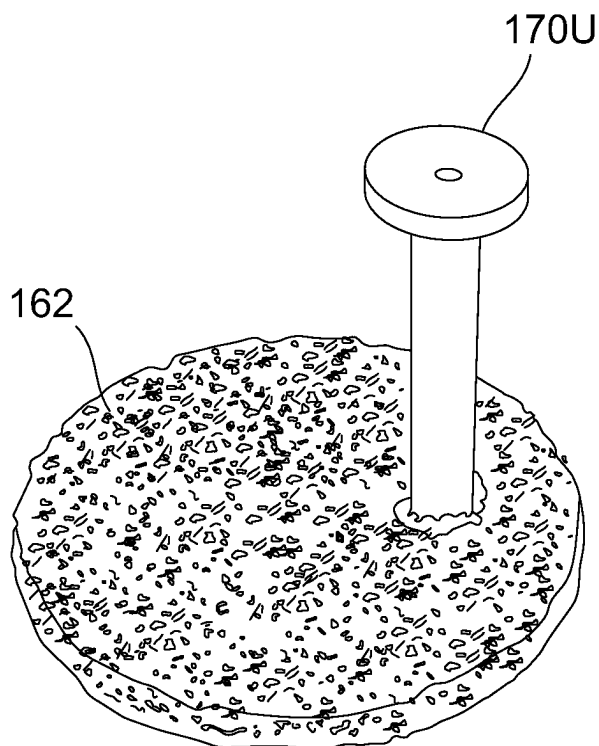


FIG. 55

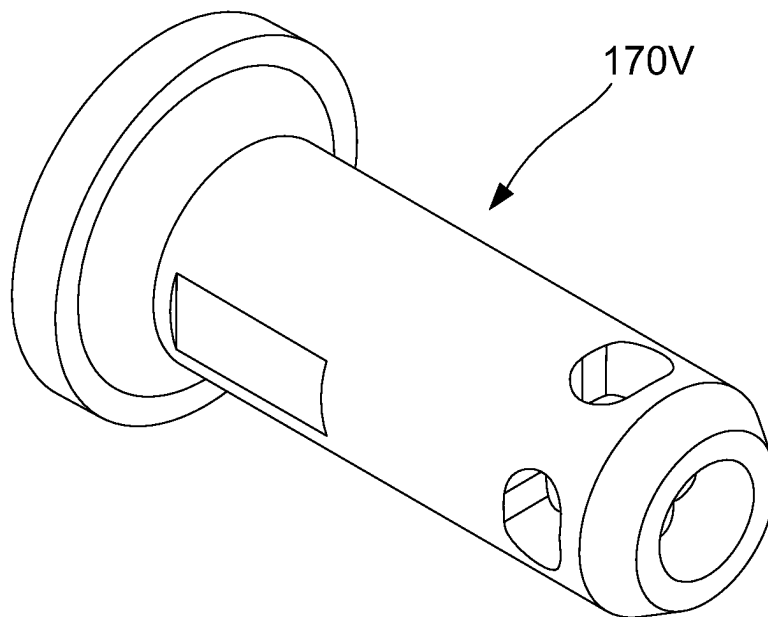


FIG. 56

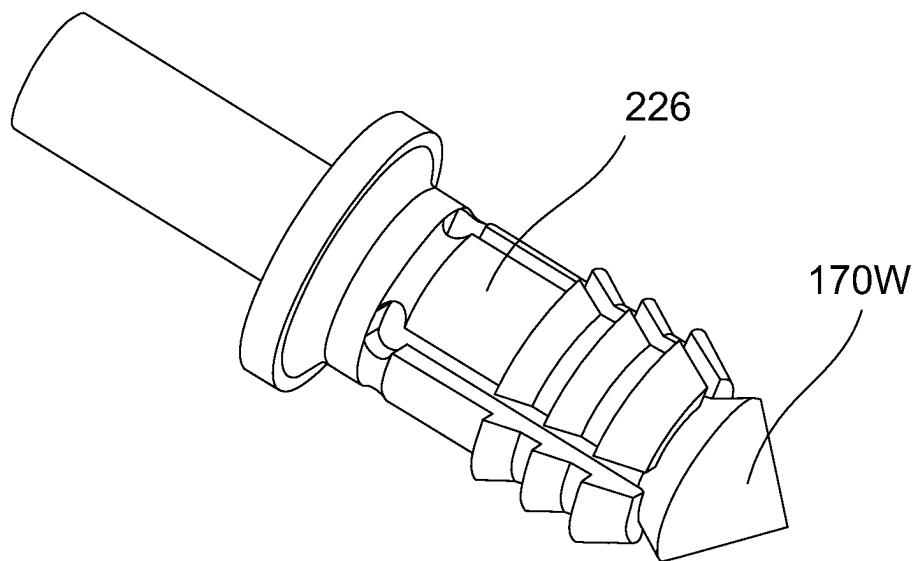


FIG. 57

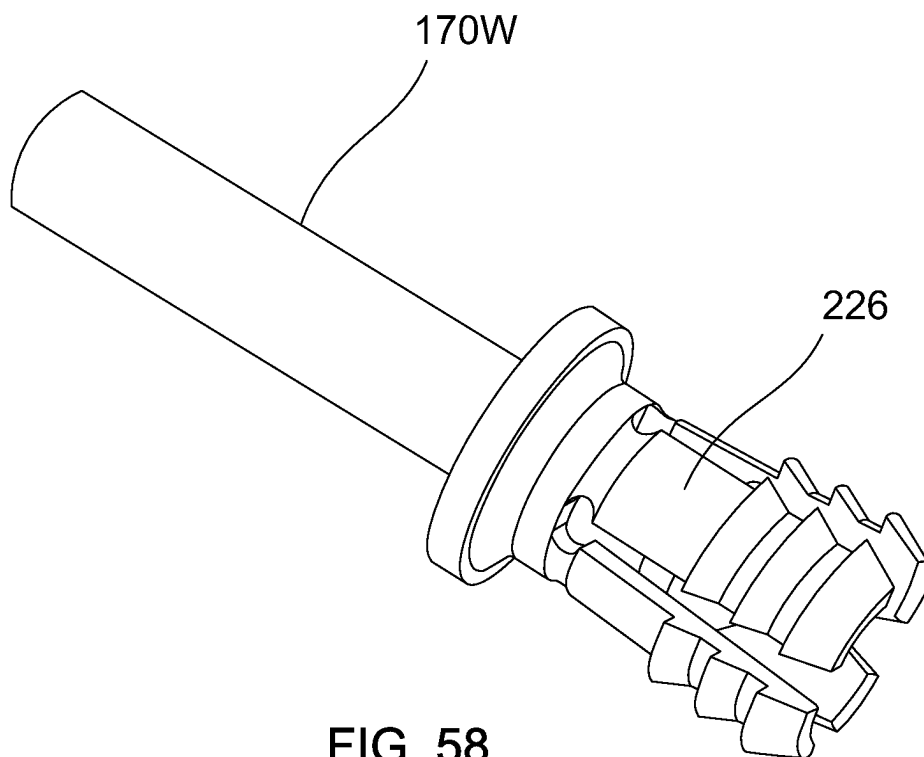


FIG. 58

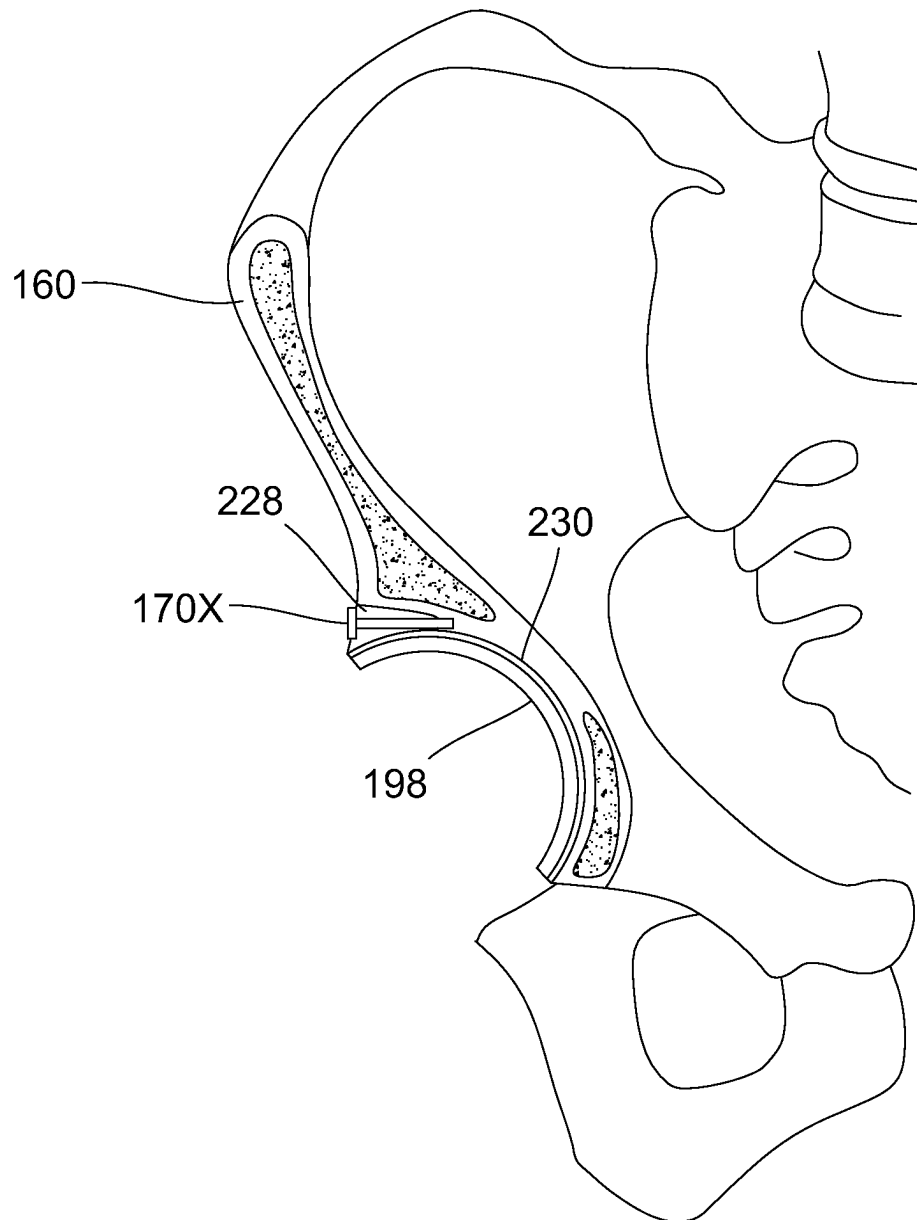


FIG. 59

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METHODS AND DEVICES FOR UTILIZING BONDABLE MATERIALS

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/581,719 filed Jan. 21, 2022, which is a continuation of U.S. patent application Ser. No. 15/224,823 filed Aug. 1, 2016, issued as U.S. Pat. No. 11,246,638 on Feb. 14, 2022, which is a continuation of U.S. patent application Ser. No. 12/711,540 filed Feb. 24, 2010, issued as U.S. Pat. No. 9,439,642 on Sep. 13, 2016, which is based upon and herein claims priority to U.S. Provisional No. 61/155,133 filed Feb. 24, 2009, and is a Continuation-in-Part of: U.S. patent application Ser. No. 12/202,210 filed Aug. 29, 2008, issued as U.S. Pat. No. 8,496,657 on Jul. 30, 2013; U.S. patent application Ser. No. 11/689,670 filed Mar. 22, 2007, issued as U.S. Pat. No. 9,610,073 on Apr. 4, 2017; U.S. patent application Ser. No. 11/671,556 filed Feb. 6, 2007, issued as U.S. Pat. No. 9,421,005 on Aug. 23, 2016; and U.S. patent application Ser. No. 11/416,618 filed May 3, 2006, issued as U.S. Pat. No. 7,967,820 on Jun. 28, 2011. The entire contents of each of the above are hereby expressly incorporated by reference into this disclosure as if set forth fully herein.

FIELD OF THE INVENTION

The invention relates to the fastening and stabilizing of tissues, implants, and bondable materials, such as the fastening of a tissue and/or implant to a bondable material, the fastening of an implant to tissue, and/or the fastening of an implant to another implant. This may involve using an energy source to bond and/or mechanically stabilize a tissue, an implant, a bondable material, and/or other biocompatible material. The present invention also relates to the use of an energy source to remove and/or install an implant and/or bondable material or to facilitate solidification and/or polymerization of bondable material.

BACKGROUND OF THE INVENTION

Body tissue often requires repair and stabilization to address weak or fractured bone, torn ligament or tendon, ripped muscle, or separation of soft tissue from bone. There are numerous methods to facilitate this repair and stabilization. For example, weak or fractured bones can be reinforced with bondable material, i.e. bone cement. Over time, these bondable materials may loosen due to tissue deterioration, improper installation of bondable materials, or deterioration of the bondable materials over potentially damaging to the tissue. After the bondable material is sufficiently removed using conventional methods, bondable material is reapplied to the tissue. Therefore, previous stabilization methods provided for the reapplication of bondable materials and did not utilize existing bondable materials. There is a need for an improved method to utilize existing bondable materials to stabilize tissue and implants.

In another example, bondable materials are used for the installation of implants, i.e. example bone cement. However, some implants loose stability over time. Previous stabilization methods require removal of the implant and the remaining bondable material left on the bone. After the bondable materials are removed, new bondable material is applied to the implant and/or bone. Again, this is a time consuming

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process, potentially damaging the surrounding tissue during the removal of the implant and remaining bondable material.

Bone plates may be positioned internal to the skin, i.e. positioned against the fractured bone, or may be positioned external to the skin with rods connecting the bone and plate. Conventional bone plates are particularly well-suited to promote healing of the fracture by compressing the fracture ends together and drawing the bone into close apposition with other fragments and the bone plate. However, one drawback with plates and screws is that with the dynamic loading placed on the plate, loosening of the screws, and loss of stored compression can result. There is a need for additional fixation devices and methods related to bone plates and other implants providing support to bone.

In addition to internal or external bone plates, surgeons sometimes use intramedullary rods to repair long bone fractures, such as fractures of the femur, radius, ulna, humerus, fibula, and tibia. The rod or nail is inserted into the medullary canal of the bone and affixed therein by screws or bolts. After complete healing of the bone at the fracture site, the rod may be removed through a hole drilled in the end of the bone. One problem associated with the use of today's intramedullary rods is that it is often difficult to treat fractures at the end of the long bone. Fastener members, such as bolts, are positioned through the cortical bone and into threaded openings in the rod. However, the number and positioning of the bolt/screw openings are limited at the tip of the rod because of the decreased surface area of the rod and the reduced strength at the tip of the rod. Fractured bone sections at the distal end of a femur, for example, may not be properly fastened to using conventional intramedullary rod stabilization techniques. Therefore, additional fixation devices and methods are required for use with intramedullary rods.

Other common methods to address weak or fractured bones use a combination of bone screws, bone plates, and intramedullary rods. Conventional methods of using bone screws required a sufficient depth within the bone to stabilize a bone plate. However, weak or fracture bones have limited purchase, as portions of the bone may be unfit for the use of bone screws. Furthermore, if an intramedullary rod has been used to stabilize the bone, the fixation area is further limited as surgeons generally avoid tapping into areas of bone with an underlying intramedullary rod. An improved method of stabilizing existing bone plates and intramedullary rods is needed.

Existing systems and techniques for repairing tissue, like the ones previously described, can be complex, time consuming, lack the characteristic of being employed with precision, be damaging to tissue, and/or fail to provide a robust fastening of tissue. Therefore, there is a need for an apparatus and method for the fastening of tissue that involves a reduction in completion time, greater strength and precision, utilization of previously implanted materials, and preservation of living tissue. There is a need for a system that utilizes of previously installed fixation devices and techniques.

BRIEF DESCRIPTION OF THE DRAWINGS

A more complete understanding of the present invention, and the attendant advantages and features thereof, will be more readily understood by reference to the following detailed description when considered in conjunction with the accompanying drawings wherein:

FIG. 1 is a perspective view of an exemplary vibratory energy device;

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FIG. 2 illustrates an embedding implant and end effector of the invention;

FIG. 3 illustrates an alternative view of FIG. 2;

FIG. 4 illustrates an embedding implant connected to an end effector and embedded in a bondable material;

FIG. 5 illustrates an alternative welding horn or embedding implant of the invention;

FIG. 6 illustrates an alternative configuration of the end effector of FIG. 3;

FIG. 7 illustrates a cross section through the center of a long axis of the end effector of FIG. 6;

FIG. 8 illustrates a cross section through the center of a long axis of a fastening implant;

FIG. 9 illustrates an alternative view of FIG. 8;

FIG. 10 illustrates the embedding implant embedded in bondable material and engaged with a fastening implant;

FIG. 11 illustrates an alternative configuration of the fastening implant of FIG. 9;

FIG. 12 illustrates a cross section through the center of a long axis of the fastening implant of FIG. 11;

FIG. 13 illustrates a washer for use with implants of the invention;

FIG. 14 illustrates a cross section through the center of a long axis of FIG. 13;

FIG. 15 illustrates embedding implant embedded in bondable material;

FIG. 16 illustrates the fastening implant of FIG. 11 disposed in the washer of FIG. 13 and engaged with the embedding implant of FIG. 15;

FIG. 17 illustrates the fastener of FIG. 11 bonded and/or staked to the washer of FIG. 13;

FIG. 18 illustrates alternative configurations of implant of the invention;

FIG. 19 illustrates alternative configurations of FIG. 18;

FIG. 20 illustrates an alternative configuration of an embedding implant;

FIG. 21 illustrates a cross section through the center of a long axis of FIG. 20;

FIG. 22 illustrates an alternative configuration of the embedding implant of FIG. 2;

FIG. 23 illustrates a cross section through the center of a long axis of FIG. 22;

FIG. 24 illustrates an alternative configuration of the embedding implant of FIG. 2;

FIG. 25 illustrates a cross section through the center of a long axis of FIG. 24;

FIG. 26 illustrates the use of a fastening implant and an embedding implant to secure a supporting implant;

FIG. 27 illustrates the assembled configuration of FIG. 26;

FIG. 28 illustrates devices and methods related to utilization of implants and bondable materials;

FIG. 29 illustrates alternatives for the devices and methods of FIG. 28;

FIG. 30 illustrates alternatives for the devices and methods of FIG. 29 and illustrates a method of connecting an instrument to an implant;

FIG. 31 illustrates an alternative method of connecting an instrument to an implant;

FIG. 32 illustrates the affixing an attachment to an implant;

FIG. 33 illustrates the use of a reservoir.

FIG. 34 illustrates an alternative configuration of FIG. 33.

FIG. 35 illustrates alternative types of reservoirs in an alternative type of implant.

FIG. 36 illustrates an additional alternative type of implant.

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FIG. 37 illustrates fixation to previously implanted bondable material.

FIG. 38 illustrates an alternative view of FIG. 37.

FIG. 39 illustrates an alternative configuration of the instrument of FIG. 1.

FIG. 40 illustrates an exploded view of the instrument of FIG. 39.

FIG. 41 illustrates a method of utilization for the instrument of FIG. 39.

FIG. 42 illustrates an alternative configuration of a fastening implant.

FIG. 43 illustrates an alternative view of FIG. 42.

FIG. 44 illustrates an alternative configuration of an embedding implant.

FIG. 45 illustrates an alternative view of FIG. 44.

FIG. 46 illustrates an alternative configuration of a fastening implant.

FIG. 47 illustrates an alternative view of FIG. 46.

FIG. 48 illustrates an energy signal through an end effector and implant.

FIG. 49 illustrates an alternative configuration of an energy signal through an end effector and implant.

FIG. 50 illustrates an exploded view of the fastening implant of FIG. 41.

FIG. 51 illustrates the fastening implant of FIG. 50 after staking.

FIG. 52 illustrates an exploded view of an alternative type of fastening implant.

FIG. 53 illustrates an alternative view of FIG. 52.

FIG. 54 illustrates an alternative configuration of a fastening implant.

FIG. 55 illustrates the fastening implant of FIG. 54 bonded to another implant.

FIG. 56 illustrates an alternative configuration of a fastening implant or embedding implant.

FIG. 57 illustrates an expandable configuration of a fastening implant and expanding implant.

FIG. 58 illustrates an alternative view of FIG. 57.

FIG. 59 illustrates an alternative configuration of an implant, interference implant, and fastening implant.

SUMMARY

As is described in further detail below, fasteners may be embedded within solidified bondable material, for example a grouting agent such as bone cement (PMMA) or other acrylic based material. In an embodiment in accordance with the invention, an embedding fastener may be connected to an end effector of a vibratory energy generator. The embedding fastener may be adapted to enter and engage the bondable material that has been locally melted by vibratory energy, and to be securely retained therein once the bondable material has cooled and hardened.

The end effector may be provided in any of a variety of shapes, one example being an elongated rod or shaft, connectable to a hand piece at a proximal end, and operative to transmit vibratory energy at a distal end. The fastener may be adapted to connect to the distal end of the end effector, for example by mechanical interlocking, threading, twist lock configurations, friction fitting, or adhesive attachment. The mechanical connection must be operative, however, to communicate the vibratory energy from the end effector to the fastener.

The fastener is adapted to be securely retained within the bondable material or adhesive, in one embodiment, by being provided with a shaped or contoured surface upon which the adhesive may grip once hardened. A roughened or porous

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surface may be provided alone or in combination with a shaped surface to increase purchase in bondable material and/or facilitate an interference fit.

The fastener may further be provided with a taper at a leading end which first enters the adhesive. The taper improves performance, at least, by promoting accurate tracking and movement of the fastener into the adhesive, piercing tissue, and facilitating initial melting by concentrating vibratory energy over a smaller surface region.

In a further embodiment of the invention, the embedded bone cement fastener (also referred to as an embedding fastener or embedding implant) described above is provided with one or more radial gaps, chambers, or ports, extending from a central bore. A polymeric fastener is inserted within the central bore, and vibratory energy is applied to the polymeric fastener, whereby polymer at the interface between the embedded fastener and the polymeric fastener melts. When the polymer melts, and particularly as pressure is applied to the polymeric fastener in the direction of insertion, polymer enters the ports, flowing in a direction away from the central bore. When vibratory energy is discontinued, the polymer solidifies, and the polymer fastener is thereafter secured within the embedding fastener.

The embodiments of the present invention may be utilized with limitless fixation techniques and in conjunction with other fasteners and implants. Furthermore, the embodiments herein may assist in the installation and removal of tissue and implants. Moreover, the embodiments of the present invention may assist in the delivery of therapeutic agents and employ methods that facilitate tissue growth and repair. In addition, the embodiments herein may be used to apply vibratory energy to remove and/or install an implant in bondable material or to facilitate solidification and/or polymerization of the bondable material.

DETAILED DESCRIPTION

The invention relates to the devices and methods for the utilization of bondable materials and bondable materials, fixation and fastening of tissue to tissue, an implant to tissue, and an implant to an implant both inside and outside the body. The invention additionally relates to removing and anchoring implants to bondable materials and/or other biocompatible materials, anchoring implants using previously implanted and hardened bondable materials, and fixation using vibratory energy, mixing, solidifying, bonding, and/or mechanical interlocking of materials. The present invention also relates to the use of an energy source to install and/or remove an implant or bondable material or to facilitate the solidification and/or polymerization of a bondable material

The methods and devices disclosed herein may be used in conjunction with any medical procedure on the body. The stabilization, fastening, and/or repair of tissue or an implant may be performed in connection with any medical procedure related to a joint, bone, muscle, ligament, tendon, cartilage, capsule, organ, skin, nerve, vessel, or other body parts. For example, tissue may be stabilized during intervertebral disc surgery, kyphoplasty, knee surgery, hip surgery, organ transplant surgery, bariatric surgery, spinal surgery, anterior cruciate ligament (ACL) surgery, tendon-ligament surgery, rotator cuff surgery, capsule repair surgery, fractured bone surgery, pelvic fracture surgery, avulsion fragment surgery, shoulder surgery, hernia repair surgery, and surgery of an intrasubstance ligament tear, annulus fibrosis, fascia lata, flexor tendons, etc.

Also, an implant may be inserted within the body and fastened to tissue with the present invention. Such implant

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insertion procedures include, but are not limited to, partial or total knee replacement surgery, hip replacement surgery, shoulder replacement surgery, bone fastening surgery, etc. The implant may be an organ, partial organ grafts, tissue graft material (autogenic, allogenic, xenogenic, or synthetic), collagen, a malleable implant like a sponge, mesh, bag/sac/pouch, collagen, or gelatin, or a rigid implant made of metal (porous or nonporous), polymer, composite, or ceramic. Collagen may provide the benefit of bolstering tissue growth. Additionally, a desiccated collagen may be used to absorb surrounding fluid, which may provide the additional benefit of applying pressure on the tissue being repaired. Other implants include breast implants, biodegradable plates, porcine or bovine patches, metallic fasteners, compliant bearing for medial compartment of the knee, nucleus pulposus prosthetic, stent, suture, suture anchor, tissue graft, tissue scaffold, biodegradable collagen scaffold, and polymeric or other biocompatible scaffold. The scaffold may include fetal cells, stem cells, embryonic cells, enzymes, and proteins.

In this application, the term "bondable" or "bondable material" is used to refer to the materials discussed herein, as well as any material, suitable for in vivo applications, which can be softened and made flowable by the application of heat (such as heat produced with vibratory energy such as ultrasonic energy), and which, when softened, may become tacky and will bond to other materials and will flow to fill available space. Thus, the material may be thermoplastic, but it may also exhibit tackiness or bonding ability when in its plastic form. Many materials suitable for in vivo applications are made of or incorporate such bondable materials. Generally speaking, the amount of heat needed to soften and make flowable should be within a temperature range that does not produce substantial thermal tissue necrosis. Alternatively stated, the amount of heat required to soften the bondable material during vibratory bonding is substantially confinable, due to the thermal properties of the bondable material, to an area of contact between the objects which are being bonded, thereby protecting living body tissue near the contact between the two objects from substantial thermal tissue necrosis. Any embodiment herein may be used with any of the materials and/or applications disclosed herein or known in the art.

The fixation and fastening system and other embodiments of the present invention contemplates the use any materials that may include or be used in conjunction with bondable materials for bonding and/or staking within the human body. Implants that may be used as fasteners may also be referred to as fastening implants. Bondable material used may include, but are not limited to, biocompatible, degradable, biodegradable, bioerodible, bioabsorbable, mechanically expandable, hydrophilic, bendable, deformable, malleable, riveting, threaded, toggling, barded, bubbled, laminated, coated, blocking, pneumatic, one-piece, multi-component, solid, hollow, polygon-shaped, pointed, self-introducing, and combinations thereof. Also, the devices may include, but are not limited to, metallic material, polymeric material, ceramic material, composite material, body tissue, synthetic tissue, hydrophilic material, expandable material, compressible material, bondable material, and combinations thereof. Bondable material may also include polymethyl methacrylate (known as "PMMA" or "bone cement"), glue, adhesive, and/or other grouting agents or acrylic materials used for fixation.

In this application, "bond", "bonded", and "bonding" includes, but is not limited to, attaching, engaging, connecting, binding, adhering, and/or fastening one or more mate-

rials through resistive heating, mechanical interlocking, application of force, application of grouting agents (i.e. bone cement), adhesives and/or solvents, spraying, radiofrequency, vibratory energy (i.e. ultrasound), microwave, laser, electromagnetic, electro shockwave therapy, plasma energy (hot or cold), and other suitable method described herein or known in the art.

Preferably, materials of the present invention can melt with the application of energy, becoming gel-like, tacky, and/or soft. The energy source and the technique used to bond and/or stake the material within the body can be selected to minimize or avoid damage to surrounding body tissue. Exemplary materials that may be used may include polymers, ceramics, composites, and metals, although other materials may also be suitable for use with the invention. While the present invention contemplates the use of any of these materials in any of the following embodiments, polymeric material is used in the following examples and description simply to illustrate how the invention may be used.

There are a limitless number of materials may be used for the present invention. Examples of amorphous polymers are polycarbonate (LEXAN), polystyrene, polysulfone (ULDALL), and acrylics polycarbonate (ABS and styrenes). Examples of semi-crystalline polymers include acetyl (DELIRIN), nylon, polyester, polyethylene, polyether ether ketone, polypropylene, polyvinylchloride (PVC), and Caprolactam. Biodegradable semi-crystalline polymers may include polylactic acid and polyglycolic acid. Copolymers of PGA and PLA may also be used. Poly-L-lactide (PLLA) or other forms of PLA may also be used. Other polymers which may be used with the present invention, either as a thermoplastic or non-thermoplastic, are polyethylene glycol (PEG)-copolymers and D,L-lactide-co-glycolide polyesters. Some semi-crystalline materials are particularly suitable for surgical bonding and/or staking, especially vibratory bonding and staking. Examples of such materials include PAEK (polyaryletherketone), including PEEK (polyetheretherketone) and PEKK (polyetherketoneketone).

In addition to PEEK and the other polymers described herein, the implants, devices, and methods of the present invention may use keratin, a naturally occurring polymer. Keratin may be vibratory bonded or staked to itself, to other implants, or within tissue. This may be performed in the operating room or intracorporeally. Keratin may be bonded to collagen or to other known polymers. In an exemplary application, keratin may be used to fasten tissue to bone since keratin has BMP and tissue scaffold properties. It is contemplated that any of devices and methods disclosed herein may utilize keratin alone or in combination with PEEK, polylactic acid, or other polymer. Keratin may be used to make fasteners, disc replacements, joint replacement components, stents, cell scaffolds, drug reservoirs, etc. Also, joint bearing surfaces may include keratin with or without collagen or chondrocytes. The bearing surfaces may be fastened to a joint component using PEEK or PLA fasteners.

Another polymer that can be used with the present invention is a class of natural materials, called polyhydroxyalkanoates, or PHA polymers. These polymers are synthesized in nature by numerous microorganisms, and they have been recently recognized as the fifth class of naturally occurring biopolymers (along with the polyamino acids, polynucleic acids, polysaccharides, and polyisoprenoids). Unlike the other naturally occurring biological polymers, however, the PHA polymers are thermoplastic, i.e. they may be repeatedly softened with heat and hardened with cooling. As such, these polymers can be processed much like other plastics. A

specific example of a PHA polymer that could be used is poly-4-hydroxybutyrate material. Such PHA polymers are available from Tefpa Inc of Lexington, MA.

Fasteners of the invention may utilize or be coated with polymethylmethacrylate (PMMA), in order to promote bonding with PMMA used in the body, or PMMA could be incorporated into polymer of the fastener, or deposited within cavities or shapes formed in the fastener surface, including threaded, roughened, porous, or nano textures. A fastener may be thus coated with PMMA, or formed entirely of PMMA, and may be heat bonded, advantageously using ultrasound, to another PMMA surface or other adhesive surface, otherwise as described herein with respect to bone cement. Although PMMA, known generally as bone cement, and other polymers may function more as a grouting agent than a cement or adhesive. The term "bondable material" is used throughout the specification for simplicity.

In accordance with the invention, metals are advantageously connected with fasteners incorporating polymeric materials. Any of a variety of metals may be used, either smooth or formed with at least portions formed of metal, or a roughened or porous surface, or formed with cavities or other shapes upon which polymeric material may mold, enter, adhere, or otherwise affix. The polymer is softened in accordance with the invention through the application of heat, including heat created using vibratory energy, to become tacky, or sufficiently softened in order to bond on a microscopic level, or a macroscopic level through adaptation to the surface structure of the metal. For use in vivo, biocompatible metals are used, including stainless steel, nitinol or other SMA (shape metal alloy), tantalum, porous tantalum, titanium, cobalt-chrome alloys, and other metals such as are known to those skilled in the art. Additional related information, including bonding polymers and metals, and polymer to polymer bonding of implant materials, may be found in U.S. Pat. No. 5,163,960 entitled "Surgical devices assembled using bondable materials", and U.S. Pat. No. 7,104,996 entitled "Method of performing surgery", the contents of each of which being incorporated herein by reference.

The fastening device of the present invention may include therapeutic substances to promote healing. These substances could include antibiotics, hydroxyapatite, anti-inflammatory agents, steroids, antibiotics, analgesic agents, chemotherapeutic agents, bone morphogenetic protein (BMP), demineralized bone matrix, collagen, growth factors, autogenic bone marrow, progenitor cells, calcium sulfate, immo suppressants, fibrin, osteoinductive materials, apatite compositions, germicides, fetal cells, stem cells, enzymes, proteins, hormones, cell therapy substances, gene therapy substances, and combinations thereof. These therapeutic substances may be combined with the materials used to make the device. Alternatively, the therapeutic substances may be impregnated or coated on the device. Time-released therapeutic substances and drugs may also be incorporated into or coated on the surface of the device. The therapeutic substances may also be placed in a bioabsorbable, degradable, or biodegradable polymer layer or layers.

The therapeutic agents may also be placed within one or more cavities disposed in a fastening device of the present invention. Different agents may be disposed in different cavities of the device to specifically tailor the implant for a particular patient. Dosages of the therapeutic agent may be the same or different within each of cavities as well. The cavities may include a cover which may release the agent in a controlled or timed manner. The cover may be biodegradable or bioerodible to allow the agent to release to surround

ing tissue. Examples of suitable therapeutic agents include bone growth inducing material, bone morphogenic proteins, osteoinductive materials, apatite compositions with collagen, demineralized bone powder, or any agent previously listed. U.S. patent application Ser. No. 11/549,994 entitled “Drug Eluting Implant” discloses means for delivering therapeutic agents. The above-mentioned patent application is incorporated by reference herein in its entirety.

The fastening devices of this and other embodiments of the invention may be used in combination with fasteners in the prior art. Examples of fasteners, implants, and their methods of employment may be found in U.S. Pat. Nos. 5,163,960; 5,403,348; 5,441,538; 5,464,426; 5,549,630; 5,593,425; 5,713,921; 5,718,717; 5,782,862; 5,814,072; 5,814,073; 5,845,645; 5,921,986; 5,948,002; 6,010,525; 6,045,551; 6,086,593; 6,099,531; 6,159,234; 6,368,343; 6,447,516; 6,475,230; 6,592,609; 6,635,073; and 6,719,765. Other fastener types are disclosed in U.S. patent application Ser. Nos. 12/202,210; 10/102,413; 10/228,855; 10/779,978; 10/780,444; and 10/797,685. The above cited patents and patent applications are hereby incorporated by reference in their entirety.

With reference to FIG. 1, any known energy emitting instrument may be used with the surgical system of the present invention. Instrument 100 may produce energy such as resistive heating, radiofrequency, ultrasound (vibratory), microwave, laser, electromagnetic, electro shockwave therapy, plasma energy (hot or cold), and other suitable energy disclosed herein or known in the art. FIG. 1 illustrates an exemplary handpiece or instrument 100 that may be used with the present invention. The instrument 100 may be a vibratory energy generator with a sheath 102 to cover and protect the end effector 104 and engage a fastener/implant near engagement feature 106. As will be discussed below, the instrument may be used to bond and/or mechanically interlock fasteners and other embodiments the present invention. Additional embodiments of instrument 100 are disclosed in U.S. patent application Ser. No. 12/202,210 entitled “Methods and Devices for Utilizing Thermal Energy to Bond, Stake and/or Remove Implants”, which is incorporated by reference herein.

With reference to FIGS. 2-7, end effector 104 may be utilized with anchor or embedding fastener 110. Embedding fastener 110 may also be referred to as an embedding implant. FIG. 2 illustrates end effector 104 that connects to embedding fastener 110 with distal end 108. The connection between distal end 108 and embedding fastener 110 may utilize threads, magnetism, friction, taper, ball and socket, linkage, adhesive, interlocking shapes, and other connections known in the art. Additionally, distal end 108 and embedding fastener 110 may be permanently or detachably connected.

As shown in FIG. 2, embedding fastener 110 may further be provided with a taper 114, which first enters bondable material 120 in FIG. 4. Taper 114 may improve performance, for example, by promoting accurate tracking and movement of embedding fastener 110 into bondable material 120, piercing body tissue, and facilitating initial melting by concentrating vibratory energy over a smaller surface region. Although embedding fastener 110 may be made of any material described herein or known in the art, it may be preferable to use titanium.

As discussed in further detail below, embedding fastener 110 may also have feature 112 and/or feature 116, either or both may be a surface feature, recess, or pass through a portion or the entirety of embedding fastener 110.

Referring to FIG. 3, an embodiment of end effector 104 may have proximal end 118 to the other components of instrument 100. The connection between instrument 100 and proximal end 118 may be threaded, magnetic, friction, hex, ball and socket, linkage, adhesive, and other methods disclosed herein or known in the art.

As shown in FIGS. 3-7, end effector 104 may be provided in any of a variety of shapes, one example being an elongated rod or shaft, connectable to a hand piece at a proximal end 118, and operative to transmit vibratory energy at a distal end 108. While a rod shape is shown and selected for reduced manufacturing cost, end effector 104 may have the form of box or hex channel, oval or other shape, provided it communicates vibratory energy to a distal end 108, an attached fastener, or embedding fastener 110. Additionally, feature 111 of FIG. 5 may be used on embedding fastener 110 or integrated into end effector 104 (not shown).

In an additional embodiment, embedding fastener 110 is adapted to connect to distal end 108 of end effector 104 by mechanical interlocking, as by a bore in embedding fastener 110, sized to receive distal end 108 of end effector 104, optionally provided with internal or external threading (not shown), wherein post 108 has mating threads. Additionally, the connection may be threaded, magnetic, friction, hex, ball and socket, linkage, adhesive, and other methods disclosed herein or known in the art. Similarly, a bore or aperture may be provided in end effector 104, mateable with a post or projection on embedding fastener 110. Other mechanical connections are contemplated, including twist lock configurations, friction fitting, or adhesive attachment. The mechanical connection should preferably be operative to communicate vibratory energy from end effector 104 to embedding fastener 110, as by a firm mechanical connection.

As shown in an embodiment of FIG. 4, embedding fastener 110 may be adapted to be securely retained within bondable material 120 by being provided with a shaped or contoured surface upon which the softened bondable may adhere. A roughened or porous surface may be provided alone or in combination with shaped surface thereby providing for increased purchase in bondable material 120.

With reference to FIG. 4, embedding fastener 110 may be embedded within solidified bone bondable material 120, for example PMMA, acrylic based adhesive, or other bondable materials. In the present invention, embedding fastener 110 is connected to end effector 104 of an embodiment of instrument 100, such as a vibratory energy generator as shown in FIG. 1. Embedding fastener 110 is adapted to enter and engage bondable material 120 or bondable material 120 that has been locally melted by vibratory energy (as shown in FIG. 4), through contact between embedding fastener 110 and bondable material 120 during operation of instrument 100. Embedding fastener 110 is securely retained by bondable material 120 once the latter has hardened. Although the embodiment in FIG. 4 may be used under a limitless number of configurations and settings, Table 1 is being set forth with operative examples:

TABLE 1

Titanium Embedding Fastener Bonded into PMMA

Embedding Fastener Type: Shown in FIG. 20
Instrument: Handpiece SN0105 with tuning of 39,000-45,000 Hz
System Settings:
Sample 1: 40,850 Hz, 100W, 2.0 sec weld time
Sample 2: 40,750 Hz, 75 W, 1.5 sec weld time

TABLE 1-continued

Sample 3: 40,800 Hz, 75 W, 1.0 sec weld time Sample 4: 40,750 Hz, 75 W, 1.0 sec weld time				
Test Sample Number	Power (watts)	Time (sec)	Force Applied to Break (lbs.)	Deformation Depth (inches)
1	63	2.56	46.7	0.116
2	48	2.01	61.9	0.119
3	48	1.58	32.5	0.109
4	48	1.47	31.5	0.098

Once anchored, end effector **104** and embedding fastener **110**, embedded in bondable material **120**, may remain connected. Alternatively, end effector **104** may be removed and another fastener of a similar or different design may be connected to an implanted embedding fastener **110** as shown in FIGS. 8-10. In a further embodiment, fastener **124** such as described in the incorporated patents and applications may be fastened to an implanted or installed embedding fastener **110**. Fastener **124** may have fastener bore **128** as shown in FIGS. 8-9. Referring to FIG. 10, a bondable insert **134** may be secured into fastener bore **128**. Bondable insert **134** may be secured by press fitting, threading, or bonding to fastener bore **128** and/or embedding fastener **110**. The fastener **124** may be utilized as detailed in U.S. patent application Ser. No. 12/202,210, which has been incorporated by reference herein. In further embodiments, any fastener described in the related references cited in paragraph [0001] or discussed herein may be fastened to the embedding fastener **110**, then secured in its respective manner.

In an additional embodiment, embedding fastener **110** may be used to remove an implant and/or bondable material **120**. For example, the ability of conventional medical tools to remove a previously installed implant or bone cement may be limited. Embedding fastener **110** may be used to obtain additional fixation. Once embedding fastener **110** is secured to the implant and/or bondable material **120**, force and/or vibratory energy may be used to remove the implant and/or bondable material **120**.

FIGS. 11-17 show an additional embodiment for use with an implanted embedding fastener **110**. Fastener **136** is shown in FIGS. 11-12 and washer **146** is shown in FIGS. 13-14. Although fastener **136** and washer **146** may be made of any material disclosed herein or known in the art, it may be preferable to use PEEK. After embedding fastener **110** has been secured with respect to bondable material **120**, fastener **136** may be engaged into embedding fastener **110**. In another embodiment, washer **146** may be used in conjunction with fastener **136** as shown in FIGS. 15-17. Additionally, fastener **136** may be bonded to embedding fastener **110** and/or washer **146**.

Referring to FIGS. 18-19, fastener **136**, washer **146**, and/or tissue implant **154** may be used to secure soft tissue **152** to hard tissue **156**, for example to secure the rotator cuff tissue to the proximal humerus or for any other procedure disclosed herein. Additionally, washer **146** and/or tissue implant **154** could be made of collagen or other materials that promote tissue growth.

With reference to FIGS. 20-25, embedding fastener **110** may be provided with channel **112**. For example, channel **112A**, **112B**, **112C**, and/or **116** may be used. Channel **112** may extend through the surface of embedding fastener **110** to facilitate the bonding of embedding fastener **110** to bondable material **120**, fastener **136**, and/or any fastener

disclosed herein. Channel **112** may provide a path for softened and/or molten bondable material to be displaced, providing room for entry of embedding fastener **110**. Channel **116** may also include radial gaps, chambers, or ports. To accommodate for embedding fastener **110** displacing a substantial amount of material, channels may be extended along the entire length of embedding fastener **110**, and may further extend along end effector **104**. Channel **116** may be further operative to reduce the possibility of rotation of fastener **110** within bondable material **120**. Channel **116** is thus disposed to extend into bondable material **120** after insertion, and may extend to the face of embedding fastener **110**. Additionally, embedding fastener **110** may have feature **158** to help attach and remove it from end effector **104**.

In an embodiment shown in FIGS. 26-27, implant **162** may be coated in bondable material **164** and implanted in body tissue **160**. For example, a metal rod coated with bone cement may be placed in the intramedullary canal of a bone. In an embodiment, support **168** may be placed in a location to facilitate stabilization. Support **168** and/or washer **146** may be referred to as a supporting implant. One or more holes may be formed in body tissue **160** and up to or into bondable material **164** to coincide with the holes in support **168**. Embedding fastener **110** may be placed through the holes in body tissue **160** and secured to and/or bonded to bondable material **164** as discussed herein. Then, one or more of fastener **136** are secured to and/or bonded to the one or more embedding fastener **110**, thereby securing support **168** relative to body tissue **160**.

Referring to FIG. 28, implant **162** may be installed in a body tissue **160** with bondable material **164**, for example bone cement. Bondable material **164** may be any material described herein or known in the art. Implant **162** may require stabilization because implant **162** has become loose and/or requires stabilization due to tissue defect **166**, for example a periprosthetic fracture. Tissue defect **166** may include, but is not limited to, damaged, deformed, and/or diseased bone, muscle, ligament, tendon, cartilage, capsule, organ, skin, nerve, vessel, or other body part. For example, a femur may be fractured or contain osteoporosis. Support **168** is fixed to body tissue **160** with fastener **170** to provide stabilization. Support **168** may be an internal bone plate, an external bone plate, a spinal plate, a wedge, a cushion, a pad, or other biocompatible support used for stabilization of tissue and/or implants. Fastener **170** may be any fastener described herein or any other biocompatible fastener known in the art.

In an embodiment, implant **162** has been previously installed and requires stabilization. One or more holes are formed through body tissue **160** and up to or into bondable material **164**. Embedding fastener **110** is inserted through a hole and bonded to bondable material **164** by utilizing instrument **100** described above. Fastener **170** engages embedding fastener **110** to secure support **168** to body tissue **160**. The head of fastener **170** may be deformed and/or bonded to support **168** to reduce loosening of fastener **170**.

Embedding fastener **110** can also be bonded to bondable material **164C/D** that is within or on the surface of body tissue **160**. For example, bondable material may have been used to repair tissue defect **166**. Bondable material **164** may be within or on the surface of body tissue **160**. A hole is formed up to or into the bondable material **164**. Embedding fastener **110** is bonded into bondable material **164**. Fastener **170** passes through support **168** and into engagement with embedding fastener **110** to secure support **168** relative to body tissue **160**.

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In another embodiment, embedding fastener 110A/B can be bonded to and/or into implant 162. The procedure is performed as described above, except the embedding fastener 110 may be bonded directly to implant 162.

In another embodiment, bondable material 164 may asymmetrically cover all or a portion of implant 162. The thickness of bondable material 164 could vary in the radial direction or along the length of implant 162. An asymmetrically coated implant 162 may provide additional purchase for fastener 136 or indication of orientation or position of implant 162.

In an additional embodiment, indirect visualization may be used to identify and/or change the orientation or position of implant 162 or fastener 136. Examples of indirect visualization may include endoscopic guidance, computer assisted navigation, magnetic resonance imaging (MM), CT scan, ultrasound, fluoroscopy, X-ray, or other visualization technique disclosed in any of the references incorporated herein. Asymmetric coating, radiopaque markers, or other features identifiable with indirect visualization may be used to identify and/or adjust orientation or position. Indirect visualization may also be used to align fastener 136 with holes in implant 162 or bondable material 164. The holes may be predrilled in implant 162 or bondable material 164 or may be drilled after installation of 162. Indirect visualization may be used to create a hole or holes in tissue to align with holes in implant 162 or bondable material 164.

For example, an intramedullary rod could be asymmetrically coated with PEEK. The intramedullary rod could have predrilled holes in the PEEK coating. After the rod is installed in the intramedullary canal of the tibia, the orientation of the rod may be determined using indirect visualization to locate the area with a thicker coating. The orientation of the rod may be adjusted to the appropriate location for holes to be made through the tissue.

With further reference to FIG. 28-29, various types of fastening devices are used to position support 168 along body tissue 160. Alternatively, support 168 may be positioned upon the surface of the skin, or at any point between the tissue surface and the skin, according to the requirements of the surgical procedure. Further, support 168 may be placed within the bone, for example in an intramedullary canal.

Referring to FIG. 29, fastener 170 may be used in intramedullary, percutaneous, and/or retrograde approaches. Fastener 170 may be bonded to bondable material 164, or a surface of implant 162. The head of fastener 170 may be provided, or may be formed using vibratory energy. A head may also be formed on the distal end of fastener 170. Fasteners 170E are shown to be bonded into the bondable material 164. Fastener 170F is shown to be bonded at the distal end and/or to bondable material 164 within the body tissue and is placed through tissue defect 166. Fasteners 170G are shown passing directly through body tissue 160, which may be fastener 170T and sleeve 171T in FIGS. 52-53 and as described below. Additional embodiments of fastener 170 are disclosed in U.S. patent application Ser. No. 12/202,210 entitled "Methods and Devices for Utilizing Thermal Energy to Bond, Stake and/or Remove Implants", which is incorporated by reference herein.

Additionally, cerclage wire 172 may be employed as known in the art, to provide further stabilization, in combination with fastener 170. For example, cerclage wire 172A may be bonded to support 168. Bondable material 164E

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could be used to affix cerclage wire 172A to support 168. In another example, cerclage wire 172B may be tied around support 168. Also, cerclage wire 172C may be fastened using a mechanical or bonded crimp 174. In additional example, cerclage wire 172D may be fastened to the side of support 168 or between support 168 and body tissue 160.

Referring to FIG. 30, end effector 104 of instrument 100 may be connected into implant 162 at recess 180. This connection may be threaded, magnetic, friction, hex, ball and socket, linkage, adhesive, and other connections suitable for transferring vibratory energy as disclosed herein or known in the art. Also, other vibratory energy devices as disclosed herein or known in the art may be utilized.

FIG. 30 also shows additional methods of stabilizing a loose implant and/or facilitating the solidification and/or polymerization of bondable material 164. For example, fastener 170H may be a metal and/or polymer fastener, which may be affixed to the bondable material 164 and/or implant 162 to stabilize implant 162. In another example, fastener 170I may be metal coated with bondable material. Upon the application of vibratory energy and/or heat, distal end 182A deforms thereby stabilizing the gap between implant 162 and body tissue 160. In an additional example, fastener 170J may be made of bondable material. Upon the application of vibratory energy and/or heat, distal end 182B deforms, thereby stabilizing the gap between implant 162 and body tissue 160.

Referring to FIG. 31, end effector 104 of instrument 100 may be connected into implant 162 with coupler 184 to stabilize previously hardened and/or polymerized bondable material 164 or to facilitate solidification and/or polymerization of bondable material 164. This connection may be threaded, magnetic, friction, hex, ball and socket, linkage, adhesive, and other connections suitable for transferring vibratory energy as disclosed herein or known in the art. Also, other vibratory energy devices disclosed herein or known in the art may be utilized.

Referring to FIG. 32, attachment 186 may be attached and/or bonded to implant 162. Attachment 186 may be made from any material described herein (i.e. collagen, graft, or growth promoter) or any other material known in the art, preferably to promote healing and/or contain bondable material 164. For example, vibratory energy may be used to bond attachment 186 to implant 162. In another example, fastener 170 may secure attachment 186 to implant 162. In an additional example, vibratory energy may be used to bond fastener 170 to attachment 186.

Referring to FIGS. 33-34, implant 162 may be manufactured with reservoir 192 or reservoir 192 may be formed during or after implantation. Additionally, therapeutic substance 194 may be incorporated in reservoir 192 of implant 162, impregnated in implant 162, or coated on or in implant 162. As shown in FIG. 33, reservoir 192 may be located in implant 162. Alternatively, reservoir 194 may be formed in body tissue 160, as shown in FIG. 34. Cap 188 may be made of bondable material. Additionally, cap 188 may be attached and/or bonded to enclose reservoir 192. Implant 162, cap 188, and/or body tissue 160 may contain attachment feature 190 and/or attachment recess 196 to facilitate mechanical attachment and/or bonding with end effector 104. Additionally, implant 162, cap 188, and/or bondable material 164 may be porous to facilitate the delivery of therapeutic substance 194.

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Referring to FIG. 35, therapeutic substance **194** may be contained in implant **198**, for example drugs or antibiotics contained in an acetabular cup. Implant **198** may be manufactured with reservoir **192A** and/or reservoir **192B** or the reservoirs may be formed during implantation. Any combination of one or more reservoir **192A** and/or reservoir **192** may be used. Cap **188** may be coated with bondable material. Additionally, any of caps **188A-D** may be attached and/or bonded to enclose reservoir **192A** or **192B**, which may provide the potential benefit of multiple release times for therapeutic substance **194**. Caps **188A-D** may contain attachment features **190A-D** to facilitate mechanical attachment and/or bonding. Additionally, implant **198** and/or any of caps **188A-D** may be porous to facilitate the delivery of therapeutic substance **194**.

Referring to FIG. 36, fastener **170** may be used to stabilize implant **162**, for example a tibial component of a total knee arthroplasty (TKA). In an embodiment, fastener **170K** may be bonded to bondable material **164** on the underside of the implant or bonded directly to implant **164**. In another embodiment, fastener **170L** may be bonded to the portion of implant **162** that is within body tissue **160**. In an additional embodiment, fastener **170** may be used to secure tissue graft **202** to implant **162** and/or body tissue **160**. For example, tissue graft **202** may be an allograft. Any embodiment of fastener **170** that has been described herein or known in the art may be used.

Referring to FIGS. 37-38, bondable material **164** may be utilized to stabilize body tissue **160**. For example, bone cement in previously performed kyphoplasty may become loose and require additional stabilization. In this example, the fasteners may utilize the previously implanted bone cement to stabilize the spine instead of removing and reapplying bone cement. In an embodiment, body tissue **160** has been previously implanted with bondable material **164**. Fastener **170N** is passed through support **168** and bonded to bondable material **164**. As shown in FIG. 37, one or more fastener **170** is passed through support **168** and secured and/or bonded to surrounding tissue **200**. Fasteners **170M**, **170N**, and **170P** may be any embodiment disclosed herein or known in the art. Additionally, fasteners **170M**, **170N**, and **170P** may be used with embedding fastener **110** as described above. Also in these embodiments, vibratory energy may be used to stabilize previously hardened and/or polymerized bondable material or to facilitate the solidification and/or polymerization of bondable material.

Referring to FIGS. 39-41 and 50-51, an additional embodiment of instrument **100** includes guide sheath **102**, spring **204**, and/or force regulator **206**. In an embodiment in FIG. 41, guide sheath **102** may align washer **146** when the tip of end effector **104** is placed in contact with fastener **136**. As fastener **138** is staked and the tip of fastener **136** is shaped with the application of vibratory energy, such as ultrasonic energy, guide sheath **102** may allow end effector **104** to advance while applying force to washer **146**, support **168**, body tissue **160**, and/or bondable material **164** (not shown in FIG. 41). In a further embodiment in FIG. 41, guide sheath **102** may hold, guide, align, and/or deliver washer **146**, fastener **136**, or other fasteners referenced herein or known in the art. In an additional embodiment, regulating sheath **206** may have a spring **204**, for example any spring, cushion, or other material or device known in the art for spring/damping applications. Additionally, instrument **100** may have regulating tab **206** for manually applying and/or regulating the movement of guide sheath **102**. Although the embodiment in FIG. 41 may be used under a

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limitless number of configurations and settings, Table 2 is being set forth with operative examples:

TABLE 2

Polycarbonate Fastener Bonding				
Instrument: Handpiece P05 with tuning of 39,000-45,000 Hz System Settings: 39,500 Hz, SOW, 1.0 sec weld time				
Test Sample Number	Power (watts)	Energy Application Time (sec)	Force Applied to Break (lbs.)	Deformation Depth (inches)
1	30	1.69	93.2	0.115
2	28	1.68	86.4	0.110
3	31	1.78	98.2	0.111
4	27	1.80	91.3	0.108
5	31	1.69	109.2	0.109

In additional embodiments, frequency may preferably be between 20 to 80 khz, power may preferably be between 5 to 200 watts, and energy application time may be preferably between from 0.1 to 5 seconds.

In an embodiment, a sensor may be included in instrument **100**. For example, a force, pressure, or temperature sensor may be used to measure bonding and/or staking. In another example, a visual and/or audio indicator may be operatively connected to the sensor, which may be used to indicate a proper bond/stake. In another embodiment, a visual and/or audio indicator may be connected to instrument **100** or the energy generator, which may be used to illustrate and/or teach proper technique during bonding and/or surgery. In another example, the visual and/or audio indicator may indicate completion of a proper bond/stake, over/under application of force, or expiration of desired energy application time.

In another embodiment, a vacuum may be operatively connected to **100**. For example, the vacuum may be communicatively connected between the guide sheath **102** and end effector **104**, which may be used for the removal of debris from instrument **100**.

Referring to FIGS. 42-43, an additional embodiment of fastener **170** may include one or more feature **208**. Feature **208** may increase or decrease the transfer of energy across fastener **170**. Feature **208** may pass into the surface or surfaces of fastener **170**. Feature **208** may be on any surface or surfaces of fastener **170** and/or contain therapeutic substances. Fastener **170** may include effector interface **210**, preferably for engagement with end effector **104**.

Referring to FIGS. 44-45, embedding fastener **110** may also include thread **212**. Embedding fastener **110** may be screwed and/or engaged into bondable material, tissue, and/or any other material disclosed herein, preferably by engaging interface **214** with a screw driver or other tool. Embedding fastener **110** may include an interface **214** which may be radiused, chamfered, funnel-shaped, threaded, or any other shape, for example square, rectangular, circular, elliptical, triangular, hexagonal, or asymmetrical shape. Embedding fastener **110** may be made of any metal, polymer, or other material disclosed herein.

Referring to FIGS. 46-47, fastener **170** may include effector interface **216**. Preferably for engagement with end effector **104**, effector interface **216** may be radiused, chamfered, funnel-shaped, threaded, or any other shape, for example square, rectangular, circular, elliptical, triangular, hexagonal, or asymmetrical shape.

Referring to FIGS. 48-49, instrument **100** may include end effector **104** dimensioned and configured for a point of

maximum displacement at or near the middle of fastener **170** at or near point **218B** (FIG. **49**). In another embodiment, it may be preferable for instrument **100** to include end effector **104** dimensioned and configured for a point of maximum displacement at or near the end of fastener **170** at or near point **218A** (FIG. **48**). For most applications, a point of maximum displacement at or near point **218B** is preferred. Fastener **170** may be threaded or have an interference fit with end effector **104**.

A point of maximum displacement along end effector **104** may occur at increments of about half its wavelength, which may be determined by the ratio of the speed of sound through the material of end effector **104** to the frequency of the wave propagated through end effector **104**. The end effector **104** may be made of titanium or any material disclosed herein. For example, at a frequency of 20 khz, points of maximum displacement along end effector **104** made of titanium may be in increments of about 4 to 6 inches, preferably 4.8 to 5.1 inches. For example, at a frequency of 40 khz, points of maximum displacement along end effector **104** made of titanium may be in increments of about 2 to 3 inches, preferably 2.4 to 2.5 inches.

A point of maximum displacement along fastener **170** may also occur at increments of about half its wavelength. Fastener **170** may be made of PEEK, PLLA, or any material disclosed herein. As an example for PEEK, at a frequency of 20 khz, points of maximum displacement along end effector **104** may be in increments of about 1 to 2 inches, preferably 1.6 to 1.7 inches. As another example for PEEK, at a frequency of 40 khz, points of maximum displacement along end effector **104** may be in increments of about 0.5 to 1 inch, preferably 0.8 inches. As an example for PLLA, as an example at a frequency of 20 khz, points of maximum displacement along end effector **104** may be in increments of about 1 to 2 inches, preferably 1.3 to 1.4 inches. As an example for PLLA, as an example at a frequency of 40 khz, points of maximum displacement along end effector **104** may be in increments of about 0.5 to 1 inch, preferably 0.7 inches.

To optimize bonding and/or reduce the stress applied, the desired point of bonding on fastener **170** should be at or near a point of maximum displacement. For example, the desired point of bonding on fastener **170** may be along half its length or at its tip (see **218A** of FIG. **48**). If the desired point of bonding is at a point of minimal or zero displacement (see **218B** of FIG. **49**), bonding may be difficult. To facilitate bonding, it may be preferable to increase power or amplitude of the signal, thereby increasing energy applied to fastener **170**.

Referring to FIGS. **52-53**, fastener **170T** may be used with sleeve **171T**, potentially to contain and/or release therapeutic substances into a body. Fastener **170T** may engage with sleeve **171T** by mechanical interlock, thread, or vibratory energy bond. End effector **104** may engage with fastener **170T** for vibratory energy bonding. Any fastener **170** disclosed herein may have energy director **222** to facilitate bonding by directing energy to the desired location of bonding.

Referring to FIGS. **54-55**, fastener **170U** may have energy director **224**. Energy director **224** may facilitate bonding with implant **162**, especially if a portion or the entirety of implant **162** includes a porous material. For example, implant **162** may include a porous metal. End effector **104** may engage with fastener **170U** for vibratory energy bonding.

Referring to FIG. **56**, fastener **170F** may be embedded into a bondable material with its leading end and stabilize a

support **168** (i.e. plate) with its trailing end. End effector **104** may engage with fastener **170F**, preferably near the trailing end, for vibratory energy bonding. Fastener **170F** may include any material disclosed herein, but preferably titanium or titanium with at least a portion coated with PEEK or PLLA.

Referring to FIGS. **57-58**, fastener **170W** may be used with, preferably disposed within, expanding anchor **226**. End effector **104** may engage with fastener **170W**, preferably near the trailing end, for vibratory energy bonding. A hole in tissue and/or bondable material may be formed or drilled into body tissue **160** prior to or during the implantation of fastener **170W** and expanding anchor **226**. Fastener **170W** and expanding anchor **226** may pass into body tissue **160**, for example in the configuration shown in FIG. **57**. Fastener **170W** may be retracted into expanding anchor **226**, preferably after being positioned in body tissue **160**. Expanding anchor **226** may expand outwards (shown in FIG. **58**), thereby engaging and/or exerting a radially outward force on body tissue **160**. Preferably after fastener **170W** and expanding anchor **226** are in an expanded condition, vibratory energy may be applied to fastener **170W**, preferably near the trailing end, to bond fastener **170W** and expanding anchor **226** together. In another embodiment, fastener **170W** and/or expanding anchor **226** may be configured to bond into a bondable material. After fastener **170W** and expanding anchor **226** have been stabilized in the expanded configuration, the excess length of the trailing end of fastener **170W** may be removed to be substantially flush with the trailing end of expanding anchor **226**.

Referring to FIG. **59**, fastener **170** may be used to increase interference between body tissue **160** and implant **198**. Fastener **170** may be an interference screw and/or used in conjunction with interference implant **228** to position and/or stabilize implant **198**. Fastener **170** may pass through all or a portion of implant **198**. In additional embodiments, fastener **170** may stabilize body tissue **160** (i.e. ACL graft) against the side of a hole in body tissue **160** (i.e. bone) (not shown), stabilize body tissue **160** (i.e. soft tissue) to another body tissue **160** (i.e. bone), or stabilize interference implant **228** (or tissue graft **202**) to body tissue **160** (i.e. bone) and/or implant **198** (FIG. **59**). In another embodiment, fastener **170** may be part or entirely made of a biodegradable and/or bondable material. In an embodiment, fastener **170** may have a snap that would overlay part of implant **198** for interference. In another embodiment, implant **198** may include porous surface **230** (FIG. **59**) or a coating of bondable material. In another example, implant **198** may be bonded with vibratory energy, hydrophilic, and/or mechanically expandable against body tissue **160** (i.e. bone), which may allow the implant to sequentially expand and provide interference against body tissue **160** (i.e. bone) or another implant.

In an embodiment related to hip (or shoulder) resurfacing, implant **198** may be an acetabular component or cup (or glenoid component), which is commonly stabilized using screws through its center. To replace the use of these screws or to provide additional stabilization, implant **198** may be stabilized by positioning fastener **170** between implant **198** and body tissue **160** (i.e. acetabulum or glenoid), which may urge implant **198** to the desired position and/or enhance interference with body tissue **160**. In an embodiment, implant **198** may be free of holes, as fastener **170** may provide the majority of interference.

There are many different features to the present invention and it is contemplated that these features may be used together or separately. Thus, the invention should not be

limited to any particular combination of features or to a particular application of the invention. Further, it should be understood that variations and modifications within the spirit and scope of the invention may occur to those skilled in the art to which the invention pertains. Accordingly, all expedient modifications readily attainable by one versed in the art from the disclosure set forth herein that are within the scope and spirit of the present invention are to be included as further embodiments of the present invention.

The invention claimed is:

1. A vibratory energy system for bonding a first component to a second component, comprising:

a hand-held instrument comprising:

a proximal end configured to couple to an energy source to enable the hand-held instrument to receive energy from the energy source;

a vibratory energy generator configured to generate mechanical vibratory energy from the energy received from the energy source and to control application of the mechanical vibratory energy;

a force regulating mechanism configured to control application of force applied by the vibratory energy generator;

a sensor configured to measure at least one of force or pressure applied by the vibratory energy generator during the bonding of the first component to the second component;

a housing enclosing the vibratory energy generator, the force regulating mechanism, and the sensor, the housing having an external gripping surface for a human hand; and

a switch coupled to the housing and operatively connected to the vibratory energy generator, the switch configured to activate the vibratory energy generator; and

an end effector configured to attach to a distal end of the hand-held instrument and operatively connected to the vibratory energy generator, wherein the end effector is configured to contact the first component; and

an indicator operatively connected with the sensor and configured to provide one or more of a visual indication of the bonding or an audible indication of the bonding based on the at least one of force or pressure measured by the sensor during the bonding of the first component to the second component,

wherein during contact of the first component with the second component, the hand-held instrument is operable to apply the mechanical vibratory energy from the vibratory energy generator through the end effector to the first component to bond the first component to the second component.

2. The vibratory energy system of claim 1, wherein the first component is formed, at least in part, from a first polymer and the second component is formed, at least in part, from a second polymer that is chemically the same as the first polymer.

3. The vibratory energy system of claim 1, wherein the first component is formed, at least in part, from a first polymer and the second component is formed, at least in part, from a second polymer that is chemically different from the first polymer.

4. The vibratory energy system of claim 1, wherein the first component is formed, at least in part, from a polymer and the second component is formed, at least in part, from a metal.

5. The vibratory energy system of claim 1, wherein the mechanical vibratory energy is ultrasonic energy, and

wherein the force regulating mechanism comprises a dampening component.

6. The vibratory energy system of claim 1, wherein the end effector is formed, at least in part, from titanium.

7. The vibratory energy system of claim 1, wherein the end effector comprises a connector to couple to the first component prior to bonding and to decouple the end effector from the first component after the bonding.

8. The vibratory energy system of claim 1, wherein the end effector is configured to be detachable from the distal end of the hand-held instrument, and wherein the hand-held instrument is configured to employ the end effector dependent on material properties of at least one of the first and second components.

9. The vibratory energy system of claim 1, further comprising a guide sheath configured to align with the end effector during the bonding of the first component to the second component and to apply a force to the first component,

wherein the guide sheath is further configured to align the end effector at a desired point of bonding of the first component to the second component, and wherein the hand-held instrument comprises a manual regulating tab for manually regulating movement of the guide sheath.

10. A vibratory energy system for bonding a first component to a second component, comprising:

an energy source;

a user interface coupled to the energy source; and

a hand-held instrument comprising:

a proximal end configured to couple to the energy source to enable the hand-held instrument to receive energy from the energy source;

a vibratory energy generator configured to generate mechanical vibratory energy from the energy received from the energy source and to control application of the mechanical vibratory energy;

a force regulating mechanism configured to control force applied by the hand-held instrument;

a sensor configured to measure at least one of force or pressure applied by the hand-held instrument during the bonding of the first component to the second component;

a housing enclosing the vibratory energy generator, the force regulating mechanism, and the sensor, the housing having an external gripping surface for a human hand; and

a switch coupled to the housing and operatively connected to the vibratory energy generator, the switch configured to activate the vibratory energy generator; and

an end effector configured to attach to a distal end of the hand-held instrument and operatively connected to the vibratory energy generator, wherein the end effector is configured to contact the first component;

an indicator operatively connected with the sensor and configured to provide one or more of a visual indication of the bonding or an audible indication of the bonding based on the at least one of force or pressure measured by the sensor during the bonding of the first component to the second component,

wherein during contact of the first component with the second component, the hand-held instrument is operable to apply the mechanical vibratory energy from the vibratory energy generator through the end effector to the first component to bond the first component to the second component.

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11. The vibratory energy system of claim 10, wherein the mechanical vibratory energy is ultrasonic energy.

12. The vibratory energy system of claim 10, wherein the force regulating mechanism comprises a dampening component, and

wherein the end effector comprises a connector to couple to the first component prior to bonding and to decouple the end effector from the first component after the bonding.

13. The vibratory energy system of claim 10, wherein the end effector is detachable, and wherein the hand-held instrument is configured to the end effector dependent on material properties of at least one of the first and second components.

14. The vibratory energy system of claim 10, wherein the end effector is detachable, and wherein the hand-held instrument is configured to employ the end effector dependent on configurations of at least one of the first and second components.

15. The vibratory energy system of claim 10, further comprising a guide sheath configured to align the end effector centered on a desired point of bonding of the first component to the second component.

16. A vibratory energy system for bonding two or more materials, comprising:

an energy source;

a hand-held vibratory energy application device comprising a housing having an external surface configured to be held by a human hand, the vibratory energy application device coupled to the energy source and comprising a vibratory energy generator configured to generate mechanical vibratory energy, wherein the vibratory energy generator is enclosed within the housing;

a switch coupled to the hand-held vibratory energy application device and configured to control activation of the mechanical vibratory energy for the bonding of the two or more materials;

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a second control mechanism incorporated into the hand-held vibratory energy application device and configured to control a force transmitted by the hand-held vibratory energy application device; and

at least one of a force sensor or a pressure sensor configured to output one or more signals indicative of at least one of the force or the pressure applied by the hand-held vibratory application device during the bonding between the two or more materials; and

an indicator operatively connected with the sensor and configured to provide one or more of a visual indication of the bonding or an audible indication of the bonding based on the at least one of force or pressure measured by the sensor during the bonding between the two or more materials.

17. The vibratory energy system of claim 16, comprising a regulatory mechanism incorporated into the vibratory energy application device and configured to regulate force applied through the vibratory energy application device to the two or more materials.

18. The vibratory energy system of claim 16, wherein the hand-held vibratory energy application device is configured for at least one detachable end effector, wherein the detachable end effector is configured to provide bonding according to configurations of the two or more materials.

19. The vibratory energy system of claim 16, wherein the hand-held vibratory energy application device is configured for at least one detachable end effector, wherein the detachable end effector is configured to provide bonding according to chemical properties of the two or more materials.

20. The vibratory energy system of claim 16, further comprising:

a coupler configured to fix the first material relative to the second material, and

wherein the second control mechanism comprises a damping component incorporated into the hand-held vibratory application device.

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